Levels of adaptivity and spectral worst-case bounds for labeled triangle joins

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Abstract. We ask how much adaptivity a variable-at-a-time join needs to be worst-case optimal for the labeled triangle $Q(x,y,z) = R(x,y) \wedge S(y,z) \wedge T(z,x)$, and for which statistics. We compare a fixed attribute order, the best of the six orders per instance, a fixed root with a per-binding choice of the next variable, and a root chosen per instance with the next variable chosen per binding. For the cost C of the best such choice, with every intersection charged at its shorter run, and the cyclic ℓ_2 bound B_2 we prove $C \le N + (N/3)^{3/8} B_2^{3/4}$ and $C \le N + 3^{1/8} N^{-1/8} B_2^{5/4}$, and explicit families attain their minimum up to a constant factor at every slope $B_2 \asymp N^{\kappa}$, $\kappa \in [9/10, 3/2]$. So this unpartitioned class cannot attain $O(N + B_2)$, even up to polylogarithmic factors (exponent $9/8$ at $B_2 \asymp N$), while known degree-partitioned algorithms can; the first known violation has $N \approx 2 \cdot 10^7$. When every tuple lies in a triangle the gap exponent lies in $[1/10, 1/8]$. Against $N + B_2$ one fixed order loses exactly $\Theta(N^{1/4})$ on general inputs and the best of the six orders $\Theta(N^{1/5})$. The lower bounds also hold for a single directed predicate and for fresh comparison-based sorted intersections under adversarial identifiers, where the worst case at $B_2 \asymp N$ is $\Theta(N^{9/8}(\log N)^{1/4})$. Under relation sizes and operator norms the worst-case triangle output lies in $(U/4, U]$ for an explicit nine-term U, with a sharp factor 4 (checked in Lean 4), and the best driver of the kernel does work at most U; for the labeled 4-cycle an explicit 38-term U_4 gives $(U_4/16, U_4]$. Engine measurements on knowledge-graph triangle queries are reported with their limits.

1. Introduction

RDF engines answer basic graph patterns over a single ternary relation of triples, stored as several sorted permutations (for example PSO and POS). A pattern with a constant predicate p is then a binary relation $p \subseteq V \times V$ that can be read in either direction, and the out- and in-degree of any term is a constant-time read of a compressed-sparse-row (CSR) offset array. The simplest cyclic pattern, the triangle

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?x p1 ?y .  ?y p2 ?z .  ?z p3 ?x
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is the canonical case where left-deep binary plans can be polynomially worse than worst-case optimal joins (WCOJ) [AGM13, NPRR18, Vel14]. Leatherback, the RDF engine this work comes from (to be open-sourced), answers triangles with a variable-at-a-time kernel that, for each tuple (x, y) of a driver relation, intersects two sorted neighbour runs from the smaller side (Appendix B).

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Levels of adaptivity. Such a kernel can adapt at several levels, and this paper asks how much adaptivity worst-case optimality needs, and for which statistics. A *fixed global attribute order* (GAO) fixes the variable order once. Choosing the *best of the six orders* per instance amounts, for the triangle, to choosing which relation drives the kernel, its *rotation*. A *fixed root* fixes the first variable and chooses the second variable separately for each binding of it; *per-binding choice* also chooses the root per instance. Finally, *partitioned* algorithms split a variable's domain, for example into degree buckets, and plan each part separately [JRR21, ANOS24, ANS25]. Table 1 places these levels against two comparators: the cost $C + N$ of the best per-binding choice, and the cyclic ℓ_2 bound B_2 of [ANOS24, JRR21] (plus N). The headline is the last two rows: per-binding choice costs at most $N + (N/3)^{3/8} B_2^{3/4}$ and no better, so for this unpartitioned shorter-side kernel the worst-case exponent at $B_2 \asymp N$ is 9/8, not 1; every coarser level loses $\Theta\big(N^{1/2}\big)$ against $C + N$. The lower bounds against $N + B_2$ also hold when intersections are fresh comparison-based calls on sorted runs and the adversary assigns the identifiers (Section 3.6).

Table 1: The levels of adaptivity and their worst-case gaps δ (cost at most $N^{\delta+o(1)}$ times the comparator, and attained). "Reduced" means pairwise semijoin-reduced; "fully reduced" means that every tuple lies in a triangle. Upper bounds 1/2 come from Lemma 3.1. Every lower bound against $N + B_2$ also holds for fresh comparison-based sorted-intersection calls under adversarial identifiers (Section 3.6), where per-binding choice gains exactly a factor $(\log N)^{1/4}$ at $B_2 \asymp N$; against $C + N$ the losses of $\mathcal{F}_a$ and $\mathcal{G}_a$ are established there only up to a factor $\log N$ (Remark 3.27). For the spectral statistics (r, σ) of Section 4, the best of six GAOs and per-binding choice have gap 0 against $N + U$ (Corollary 5.3, Lemma 3.1), and one fixed GAO has gap at least 1/4 (Proposition 5.5).

Level	**Against $C + N$**	**Against $N + B_2$**
One fixed GAO	1/2 ($\mathcal{F}_a$, reduced)	exactly 1/4 (Theorem 3.18, GMR); reduced: $[1/5, 1/4]$
Best of six GAOs (rotation per instance)	1/2 ($\mathcal{G}_a$, reduced)	exactly 1/5, also reduced (Theorem 3.17)
Fixed root, per-binding option	1/2 ($\mathcal{H}_a$, padded: reduced)	$[1/8, 1/4]$ (G_8; Corollary 3.19)
Per-binding (root and option)	0	exactly 1/8 (Theorem 3.3, Theorem 3.6); 0 for $B_2 \le N^{9/10}$; fully reduced: $[1/10, 1/8]$ (Theorem 3.14)
Partitioned [JRR21]	—	0 (outside our cost model)

Why sizes and operator norms. Besides the ℓ_2 statistics we study one statistic beyond sizes: the operator norm $\sigma = \|A\|_2$ of each relation's 0/1 incidence matrix A. The class (r, σ) of sizes and norms is worth a section for four reasons. It is *cheap*: one number per predicate, computed at compaction time. It is *certifiable*: a few Collatz–Wielandt steps give a guaranteed upper bound on σ, and the resulting bound is at most 2% above U on 90% of our queries (Section 6). It gives a *guarantee, not an estimate*, and on real queries rarely the tightest one (Section 6): the nine-term U bounds the output of every admissible instance with constant 1 and is attained up to a factor 4. And, among the classes studied here, it is the one in which *per-instance rotation is provably the right lever*: under sizes alone every rotation is already worst-case optimal (each does work at most AGM [NRR13]); under

ℓ_2 statistics no unpartitioned level is (Corollary 3.7); under (r, σ) the best rotation does work at most $U < 4W$ (Corollary 5.3), while a fixed rotation loses a polynomial factor (Proposition 5.5). The statistic sits between degrees and sizes: every degree is at most σ^2 (Lemma 4.2), and σ^2 is at most the size, because $\|A\|_2^2 \le \|A\|_F^2$ (Section 2).

Contributions

1. **Per-binding choice and the cyclic ℓ_2 bound (Section 3).** For the per-binding cost C (root variable chosen per instance, second variable chosen per binding, every intersection charged at its shorter run), $C \le N + (|R|\ |S|\ |T|)^{1/8}\min(\Lambda, \Lambda')^{1/2} \le N + (N/3)^{3/8}B_2^{3/4}$ with constant 1 (Theorem 3.3), and $C \le N + 3^{1/8}N^{-1/8}B_2^{5/4}$ (Theorem 3.4). An explicit family with $C/(N + B_2) \ge (N/9)^{1/8}/14$ refutes $C = O(N + B_2)$ and its polylogarithmic relaxation (Theorem 3.6, Corollary 3.7), also on inputs that are both semijoin-reduced and connected (Remark 3.9). The minimum of the two bounds is the exact worst case, up to a constant factor, at every real slope $B_2 = N^\kappa$, $\kappa \in [9/10, 3/2]$ (Theorem 3.10, Corollary 3.11, Theorem 3.12), and $C = O(N)$ below $9/10$. When every tuple lies in a triangle the gap lies in $[1/10, 1/8]$, and the fully reduced worst case equals the general one at every slope in $[11/10, 3/2]$ (Theorem 3.14). The refutation is asymptotic: the first violation we know needs $N \approx 2 \cdot 10^7$, and on the refuting family at the sizes we evaluated exactly ($N \le 4.6 \cdot 10^5$), $C/(N + B_2)$ is between 0.25 and 0.42 (Table 4). Per-binding choice itself is prior art [NRR13, ABGHNR24, ABFGN25]; the contribution is the bound for this cost model. **[Proved]**
2. **Exact gaps of the coarser levels (Section 3.5).** Against $N + B_2$, one fixed order has gap exactly $1/4$ on general inputs (between $1/5$ and $1/4$ on semijoin-reduced inputs), the best of the six exactly $1/5$, also on a semijoin-reduced family with $C = \Theta(N)$, and a fixed root between $1/8$ and $1/4$, against $1/8$ for per-binding choice (Theorem 3.18, Theorem 3.17, Corollary 3.19, Table 1). Against $C + N$, one fixed GAO, the best of the six and a fixed root can each lose $\Theta\big(N^{1/2}\big)$ on pairwise semijoin-reduced inputs (Theorem 3.16). A deterministic price-then-run scheduler selects a root of cost exactly C in $C + \min(6C, 2N) + 4n + O(1)$ per-edge units without output buffering (Proposition C.2). **[Proved]**
3. **Fresh sorted-intersection calls and a single predicate (Section 3.6, Proposition 3.21).** When intersections are sorted runs charged by key comparisons, every call is fresh (it knows only its two runs) and the adversary assigns the identifiers, every lower bound against $N + B_2$ survives without logarithmic loss (Theorem 3.25, Theorem 3.26); against $C + N$ some ratios are established only up to a factor $\log N$ (Remark 3.27). The upper bound survives up to a factor $(\log N)^{1/4}$ (Theorem 3.29), and that factor is necessary, also for randomized routines (Theorem 3.32), so the worst case at $B_2 \asymp N$ is $\Theta\big(N^{9/8}(\log N)^{1/4}\big)$. The certificate and doubling-search ingredients are classical [DLM00, BK02, BY76]; what is new is their simultaneous realization inside these join constructions and the resulting tradeoff. The $9/8$ lower bound and the order separations also hold for one directed predicate, while a symmetric predicate has $C \le |P| + B_2$ (Proposition 3.21). **[Proved]**

4. **Spectral worst-case output bounds (Section 4).** For regular statistics (r_i, σ_i) the worst-case triangle output W satisfies $U/4 < W \le U$, with U the minimum of nine monomials. The upper bound is textbook (Schatten–Hölder and Lemma 4.2). The content is that the nine terms are exactly the feasibility conditions of one block family, so one family witnesses the lower bound in every regime, and that $\inf W/U = 1/4$ (Theorem 4.7; this family, in rational form, was found by a model-based review, Section 9). **[Proved; Lean-checked]** Two measured norms give the same factor-4 characterization with six terms (Corollary 4.9), and none of the nine terms can be dropped (Proposition 4.10). For the labeled 4-cycle we prove $U_4/16 < W_4 \le U_4$ for an explicit minimum U_4 of 38 monomials, each needed, with the best constant in $[8, 16]$ (Theorem 4.12, Proposition 4.14); the new ingredient is a lifted Gram bound (Lemma 4.13), and the witnesses are matchings along arcs of the cycle rather than complete blocks. **[Proved]**
5. **Per-instance rotation is optimal for (r, σ) (Section 5).** The probe work M_π of the kernel driven by rotation π equals the output of an instance with the same driver relation, the same relevant one-sided degree vectors and the same sizes (Lemma 5.1), and $\min_\pi M_\pi \le U$, which is $< 4W$ for regular statistics (Corollary 5.3). **[Proved; Lean-checked]** No fixed rotation, and not their sum, is $O\left(\sum_i r_i + W\right)$ (Proposition 5.5). **[Proved]**
6. **Engine and experiments (Section 6; details in Appendix A and Appendix B).** On 3,124 selected canonical triangle queries with nonzero output, U is below AGM by a median factor 9.6 but tighter than all of AGM, the cyclic ℓ_2 bound and the degree bound on only 6.5% of them. On 1,241 knowledge-graph triangles run under six driver policies, $\min_\pi M_\pi \le U$ holds on every query; the exact planner objective picks a different driver than its 64-row estimate on only 3–6% of queries and is slower end to end, although no fixed sampling grid has a worst-case guarantee (Proposition 6.1). The triangle operator is faster on skewed triangle queries (32 × for rows and 114 × for `COUNT(*)` on a Zipf store; a median 8.8–10.2 × where the first admission stage admits it on the knowledge graphs) and neutral on a uniform store. A calibrated second admission stage cuts total time by 21–49% against the first stage measured in the same run; the rule was fitted to these queries (in-sample), and both stages together still decline 29–49% of them. At the time of writing the operator and both stages are on a development branch of Leatherback.

The results of Section 4 and Section 5 marked Lean-checked are kernel-checked in Lean 4 (Appendix D); everything else is proved on paper, and every construction found during the work was also rebuilt in exact arithmetic by at least two separately run model-based checks (Appendix G). AI tools were used throughout; Section 9 states how. Appendix E recalls two classical

spectral facts used for certification; they are not contributions.

What is and is not new

All proofs here are elementary: Cauchy–Schwarz, the Schatten–Hölder inequality, the Schur test, Hölder's inequality, and counting. Several ingredients are known and we do not claim them: for one symmetric relation the Hölder term of U is the folklore spectral triangle bound $6t = \operatorname{tr}(A^3) \le \lambda_1 \operatorname{tr}(A^2) = 2m\lambda_1$; the degree terms are the ℓ_1/ℓ_∞ bounds of [ANOS24] with

maximum degrees bounded by σ^2; the smaller-side kernel is the Generic Join intersection of [NRR13]; choosing the next variable separately for each binding is already done in [NRR13, ABGHNR24, ABFGN25]; and comparison certificates, alternation and doubling search for sorted-set intersection are classical [DLM00, BK02, BK08, BY76]. What is new, to our knowledge, are the *bounds*: for the per-binding cost model of Section 3 (one root per instance, an option per binding chosen by exact cost, smaller-side intersections, no partitioning), the upper bounds of Theorem 3.3 and Theorem 3.4 with matching lower bounds at every slope in $[9/10, 3/2]$, the fully reduced interval, the exact gaps of the orders, and their survival under fresh sorted intersections with adversarial identifiers; and, for sizes and operator norms, the nine-term characterization $U/4 < W \le U$ for three labeled relations with an asymptotically sharp factor 4 (Theorem 4.7), its 38-term analogue for the 4-cycle, and the worst-case optimality of choosing the kernel's driver per instance for these statistics. Partitioned algorithms meet B_2 [JRR21, ANS25]; our lower bounds are for the unpartitioned class only. Section 7 compares each claim with the closest prior work.

2. Preliminaries

Instances. Fix finite sets X, Y, Z and relations $R \subseteq X \times Y$, $S \subseteq Y \times Z$, $T \subseteq Z \times X$ (sets, not bags). Their incidence matrices are $A_R \in \{0,1\}^{X\times Y}$, $A_S \in \{0,1\}^{Y\times Z}$, $A_T \in \{0,1\}^{Z\times X}$, and

$$\mathrm{Out} = \mathrm{tr}(A_R A_S A_T) = |\{(x,y,z) : (x,y) \in R, (y,z) \in S, (z,x) \in T\}|.$$

Every RDF triangle pattern with three distinct variables and constant predicates is of this form after transposing some predicates, because each predicate slice is readable in both directions. We use the cyclic order $R \to S \to T \to R$ of the labels throughout.

Degrees. For a set P of pairs, $d_P^+(u)$ is the number of pairs leaving u and $d_P^-(v)$ the number entering v; Δ_P^+ and Δ_P^- are their maxima. We write $N = |R| + |S| + |T|$ and $\|A\|_2$ for the largest singular value.

Statistics. Given reals r_i, σ_i for $i \in \{R, S, T\}$, an instance is *admissible* if $|i| \le r_i$ and $\|A_i\|_2 \le \sigma_i$ for every i, and $W(r,\sigma) = \sup\{\mathrm{Out} : \text{admissible instances}\}$, with the convention $\sup \varnothing = 0$. Some instance is admissible iff all six statistics are nonnegative (the empty instance is then admissible; Lean: `admissible_exists_iff_nonneg`); if a statistic is negative the class is empty and $W = 0$ only by that convention. A nonzero 0/1 matrix has $\|A\|_2 \ge \|Ae_y\|_2 \ge 1$, so if some $r_i < 1$ or $\sigma_i < 1$ that relation is empty and every admissible instance has $\mathrm{Out} = 0$ (Lean: `theoremB_degenerate`, which assumes no sign condition and is vacuous when a bound is negative); for nonnegative statistics the value $W = 0$ is attained by an admissible instance (`theoremB_degenerate_attained`). We call (r,σ) *regular* when all $r_i, \sigma_i \ge 1$.

Clipping. Put $s_i = \min(\sigma_i^2, r_i)$. Nothing is lost, because $\|A_i\|_2^2 \le \|A_i\|_F^2 = |i| \le r_i$ (Lean: `opNorm_adj_sq_le_card`).

The nine terms. For $\{i,j,k\} = \{R,S,T\}$ and $i \ne j$ let

$$H_i = \sqrt{s_i r_j r_k} \quad \text{(Hölder)}, \quad D_{i,j} = r_i s_j \quad \text{(degree)}, \quad U = \min\big\{H_R, H_S, H_T, D_{R,S}, D_{R,T}, D_{S,R}, D_{S,T}, D_{T,R}, D_{T,S}\big\}.$$

AGM and the trivial bounds $r_j r_k$ are redundant: $\sqrt{r_R r_S r_T} \geq H_i$ and $r_j r_k \geq D_{j,k}$ because $s_i \leq r_i$. In particular $U \leq \mathrm{AGM}$ always. With unclipped σ^2, U is the minimum of thirteen monomials. The problem and U are invariant under S_3 acting on the labels (rotation by cyclicity of the trace; reflection by $\mathrm{tr}(A_R A_S A_T) = \mathrm{tr}\big(A_T^\top A_S^\top A_R^\top\big)$).

Kernel work. The adaptive kernel driven by rotation R scans R and, for each $(x, y) \in R$, intersects $S(y, \cdot)$ with $T(\cdot, x)$ from the smaller side. Its probe work is

$$M_R = \sum_{(x,y)\in R} \min(d_S^+(y), d_T^-(x)),$$

and M_S, M_T are its rotations: $M_S = \sum_{(y,z)\in S} \min(d_T^+(z), d_R^-(y))$, $M_T = \sum_{(z,x)\in T} \min(d_R^+(x), d_S^-(z))$. We write π for a rotation, i.e. the relation that drives the kernel. Clearly $\mathrm{Out} \leq M_\pi$ for each π.

Cost model. We charge one unit for reading a CSR offset (a degree), one for reading one neighbour, and one for one membership test. With hash or bitmap membership the intersection of two runs of lengths $p \leq q$ costs at most $2p$ (read each element of the shorter run and test it). This *min-side charge* is the cost model of Section 3, Section 4 and Section 5. Sorted runs intersected by comparisons are a different model: with contiguous identifiers leapfrog finishes after a few seeks (on G_8 of Section 3.3 the two runs are disjoint), so lower bounds there must let the adversary assign the identifiers. Section 3.6 defines that model and shows that every lower bound of Section 3 against $N + B_2$ survives in it without logarithmic loss (against $C + N$ some survive only up to a factor $\log N$, Remark 3.27), while its worst case at $B_2 \asymp N$ gains exactly a factor $(\log N)^{1/4}$.

Comparators. AGM is $\sqrt{r_R r_S r_T}$. The *degree bound* is

$$\min\{r_R \min(\Delta_S^+, \Delta_T^-),\, r_S \min(\Delta_T^+, \Delta_R^-),\, r_T \min(\Delta_R^+, \Delta_S^-)\},$$

the form $|R| \cdot \|\deg_S(Z|Y)\|_\infty$ of [ANOS24, Eq. (3)] applied to each relation. The *cyclic ℓ_2 bound* is

$$B_2 = \min\left\{ \left(\sum_x d_R^+(x)^2 \sum_y d_S^+(y)^2 \sum_z d_T^+(z)^2\right)^{1/3}, \left(\sum_y d_R^-(y)^2 \sum_z d_S^-(z)^2 \sum_x d_T^-(x)^2\right)^{1/3} \right\},$$

an upper bound on Out from [ANOS24, Eq. (4)] and [JRR21]. In Section 3, B_2 is the comparator for the per-binding cost: Theorem 3.3 bounds C in terms of the squared-degree sums behind B_2, and Theorem 3.6 shows that C can exceed B_2 by a polynomial factor. No result of Section 4 or Section 5 depends on B_2.

Notation. Table 2 collects the symbols used in more than one section.

Table 2: Notation.

Symbol	Meaning
A_R, A_S, A_T	0/1 incidence matrices of R, S, T
r_i, σ_i, s_i	size bound, norm bound, clipped statistic $\min(\sigma_i^2, r_i)$

Symbol	Meaning
H_i, $D_{i,j}$, U	Hölder terms, degree terms, their minimum (Section 2)
W	worst-case output over admissible instances ($w(x)$ in Lemma 3.2 is unrelated)
$\mathrm{Blk}(a,b,c,m)$	block instance: m disjoint complete block triangles (Section 4.2)
$\hat{a},\hat{b},\hat{c}$	real block shape; a,b,c are their floors (Lemma 4.3)
π, M_π	rotation (driving relation) and the kernel's probe work
$c_P(u)$, $c^*(u)$	cost of option P at binding u; $c^*(u)$ the smaller option cost (Section 3)
C_x, C_y, C_z, C	per-binding cost for a fixed root; $C = \min(C_x, C_y, C_z)$
n_v, n	number of bindings of root variable v with an incident tuple; $n = n_x + n_y + n_z$
B_2	cyclic ℓ_2 bound
α,β,γ; α',β',γ'	forward and reverse squared-degree sums (Section 3)
Λ, Λ'	$(\alpha\beta\gamma)^{1/2}$, $(\alpha'\beta'\gamma')^{1/2}$; $B_2 = \min(\Lambda, \Lambda')^{2/3}$
$G_8(m)$, $\mathcal{U}(m)$	lower-bound gadget and its three-rotation union (Theorem 3.6)
$F_{i,j}(u)$, κ	slope family; slope κ with $B_2 = \Theta(N^\kappa)$ (Theorem 3.10); every real $\kappa \in [1, 3/2]$ (Corollary 3.11)
U_2, U_1'	two- and one-norm specializations of U (Corollary 4.9)
$D(L,a,h)$, $E(L,a,h)$	gadgets of Theorem 3.12, Theorem 3.18
$L(p,q,t,g,k,w)$, $\mathcal{L}(v)$	live gadget and its three-rotation union (Lemma 3.13, Theorem 3.14)
U_4, W_4, K	4-cycle bound, its worst case, the lifted Gram matrix (Section 4.5)
Q_A, $h(p,q)$, L_N, $\hat{C}$	sorted-model cost, galloping price, $1 + \lceil \log_2 N \rceil$, priced per-binding cost (Section 3.6)
$\mathcal{F}_a, \mathcal{G}_a, \mathcal{H}_a$, GMR, GMIN	separation families (Theorem 3.16)
padding	attaching one fresh triangle per half-dangling vertex (Lemma 3.8)
L_v	running lower bound on C_v in the price-then-run scheduler (Appendix C.1)
N	$\lvert R\rvert + \lvert S\rvert + \lvert T\rvert$

3. Per-binding choice: a tight ℓ_2 bound and the exact gaps of coarser levels

For sizes and operator norms, choosing the kernel's driver per instance is already worst-case optimal (Corollary 5.3). Degree-sequence statistics ask for finer adaptivity, and this section studies a per-binding variant of the kernel. Its cost C obeys a worst-case bound in terms of the degree-sequence norms behind B_2, and a second bound that is sharper below slope 1 (Theorem 3.3, Theorem 3.4). Their minimum is tight at $B_2 \asymp N$ (Theorem 3.6) and, up to a constant factor, at every real slope $B_2 = N^\kappa$, $\kappa \in [9/10, 3/2]$ (Theorem 3.10, Corollary 3.11, Theorem 3.12). In particular $C = O(N + B_2)$ is false, so this unpartitioned class, which fixes one root per instance and intersects at the shorter run, is not optimal for the cyclic ℓ_2 bound; whether every partition-free algorithm fails is open (Section 8). We then separate the per-binding strategy from fixed attribute orders and fixed roots, with exact gaps for the orders (Theorem 3.16, Theorem 3.17, Theorem 3.18, Corollary 3.19), and show that the lower bounds survive fresh sorted intersections under adversarial identifiers (Section 3.6). A deterministic scheduler that pays for the statistics the choice needs (Proposition C.2) and a negative result for classes defined by degree metadata (Proposition C.4) are in Appendix C.

3.1. Per-binding cost

Fix a root variable, say x. A binding x has two options. Option R walks $y \in R(x,\cdot)$ and intersects $S(y,\cdot)$ with $T(\cdot,x)$; option T walks $z \in T(\cdot,x)$ and intersects $R(x,\cdot)$ with $S(\cdot,z)$. Their costs are

$$c_R(x) = \sum_{y:(x,y)\in R} (1 + \min(d_S^+(y), d_T^-(x))), \quad c_T(x) = \sum_{z:(z,x)\in T} (1 + \min(d_R^+(x), d_S^-(z))),$$

$c^*(x) = \min(c_R(x), c_T(x))$, and $C_x = \sum_x c^*(x)$. Roots y (options R and S) and z (options S and T) are defined symmetrically, with per-edge costs $1 + \min(d_S^+(y), d_T^-(x))$ on R-tuples, $1 + \min(d_R^-(y), d_T^+(z))$ on S-tuples and $1 + \min(d_R^+(x), d_S^-(z))$ on T-tuples. The per-binding cost is $C = \min(C_x, C_y, C_z)$. A *fixed global attribute order* (GAO) fixes the root and, for every binding, the option; the option's relation is the GAO's *second relation*. A *fixed root* fixes the root variable and chooses the option per binding. We measure C in *per-edge cost units*; by the cost model of Section 2, executing an edge of cost $1 + k$ (one neighbour read, one degree read to find the smaller side, and an intersection of cost at most $2k$) takes at most $2(1 + k)$ unit operations.

Lemma 3.1 (sum identities)**.** *$\sum_x c_R(x) = |R| + M_R$, and likewise for the other five (root, option) pairs. A GAO whose second relation is π costs $|\pi| + M_\pi$, so the six GAOs have only three costs. Moreover* $\mathrm{Out} \le C \le \min_\pi(|\pi| + M_\pi) \le N + \min(M_R, M_S, M_T)$*, and* $C \le N + \sqrt{|R|\ |S|\ |T|}$.

[Proved]

Proof. Each R-tuple (x,y) appears once in $\sum_x c_R(x)$ with cost $1 + \min(d_S^+(y), d_T^-(x))$; this also holds in $\sum_y c_R(y)$ for root y. For root x, $C_x \le \min\left(\sum_x c_R, \sum_x c_T\right) = \min(|R| + M_R, |T| + M_T)$; for root y, $C_y \le |S| + M_S$. Every triangle through binding u is found by either option at u, at cost at least one per triangle, so $\mathrm{Out} \le C_v$ for each root v. The last bound follows from $M_R \le \|A_R\|_2 \sqrt{|S|\ |T|} \le \sqrt{|R|\ |S|\ |T|}$ (Proposition 5.2). □

So per-binding choice is never asymptotically worse than the best rotation, or than AGM.

Degree-sequence norms. Write

$$\alpha = \sum_x d_R^+(x)^2, \quad \beta = \sum_y d_S^+(y)^2, \quad \gamma = \sum_z d_T^+(z)^2, \quad \alpha' = \sum_y d_R^-(y)^2, \quad \beta' = \sum_z d_S^-(z)^2, \quad \gamma' = \sum_x d_T^-(x)^2,$$

and $\Lambda = (\alpha\beta\gamma)^{1/2}$, $\Lambda' = (\alpha'\beta'\gamma')^{1/2}$, so that $B_2 = \min(\Lambda, \Lambda')^{2/3}$ (Section 2). The sizes, N, Out, the probe works M_π, the per-root costs C_v and the six squared-degree sums are sums over vertices or tuples of functions of local degrees, so they add over vertex-disjoint unions. Their minima and products do not in general: $C = \min_v C_v$, Λ, Λ' and B_2 are computed from the summed quantities, not summed themselves. *Rotating* an instance means relabelling (R,S,T) on (X,Y,Z) as (S,T,R) on (Y,Z,X); this preserves Out and N and permutes (M_R, M_S, M_T), (C_x, C_y, C_z), $(|R|, |S|, |T|)$ and (α, β, γ) cyclically. *Reflecting* it means replacing (R,S,T) on (X,Y,Z) by $(T^\top, S^\top, R^\top)$ on (X,Z,Y); this exchanges the two options of every root, so it preserves C, and it exchanges (α,β,γ) with (γ',β',α'), so $\Lambda \leftrightarrow \Lambda'$.

3.2. An ℓ_2 upper bound for the per-binding cost

Lemma 3.2 (per-root bound)**.** *For root x,*

$$C_x \le |R| + |T| + |S|^{3/8}\ \alpha^{1/4}\beta^{1/8}\gamma^{3/8} \quad \text{and} \quad C_x \le |R| + |T| + |S|^{3/8}\ \gamma'^{1/4}\beta'^{1/8}\alpha'^{3/8}.$$

By rotation, $C_y \le |R| + |S| + |T|^{3/8}\ \beta^{1/4}\gamma^{1/8}\alpha^{3/8}$ and $C_z \le |S| + |T| + |R|^{3/8}\ \gamma^{1/4}\alpha^{1/8}\beta^{3/8}$, each with a reflected (primed) form.

[Proved]

Proof. Write $c_R(x) = d_R^+(x) + P(x)$ and $c_T(x) = d_T^-(x) + Q(x)$, where $P(x) = \sum_{y\in R(x,\cdot)} \min(d_S^+(y), d_T^-(x))$ and $Q(x) = \sum_{z:(z,x)\in T} \min(d_R^+(x), d_S^-(z))$. Since $\min(u+P, v+Q) \le u + v + \min(P,Q)$, summing over x gives $C_x \le |R| + |T| + H_x$ with $H_x = \sum_x \min(P(x), Q(x))$. We bound H_x in five steps; write $a = d_R^+(x)$.

1. *Cauchy–Schwarz over the a distinct y:* $P(x) \le \sum_{y\in R(x,\cdot)} d_S^+(y) \le \sqrt{a\beta}$.
2. $\min(u,v) \le \sqrt{uv}$*:* $Q(x) \le \sqrt{a}\cdot w(x)$ with $w(x) = \sum_{z:(z,x)\in T} \sqrt{d_S^-(z)}$; the radical covers a only.
3. *Split the minimum:* $\min(P,Q) \le P^{1/4}Q^{3/4}$ (a weighted geometric mean; it also holds if P or Q is 0). Hence $\min(P(x), Q(x)) \le \beta^{1/8}a^{1/2}w(x)^{3/4}$.
4. *Hölder with exponents* $(4, 4/3)$*:* $\sum_x d_R^+(x)^{1/2}w(x)^{3/4} \le \alpha^{1/4}\big(\sum_x w(x)\big)^{3/4}$.
5. *Double counting, then Cauchy–Schwarz:* $\sum_x w(x) = \sum_z d_T^+(z)\sqrt{d_S^-(z)} \le \sqrt{\gamma}\sqrt{\sum_z d_S^-(z)} = \sqrt{\gamma\ |S|}$.

Together, $H_x \le \alpha^{1/4}\beta^{1/8}(\gamma\ |S|)^{3/8}$. The primed form exchanges the roles of P and Q: use $\min(P,Q) \le Q^{1/4}P^{3/4}$, $Q(x) \le \sqrt{d_T^-(x)\beta'}$, $P(x) \le \sqrt{d_T^-(x)}\sum_{y\in R(x,\cdot)}\sqrt{d_S^+(y)}$, Hölder against γ', and $\sum_y d_R^-(y)\sqrt{d_S^+(y)} \le \sqrt{\alpha'\ |S|}$. The bounds for roots y and z are the root-x bounds of the rotated instance. □

Theorem 3.3 (per-binding ℓ_2 bound)**.** *For every instance,*

$$C \le N + (|R|\ |S|\ |T|)^{1/8}\min(\Lambda,\Lambda')^{1/2} \le N + (N/3)^{3/8}B_2^{3/4};$$

equivalently, in integers, $(C-N)_+^8 \le |R|\ |S|\ |T| \cdot \min(\alpha\beta\gamma, \alpha'\beta'\gamma')^2$. The constant is 1 *and there are no logarithmic factors.*

[Proved] (not formalized; exact checks in Appendix G)

Proof. Let $K_x = |S|^{3/8}\ \alpha^{1/4}\beta^{1/8}\gamma^{3/8}$, $K_y = |T|^{3/8}\ \beta^{1/4}\gamma^{1/8}\alpha^{3/8}$ and $K_z = |R|^{3/8}\ \gamma^{1/4}\alpha^{1/8}\beta^{3/8}$ be the multiplicative terms of the unprimed bounds of Lemma 3.2; they are nonnegative. Since $C \le C_v$ for every root v and N is at least the sum of the two sizes walked at v, Lemma 3.2 gives $C - N \le K_v$ for each v. Hence

$$(C-N)_+ \le \min\big(K_x, K_y, K_z\big) \le \big(K_xK_yK_z\big)^{1/3},$$

the second step because the K_v are nonnegative. In the product each of α, β, γ has total exponent $1/4 + 1/8 + 3/8 = 3/4$ and each cardinality exponent $3/8$, so the cube root is $(|R|\ |S|\ |T|)^{1/8}\Lambda^{1/2}$. The primed bounds give Λ' in the same way. Finally $\min(\Lambda,\Lambda')^{1/2} = B_2^{3/4}$, AM–GM gives $|R|\ |S|\ |T| \le (N/3)^3$, and the eighth power of the first bound is the

integer form. The geometric mean must be taken of the bounds K_v, not of the differences C_v minus the two walked sizes, which can have mixed signs: on $K_{2,3,3}$ plus ten isolated R -tuples, $(|R|,|S|,|T|) = (16,9,6)$ and $C_x = C_y = C_z = 24$, so the differences are $(2,-1,9)$, whose geometric mean $-\sqrt[3]{18}$ lies below their minimum -1. □

Theorem 3.4 (a second per-binding ℓ_2 bound)**.** *For every instance with $N > 0$,*

$$C \le N + 3^{1/8} N^{-1/8} B_2^{5/4};$$

equivalently, in integers, $N^3 (C-N)_+^{24} \le 27 \min(\alpha\beta\gamma, \alpha'\beta'\gamma')^{10}$. Its main term is that of the second form of Theorem 3.3 times $\sqrt{3B_2/N}$, so it is the sharper of the two exactly when $B_2 < N/3$; at $B_2 = N^\kappa$ its exponent $(10\kappa - 1)/8$ is below $3/8 + 3\kappa/4$ by $(1-\kappa)/2$ for $\kappa < 1$, and both equal $9/8$ at $\kappa = 1$.

[Proved] (on paper; exact checks in Appendix G)

Proof. If $C \le N$ there is nothing to prove; let $C > N$. Reflection preserves C and N and exchanges the two moment products, so we may assume $\alpha\beta\gamma = \min(\alpha\beta\gamma, \alpha'\beta'\gamma') = B_2^3$. Rotation preserves C, N and the forward product and permutes $(|R|,|S|,|T|)$ and (α,β,γ) cyclically, so we may further assume that R is a largest relation. Degrees are integers, so $d^2 \ge d$ and $|R| \le \alpha$, $|S| \le \beta$, $|T| \le \gamma$; in particular $\alpha \ge |R| \ge N/3$. With H_v the sum of the minima in the proof of Lemma 3.2 (nonnegative), $C - N \le H_x$ and $C - N \le H_y$, and

$$H_x \le |S|^{3/8}\ \alpha^{1/4}\beta^{1/8}\gamma^{3/8} \le \alpha^{1/4}\beta^{1/2}\gamma^{3/8}, \quad H_y \le |T|^{3/8}\ \beta^{1/4}\gamma^{1/8}\alpha^{3/8} \le \alpha^{3/8}\beta^{1/4}\gamma^{1/2}.$$

Hence $C - N \le \min\big(H_x, H_y\big) \le H_x^{2/3} H_y^{1/3} \le \alpha^{7/24}\beta^{5/12}\gamma^{5/12} = (\alpha\beta\gamma)^{5/12}\alpha^{-1/8} \le B_2^{5/4}(N/3)^{-1/8}$. The integer form is the 24th power. The comparison of main terms is $3^{1/8}N^{-1/8}B_2^{5/4} / \Big((N/3)^{3/8}B_2^{3/4}\Big) = (3B_2/N)^{1/2}$. □

Corollary 3.5 (consequences of Theorem 3.3 and Theorem 3.4)**.**
1. $C \le N + N^{1/8}(N + B_2)$, and $C \le 2\max(N, B_2)^{9/8}$.
2. $C < 2.15N$ whenever $B_2 \le N^{9/10}$, and $C \le 2(N + B_2)$ whenever $B_2 \ge N^{3/2}$.
3. $\min(M_R, M_S, M_T) \le (\Lambda\Lambda')^{1/3}$, so $C \le N + \min\Big((\Lambda\Lambda')^{1/3}, \sqrt{|R|\ |S|\ |T|}\Big)$.

[Proved]

Proof. (1) $(N/3)^{3/8}B_2^{3/4} \le N^{1/8} \cdot N^{1/4}B_2^{3/4}$ and $N^{1/4}B_2^{3/4} \le N/4 + 3B_2/4$ by weighted AM–GM. The second form follows directly from Theorem 3.3: with $M = \max(N, B_2) \ge 1$, $C \le N + 3^{-3/8}M^{9/8} \le \big(1 + 3^{-3/8}\big)M^{9/8} < 1.67M^{9/8}$. (2) If $B_2 \le N^{9/10}$, Theorem 3.4 gives $C \le N + 3^{1/8}N^{-1/8}N^{9/8} = \big(1 + 3^{1/8}\big)N$, and $3^{1/8} < 1.15$. If $B_2 \ge N^{3/2}$ then $N^{3/8} \le B_2^{1/4}$ in Theorem 3.3. (3) $M_R \le \sum_y d_R^-(y) d_S^+(y) \le \sqrt{\alpha'\beta}$, $M_S \le \sum_z d_S^-(z) d_T^+(z) \le \sqrt{\beta'\gamma}$ and $M_T \le \sum_x d_T^-(x) d_R^+(x) \le \sqrt{\gamma'\alpha}$ by Cauchy–Schwarz; the product is $\Lambda\Lambda'$, and a minimum is at most the geometric mean. The last term is Lemma 3.1. □

So per-binding choice meets $N + B_2$ up to a constant factor for every slope $\kappa \le 9/10$, not only $\kappa \le 5/6$, and Theorem 3.3 is not tight on $(5/6, 1)$. The next subsection shows that the minimum of the two bounds is the exact worst case for every real slope in $[9/10, 3/2]$.

3.3. Per-binding choice does not meet the ℓ_2 bound

A natural conjecture is that per-binding choice is optimal for the cyclic ℓ_2 class without partitioning: that for an absolute constant c, or at least for $c = O(\log^2 N)$, every instance satisfies

$$C^3 \le c \cdot \left(N^3 + \min\left\{ \sum_x d_R^+(x)^2 \sum_y d_S^+(y)^2 \sum_z d_T^+(z)^2,\ \sum_y d_R^-(y)^2 \sum_z d_S^-(z)^2 \sum_x d_T^-(x)^2 \right\} \right),$$

i.e. $C = O(N + B_2)$. We refute it. Small-instance search found no violation, because the first violation needs $N \approx 2 \cdot 10^7$. The first member of the families below with $C/(N + B_2) > 1$ is the union of $F_{1,1}(7)$ and its two rotations (Theorem 3.10; $F_{1,1}$ is G_8 without Z_0 and has the better constant, $C/(N + B_2) \approx u/7$), with $N \approx 2.0 \cdot 10^7$ (ratio 0.93 at $u = 6$, 1.07 at $u = 7$). The family $\mathcal{U}(m)$ used in Theorem 3.6 needs $m = 11$, $N \approx 1.35 \cdot 10^9$ (ratio 0.97 at $m = 10$, 1.06 at $m = 11$); in the cubic form displayed above it exceeds 1 from $m = 7$, $N \approx 3.7 \cdot 10^7$.

The gadget $G_8(m)$. For $m \ge 2$ take parts X of size m^4, Y_0 of size m^2, Y_2 of size m^8, Z_0 of size m^8, Z_1 of size m^3 and Z_2 of size m^6, and the blocks of Table 3, each complete or biregular.

Table 3: The gadget $G_8(m)$. Degrees are listed as (left part, right part) of each block, in the orientation of the relation.

Block	Shape	Degrees
$R = X \times Y_0$	complete	$d_R^+ = m^2$ on X, $d_R^- = m^4$ on Y_0
$S \supseteq Y_0 \times Z_1$	complete	$d_S^+ = m^3$ on Y_0, $d_S^- = m^2$ on Z_1
$S \supseteq Y_2 \to Z_2$	biregular	$d_S^+ = 1$ on Y_2, $d_S^- = m^2$ on Z_2
$T \supseteq Z_0 \to X$	biregular	$d_T^+ = 1$ on Z_0, contributes m^4 to d_T^- on X
$T \supseteq Z_2 \to X$	biregular	$d_T^+ = m$ on Z_2, contributes m^3 to d_T^- on X

Theorem 3.6 (the per-binding ℓ_2 worst case, and refutation of $C = O(N + B_2)$)**.** *Let $\mathcal{U}(m)$ be the disjoint union of $G_8(m)$ and its two rotations. For every $m \ge 2$, $\mathrm{Out}(\mathcal{U}(m)) = 0$ and*

$$C_x = C_y = C_z = m^9 + m^8 + m^6 + m^5, \quad N = 3(2m^8 + m^7 + m^6 + m^5) \le 9m^8, \quad \alpha = \beta = \gamma = 5m^8.$$

Hence $B_2 \le 5m^8$ and

$$C/(N + B_2) \ge m/14 \ge (N/9)^{1/8}/14, \qquad C^3/(N^3 + \min(\alpha\beta\gamma, \alpha'\beta'\gamma')) \ge m^3/854.$$

So no constant c, and no $c = O(\log^k N)$ for any fixed k, satisfies $C \le c(N + B_2)$ on all instances; the geometric-mean variant $(C_x C_y C_z)^{1/3} = O(N + B_2)$ fails as well.

[Proved] (exact values also recomputed for $m = 2, 3, 4$ by two separately written evaluators; Table 4, Appendix G)

Proof. One gadget. For $x \in X$: $d_T^-(x) = m^4 + m^3$; option R walks m^2 vertices of Y_0, each with $d_S^+ = m^3$, so $c_R(x) = m^2(1 + m^3)$; option T walks m^4 vertices of Z_0 (where $d_S^- = 0$) and m^3 of Z_2 (where $d_S^- = m^2 = d_R^+(x)$), so $c_T(x) = m^4 + m^3(1 + m^2)$. Thus $c^*(x) = m^5 +$

m^2 and $C_x(G_8) = m^9 + m^6$. For $y \in Y_0$, option S walks Z_1, where $d_T^+ = 0$, at cost m^3, and option R costs more; Y_2 has no R-tuples, so its option R is empty. Hence $C_y(G_8) = m^5$. For z: Z_0 has no S-tuples and Z_1 no T-tuples (cost 0); on Z_2, option S walks m^2 vertices of Y_2 with $d_R^- = 0$ (cost m^2) and option T costs $m(1 + m^2)$. Hence $C_z(G_8) = m^8$. Sizes: $|R| = m^6$, $|S| = m^5 + m^8$, $|T| = m^8 + m^7$. Norms: $\alpha = m^4 \cdot m^4 = m^8$, $\beta = m^2 \cdot m^6 + m^8 = 2m^8$, $\gamma = m^8 + m^6 \cdot m^2 = 2m^8$. No triangle exists: R leads into Y_0, S maps Y_0 into Z_1, and T leaves no vertex of Z_1.

The union. Costs and squared norms add over components, and rotation permutes (C_x, C_y, C_z) and (α, β, γ) cyclically, so each $C_v(\mathcal{U}) = C_x + C_y + C_z$ of G_8, i.e. $m^9 + m^8 + m^6 + m^5$, and each forward norm is $m^8 + 2m^8 + 2m^8$. $N \le 9m^8$ because $m^7 + m^6 + m^5 \le m^8$ for $m \ge 2$. Then $C/(N + B_2) \ge m^9/(14m^8)$, and $N^3 + \alpha\beta\gamma \le (729 + 125)m^{24}$. A polylogarithm of $N \le 9m^8$ is $o(m)$. □

Table 4: Exact values on $\mathcal{U}(m)$. The last column is C divided by the right-hand side of Theorem 3.3.

m	N	B_2	C	$C/(N + B_2)$	C/bound
2	2,208	1,280	864	0.248	0.182
3	48,843	32,805	27,216	0.333	0.192
4	457,728	327,680	332,800	0.424	0.200

Corollary 3.7 ($C = O(N + B_2)$ fails)**.** *There is no constant c, and no $c = O(\log^k N)$, such that every instance satisfies $C^3 \le c(N^3 + \min(\alpha\beta\gamma, \alpha'\beta'\gamma'))$. The partition-free per-binding strategy is not optimal for the cyclic ℓ_2 bound; at $B_2 \asymp N$ its worst-case exponent is exactly $9/8$ (Theorem 3.3, Theorem 3.6).*

[Proved]

Degree-bucket partitioning with multiplexing over acyclic subqueries [JRR21, Thm. 4.4] computes the triangle in time linear in the ℓ_2 bound of one orientation, given as input norm bounds (their Theorem 4.4, with a query-dependent factor $2^{(p+1)|V|}$, $O(1)$-time B-tree access in the RAM model, and $O(N \log N)$ index construction); running it on the orientation with the smaller product meets B_2 up to the same factors. Their algorithm is outside our class: it partitions every relation into degree buckets and chooses the plan per combination of buckets. So, within the per-binding cost model of Section 3 (one root per instance, a per-binding option, min-side intersections), some partitioning of a variable's domain is needed to reach the ℓ_2 bound; whether every partition-free algorithm fails is open (Section 8). Price-then-run (Proposition C.2) runs in $O\left(N + N^{3/8} B_2^{3/4}\right)$ rather than $O(N + B_2)$. Since B_2 can exceed AGM (double stars), Theorem 3.3 is useful only together with $C \le N + \sqrt{|R|\ |S|\ |T|}$ (Lemma 3.1).

The family $\mathcal{U}(m)$ has no triangles, so one may ask whether the gap is an artefact of dead tuples. It is not.

Lemma 3.8 (padding)**.** *Let I be an instance and pad it as follows: for every vertex v that has a tuple in exactly one of its two relations, add two fresh vertices and the three tuples of one triangle through v (for $v = x$: fresh y', z' with $(x, y') \in R$, $(y', z') \in S$, $(z', x) \in T$).*

Let $p \le 2N$ be the number of padded vertices. The padded instance I' satisfies: (i) it is pairwise semijoin-reduced (every vertex with a tuple has a tuple in both of its relations); (ii) $N' = N + 3p \le 7N$ and $\mathrm{Out}' = \mathrm{Out} + p$; (iii) every C_v and every M_π is nondecreasing, and $C'_v \le C_v + 9N$ for each root v; (iv) each of the six squared-degree sums is at most $2(\cdot) + 8N$.

[Proved]

Proof. (i) Padded vertices gain the missing relation; fresh vertices have both. (ii) Each fresh pair closes exactly one triangle: a fresh vertex is adjacent only to its partner and to v. (iii) Adding tuples never lowers a degree, so per-edge costs and the minimum of two option costs do not decrease. An old vertex gains tuples only from its own pad, so each old degree grows by at most 1, and each old term $1 + \min(\cdot, \cdot)$ by at most 1; keeping the old option choices, old terms add at most N. A padded vertex of root v gains one term of cost at most 2 in its chosen option (its new neighbour has degree 1), and a fresh vertex of root v has both options of cost 2; there are at most p of each, so $C'_v \le C_v + N + 4p \le C_v + 9N$. (iv) $(d+1)^2 \le 2d^2 + 2$ for each of at most p old vertices, and each of at most $2p$ fresh vertices contributes 1. □

Remark 3.9 (the gap survives reduction and connectivity)**.** (a) *Semijoin reduction.* Padding $\mathcal{U}(m)$ gives a pairwise semijoin-reduced instance with $\mathrm{Out} = \Theta(N)$ (the m^8 vertices of each copy of Y_2 are padded), $C \ge m^9$ by monotonicity, and $N + B_2 = O(m^8)$ by Lemma 3.8(ii) and (iv). So the $\Omega(N^{1/8})$ gap holds on semijoin-reduced inputs with $\Theta(N)$ triangles; the measured ratios $C/(N + B_2)$ are 0.397, 0.425, 0.453 for $m = 2, 3, 4$. (b) *Root choice per component.* $\mathcal{U}(m)$ is disconnected, and each component alone has $C(G_8) = C_y(G_8) = m^5$. Adding two R-tuples that join the three copies makes the instance connected, changes N by 2 and the squared norms by $O(m^4)$, and cannot lower any cost, so $C \ge m^9$ still holds. (c) *Both at once.* Add the two joining R-tuples after padding, between vertices that already have tuples in both of their relations. Every vertex still has tuples in both of its relations, so the instance stays pairwise semijoin-reduced; by the monotonicity of Lemma 3.8(iii), $C \ge m^9$; and $N + B_2 = O(m^8)$, since the two tuples change N by 2 and the squared norms by $O(m^4)$. What escapes the bound is choosing different roots for different regions of one variable's domain, which is partitioning.

[Proved] (the ratios in (a) are measured)

Theorem 3.10 (tightness of Theorem 3.3 across slopes)**.** *Fix integers $i \ge j \ge 1$ and $u \ge 2$. Let $F_{i,j}(u)$ have parts X of size u^{4i}, Y_0 of size u^{4i-2j}, Y_2 of size u^{8i}, Z_1 of size u^{4i-j} and Z_2 of size u^{4i+2j}, with $R = X \times Y_0$ complete, $S = (Y_0 \times Z_1) \cup (Y_2 \to Z_2)$ where the second block has $d_S^+ = 1$, and $T = Z_2 \to X$ biregular with $d_T^- = u^{4i-j}$ on X and $d_T^+ = u^{4i-3j}$ on Z_2. The union of $F_{i,j}(u)$ and its two rotations satisfies $C \ge u^{12i-3j}$, $u^{8i} \le N \le 12u^{8i}$ and $B_2 \le 4u^{12i-4j}$, hence*

$$C \ge N^{3/8} B_2^{3/4} / \left(12^{3/8} \cdot 4^{3/4}\right),$$

and $B_2 = \Theta(N^\kappa)$ with $\kappa = 3/2 - j/(2i)$. Together with the complete block triangle $K_{s,s,s}$ ($\kappa = 3/2$), Theorem 3.3 is tight up to a constant factor on a dense set of slopes $\kappa \in [1, 3/2]$, and by Corollary 3.11 at every slope in $[1, 3/2]$.

[Proved] (on paper; exact checks in Appendix G)

Proof. For $x \in X$, $c_R(x) = u^{4i-2j}(1 + u^{4i-j})$ (each $y \in Y_0$ has $d_S^+ = u^{4i-j} = d_T^-(x)$) and $c_T(x) = u^{4i-j}(1 + u^{4i-2j})$ (each $z \in Z_2$ has $d_S^- = u^{4i-2j} = d_R^+(x)$); both are at least u^{8i-3j}, so $C_x(F) \geq u^{12i-3j}$, and every root of the union contains a copy of root x of F. Each copy has $N \leq 4u^{8i}$, since $j \leq i$. Norms of one copy: $\alpha = \gamma = u^{12i-4j}$ and $\beta = u^{12i-4j} + u^{8i} \leq 2u^{12i-4j}$; in the union each forward norm is their sum. The displayed inequality follows. For the slope, Theorem 3.3 with $C \geq u^{12i-3j}$ and $12i - 3j > 8i$ forces $B_2^{3/4} = \Omega(u^{9i-3j})$, so $B_2 = \Theta(u^{12i-4j})$. For $K_{s,s,s}$, $N = 3s^2$, $B_2 = s^3$ and $C = s^2(1 + s)$. □

Corollary 3.11 (every real slope in $[1, 3/2]$)**.** *Fix a real $\kappa \in [1, 3/2)$, put $\theta = 3 - 2\kappa \in (0, 1]$, and for every integer $n \geq 1/\theta$ take the union of $F_{i,j}(2)$ and its two rotations with $i = n$, $j = \lfloor \theta n \rfloor$. Then $3 \cdot 2^{8n} \leq N \leq 12 \cdot 2^{8n}$, $3 \cdot 2^{8\kappa n} \leq B_2 < 64 \cdot 2^{8\kappa n}$ and $C \geq N^{3/8} B_2^{3/4} / \big(12^{3/8} \cdot 4^{3/4}\big)$. Hence $B_2 = \Theta(N^\kappa)$ and $C = \Theta\big(N^{3/8+3\kappa/4}\big)$ with constants independent of n, and, with $K_{s,s,s}$ at $\kappa = 3/2$, Theorem 3.3 is tight up to a constant factor at every real $\kappa \in [1, 3/2]$.*

[Proved] (exact checks in Appendix G. This extension was found by a model-based review: the GPT-6-Astra Pro review of the second draft, Section 9.)

Proof. $1 \leq j \leq i$ because $\theta n \geq 1$ and $\theta \leq 1$. By the proof of Theorem 3.10, each forward squared-degree sum of the union is $3u^{12i-4j} + u^{8i}$ and each reverse sum is $3u^{12i-2j} + u^{12i-5j}$, which is larger; so $B_2 = 3u^{12i-4j} + u^{8i}$, and $N = 3(u^{8i} + u^{8i-j} + u^{8i-2j} + u^{8i-3j})$. With $u = 2$, $i = n$ and $\varepsilon = \theta n - j \in [0, 1)$ we get $12n - 4j = 8\kappa n + 4\varepsilon$, so $3 \cdot 2^{8\kappa n} \leq 3 \cdot 2^{12n-4j} < 48 \cdot 2^{8\kappa n}$, and $2^{8n} \leq 2^{8\kappa n}$ because $\kappa \geq 1$. The bound on C is the displayed inequality of Theorem 3.10, whose constants do not depend on (i, j). The upper bound is Theorem 3.3.□

The gadget $D(L, a, h)$. For integers $L, a, h \geq 1$ with $h \mid L$, put $b = L/h$ and take tagged classes $X = [h] \times [b]$, $Y_0 = [a]$, $Y_2 = [h]^2 \times [a]$, $Z_1 = [h]$, $Z_2 = [h]^2$, with

$$R = X \times Y_0, \quad S = (Y_0 \times Z_1) \cup \{((i, j, k), (i, j))\}, \quad T = \{((i, j), (i, \ell)) : i, j \in [h], \ell \in [b]\}.$$

By direct counting, $(|R|, |S|, |T|) = (La, ah + ah^2, Lh)$, $(\alpha, \beta, \gamma) = (La^2, 2ah^2, L^2)$, $(\alpha', \beta', \gamma') = (aL^2, a^2(h + h^2), Lh^2)$, $(M_R, M_S, M_T) = (Lah, 0, Lah)$, $(C_x, C_y, C_z) = (Lah + L\min(a, h), ah, ah^2)$ and $\mathrm{Out} = 0$: an R-tuple sees degrees (h, h), a T-tuple (a, a), every S-tuple has a zero adjacent degree; at x the options cost $a(1 + h)$ and $h(1 + a)$; at Y_0 option S costs h, and Y_2 has an empty option; at Z_1 option T is empty, and at Z_2 option S costs $a < (L/h)(1 + a)$. $D(u^{4i}, u^{2i}, u^{3i})$ is $F_{i,i}(u)$.

Theorem 3.12 (the worst-case exponent at slopes in $[9/10, 1)$)**.** *Let $u \geq 2$ and integers p, q with $9/10 \leq p/q \leq 1$. Take the vertex-disjoint union of $D_1 = D(u^{8p-4q}, u^{6q-4p}, u^{6p-3q})$ in its own orientation; $D_2 = D(u^{4q}, u^{4p-2q}, u^{6p-3q})$ with its local R, S, T placed as the global S, T, R (its X, Y, Z become the global Y, Z, X); and $D_3 = D(u^{4p}, u^{6p-4q}, u^{6q-3p})$ with its local R, S, T placed as the global T, R, S. Then $(\alpha, \beta, \gamma) = (4u^{8q}, 4u^{8p}, 4u^{16p-8q})$, every reverse moment is at least its forward counterpart, so $B_2 = 4u^{8p}$; $u^{8q} \leq N \leq 12u^{8q}$; and $C = C_x =$*

$u^{10p-q} + u^{4p+2q} + u^{16p-8q} + u^{3p+2q}$, *so* $u^{10p-q} \le C \le 4u^{10p-q}$ *and* $C \ge 4^{-5/4} N^{-1/8} B_2^{5/4}$. *Consequently, for every real* $\kappa \in [9/10, 1)$ *(take* $u = 2$, $q \to \infty$, $p = \lfloor \kappa q \rfloor$, *or* $p/q = \kappa$ *if* κ *is rational) there are instances with* $B_2 = \Theta(N^\kappa)$ *and* $C = \Theta\big(N^{(10\kappa-1)/8}\big)$, *so* $C/(N + B_2) = \Theta\big(N^{(10\kappa-9)/8}\big)$*; Theorem 3.4 is attained up to the factor* $3^{1/8} \cdot 4^{5/4} < 6.5$, *and below slope* 1, $C = O(N + B_2)$ *fails, even with any fixed polylogarithmic factor, exactly on* $(9/10, 1)$ *(by Theorem 3.10 and Corollary 3.11 it also fails at every slope in* $[1, 3/2)$, *where the exponent* $3/8 + 3\kappa/4$ *exceeds* κ, *so over* $[9/10, 3/2]$ *it fails exactly on* $(9/10, 3/2)$*).*

[Proved] (on paper; exact checks in Appendix G)

Proof. Parameters. The exponent differences giving $h_i \mid L_i$ are $2p - q$, $7q - 6p$, $7p - 6q$, and those giving $a_i \le h_i$ are $10p - 9q$, $2p - q$, $10q - 9p$; all are nonnegative for $9/10 \le p/q \le 1$. *Moments.* D_1 contributes $(u^{8q}, 2u^{8p}, u^{16p-8q})$ to (α, β, γ). In D_2 the local forward sums $(La^2, 2ah^2, L^2) = (u^{8p}, 2u^{16p-8q}, u^{8q})$ land on (β, γ, α), and in D_3 the local sums $(u^{16p-8q}, 2u^{8q}, u^{8p})$ land on (γ, α, β); the totals are as stated. The reverse moments are $\alpha' = 2u^{12p-2q} + u^{9p-2q} + u^{6p+4q}$, $\beta' = u^{9q-2p} + 2u^{4p+6q} + u^{12q-2p}$, $\gamma' = 2u^{20p-10q} + u^{14p-7q} + u^{14p-4q}$; their terms with exponents $6p + 4q$, $12q - 2p$, $14p - 4q$ exceed $8q$, $8p$, $16p - 8q$ by $6p - 4q$, $12q - 10p$, $4q - 2p$, integers at least 2, so each alone is at least 4 times the forward counterpart. *Costs.* The global root x is local root x of D_1, local root z of D_2 and local root y of D_3, so $C_x = (Lah + La)_{D_1} + (ah^2)_{D_2} + (ah)_{D_3}$, which is the stated sum; likewise $C_y = u^{10p-q} + u^{4p+2q} + u^{8q} + u^{2p+3q}$ and $C_z = u^{7p+2q} + u^{10p-4q} + u^{8p} + u^{10p-5q}$. Every term of C_x is at most u^{10p-q}, so $C_x \le 4u^{10p-q}$; $C_x \le C_y$ termwise; and if $p < q$ the leading term of C_z is $u^{7p+2q} \ge u^3 u^{10p-q} > C_x$, while for $p = q$ the three costs coincide. So $C = C_x \ge u^{10p-q}$. *Size.* N is a sum of twelve powers of u with exponents at most $8q$, one of them $8q$. *Rate.* $N^{-1/8} B_2^{5/4} \le u^{-q} 4^{5/4} u^{10p} \le 4^{5/4} C$. With $u = 2$ and $p = \lfloor \kappa q \rfloor$, eventually $9/10 \le p/q < 1$ (for $\kappa > 9/10$), $8\kappa q - 8 < 8p \le 8\kappa q$, so $B_2 = \Theta(N^\kappa)$ and $C \ge 2^{10p-q} \ge 2^{-10} \big(2^{8q}\big)^{(10\kappa-1)/8}$; the upper bound is Theorem 3.4, and $N + B_2 = \Theta(N)$ for $\kappa < 1$. Below $9/10$, Corollary 3.5(2) gives $C = O(N + B_2)$. □

So the worst-case exponent of C at slope κ is $\min(3/2, \max(1, \min((10\kappa - 1)/8, 3/8 + 3\kappa/4)))$ for every real $\kappa \ge 9/10$, and at most 1 below; Figure 1 plots it.

3.4. Fully reduced inputs

Padding (Remark 3.9) leaves most tuples outside triangles. When *every* tuple lies in a triangle ("fully reduced") the low-slope families cannot be used: a triangle has one tuple of each relation, so Out $\ge \max(|R|, |S|, |T|) \ge N/3$, and Out $\le B_2$ (Section 2); hence $B_2 \ge N/3$ and every growing fully reduced family has slope $\kappa \ge 1$. The gap shrinks but survives. The gadget below keeps the two expensive options of $F_{i,j}$ at every x and adds helper vertices that close every tuple.

The live gadget $L(p, q, t, g, k, w)$. Let p, q, t, g, k, w be positive integers with $t \mid p$, $g \mid pt$, $g \le p$, $k \mid q$, and, for $z = pt/g$, $q \mid z$ and $w \mid z$. Take parts X_0 of size p, Y_0 of size q, Z_1 and Z_w of size t each, and Z_2 of size z. Split Z_2 into z/w groups W of w vertices; each group owns q fresh y-vertices Y_W, split into q/k blocks of k, and q/k fresh helper x-vertices $h_{W,1}, \ldots, h_{W,q/k}$. Then

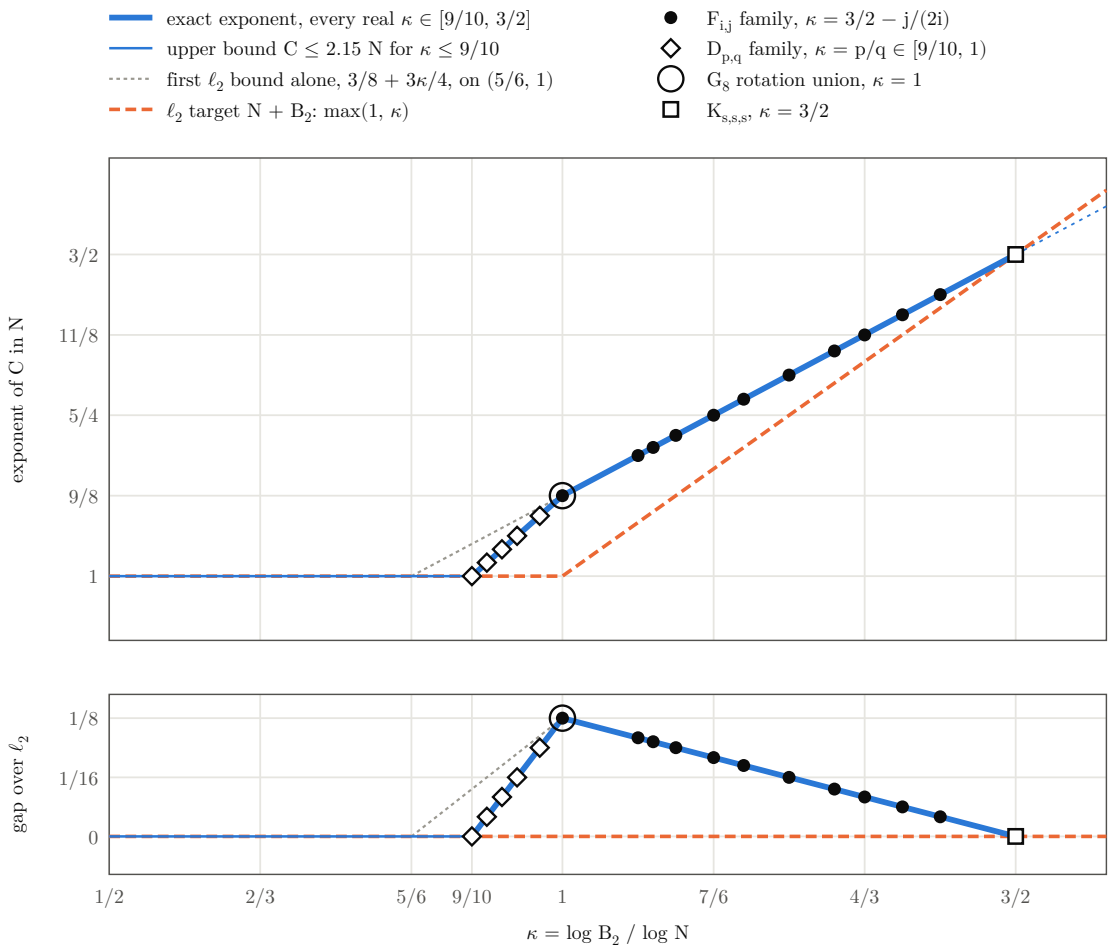


Figure 1: Worst-case exponent of the per-binding cost on instances with $B_2 = N^\kappa$. Thick solid: the exponent $\max(1, \min((10\kappa-1)/8, 3/8+3\kappa/4))$ of the bounds of Theorem 3.4 and Theorem 3.3, capped at $3/2$ by $C \le N + \sqrt{|R|\ |S|\ |T|}$ (Lemma 3.1). It is the exact worst-case exponent at every real $\kappa \in [9/10, 3/2]$ (Theorem 3.12, Corollary 3.11); thin solid, for $\kappa \le 9/10$, only the upper bound $C < 2.15N$ (Corollary 3.5). Dotted: the exponent $3/8 + 3\kappa/4$ of Theorem 3.3 alone on $(5/6, 1)$. Dashed: the exponent $\max(1, \kappa)$ of $N + B_2$. Dots: $F_{i,j}$ at $(i,j) = (1,1), (6,5), (5,4), (4,3), (3,2), (5,3), (2,1), (5,2), (3,1), (4,1), (6,1)$ ($\kappa = 3/2 - j/(2i)$, exponent $3/2 - 3j/(8i)$); diamonds: the asymmetric family of Theorem 3.12 at $p/q = 9/10, 11/12, 14/15, 19/20, 39/40$ (exponent $(10p-q)/(8q)$); ring: the union $\mathcal{U}(m)$ of Theorem 3.6 ($\kappa = 1$, exponent $9/8$); square: $K_{s,s,s}$. The lower panel plots the gap over $N + B_2$: 0 up to $9/10$, rising to $1/8$ at $\kappa = 1$ and falling to 0 at $3/2$. Generated by `make_fbeta_figure.py` with exact rational exponents.

$$R = X_0 \times Y_0 \cup \{(h_{W,m}, y) : y \text{ in block } m \text{ of } Y_W\},$$
$$S = Y_0 \times (Z_1 \cup Z_w) \cup S_0 \cup \bigcup_W Y_W \times W,$$
$$T = T_2 \cup T_w \cup T_1 \cup \bigcup_W W \times \{h_{W,1}, \ldots, h_{W,q/k}\},$$

where $S_0 \subseteq Y_0 \times Z_2$ gives every $z \in Z_2$ one predecessor and every $y \in Y_0$ exactly z/q successors, $T_2 \subseteq Z_2 \times X_0$ is biregular with out-degree g and in-degree t, $T_w \subseteq Z_w \times X_0$ is biregular with out-degree p/t and in-degree 1, and T_1 is a matching of Z_1 onto t distinct vertices of X_0.

Lemma 3.13 (the live gadget)**.** *Every tuple of $L = L(p, q, t, g, k, w)$ lies in a triangle, $\mathrm{Out}(L) = tq + pq + pt + zq$, and every $x \in X_0$ has both option costs at least qt, so $C_x(L) \geq pqt$. Its sizes are $(|R|, |S|, |T|) = (pq + zq/w,\ 2qt + z + zq,\ pt + p + t + zq/k)$, its forward squared-degree sums are $(\alpha, \beta, \gamma) = (pq^2 + zqk/w,\ q(2t + z/q)^2 + zqw,\ t + p^2/t + z(g + q/k)^2)$, and its reverse sums are $(\alpha', \beta', \gamma') = (qp^2 + zq/w,\ 2tq^2 + z(q+1)^2,\ (p - t)(t+1)^2 + t(t+2)^2 + zqw/k)$.*

[Proved]

Proof. Support. An $X_0 \times Y_0$ tuple (x, y) closes through the unique T_w-predecessor of x, since $Y_0 \times Z_w$ is complete. A tuple (y, z) with $z \in Z_1 \cup Z_w \cup Z_2$ closes through any T-successor of z in X_0, since $X_0 \times Y_0$ is complete; a T-tuple (z, x) with $x \in X_0$ closes through a Y_0-predecessor of z (every z of $Z_1 \cup Z_w \cup Z_2$ has one). For a group W, a tuple $\big(h_{W,m}, y\big)$, (y, z) or $\big(z, h_{W,m}\big)$ with y in block m of Y_W and $z \in W$ lies in the triangle $\big(h_{W,m}, y, z\big)$. *Output.* The triangles are (x_i, y, z_i) through T_1 (tq), (x, y, z) through the T_w-predecessor of x (pq), (x, y_z, z) through T_2 with y_z the S_0-predecessor of z ($zg = pt$), and the helper triangles (qw per group, zq in all). *Degrees.* On X_0, $d_R^+ = q$ and $d_T^- = t + 1 + [x \in T_1(Z_1)]$; on a helper, $d_R^+ = k$ and $d_T^- = w$; on Y_0, $d_R^- = p$ and $d_S^+ = 2t + z/q$; on Y_W, $d_R^- = 1$ and $d_S^+ = w$; on Z_1 and Z_w, $d_S^- = q$ and $d_T^+ = 1$, resp. p/t; on Z_2, $d_S^- = q + 1$ and $d_T^+ = g + q/k$. The sizes and moments follow. *Cost.* At $x \in X_0$, option R walks q vertices y with $\min\big(d_S^+(y), d_T^-(x)\big) \geq t$, and option T walks at least t vertices of Z_2 with $\min\big(d_R^+(x), d_S^-(z)\big) = q$. □

Theorem 3.14 (fully reduced inputs: gap in $[1/10, 1/8]$)**.** *For $v \geq 2$ let $\mathcal{L}(v)$ be the disjoint union of $L(v^5, v^3, v^4, v^2, v^2, v)$ and its two rotations. It is fully reduced, with*

$$N = 3\big(v^{10} + 2v^9 + 2v^8 + 3v^7 + v^5 + v^4\big) \leq 9v^{10}, \quad \mathrm{Out} = 3\big(v^{10} + v^9 + v^8 + v^7\big),$$

$$\alpha = \beta = \gamma = 13v^{11} + 2v^{10} + v^9 + v^6 + v^4 \leq 15v^{11}, \quad C = v^{12} + 4v^{10} + 3v^9 + 7v^8 + 7v^7 + v^5 + v^4,$$

and every reverse moment larger than α, so $B_2 = \alpha$, $B_2 = \Theta\big(N^{11/10}\big)$ and

$$C/(N + B_2) \geq v/24 \geq (N/9)^{1/10}/24.$$

More generally, for integers $i \geq 1$, $j \geq 0$ with $5j \leq 4i$ and $u \geq 2$, the union of three rotations of $L(u^{4i}, u^{4i-2j}, u^{4i-j}, u^{4i-3j}, u^j, 1)$ is fully reduced with $C \geq u^{12i-3j}$, $N = \Theta(u^{8i})$ and $B_2 = \Theta(u^{12i-4j})$, the constants independent of i, j, u. With $u = 2$, $i = n$, $j = \lfloor (3 - 2\kappa)n \rfloor$ this gives, for every real $\kappa \in [11/10, 3/2]$, fully reduced instances with $B_2 = \Theta(N^\kappa)$ and $C = \Theta\big(N^{3/8 + 3\kappa/4}\big)$. So on fully reduced inputs the worst-case exponent of C equals the general one (Theorem 3.3) at every slope in $[11/10, 3/2]$, the worst-case gap against $N + B_2$, over all slopes, lies in $[1/10, 1/8]$, and it is open only on the window $\kappa \in [1, 11/10)$.

[Proved] (verified by two separately run exact-arithmetic rebuilds; not in Lean; Appendix G)

Proof. $\mathcal{L}(v)$. The parameters satisfy the divisibility conditions ($z = v^7$). The union of fully reduced instances is fully reduced. By Lemma 3.13 and rotation, each root of the union contains a copy of root x of one gadget, so $C \geq pqt = v^{12}$; direct counting of the other two roots of each gadget, whose cheaper options cost $O(N)$, gives the displayed polynomial. Each forward moment of the union is $\alpha + \beta + \gamma$ of one gadget: $2v^{11} + (9v^{11} + v^{11}) + (v^4 + v^6 + v^{11} + 2v^{10} + v^9)$, and each reverse moment is $\alpha' + \beta' + \gamma' = (v^{13} + v^9) + (v^{13} + 4v^{10} + v^7) + (v^{13} + 3v^9 + 2v^8 + v^5 + 3v^4)$, which exceeds the forward one for $v \geq 2$ (their difference is $3v^{13} - 13v^{11} + 2v^{10} + 3v^9 + 2v^8 + v^7 - v^6 + v^5 + 2v^4 > 0$). $N \leq 9v^{10}$ because $2v^9 + 2v^8 + 3v^7 + v^5 + v^4 \leq 2v^{10}$ for $v \geq 2$; then $C/(N + B_2) \geq v^{12}/(9v^{10} + 15v^{11}) \geq v/24$. *The general family.* Here $z = u^{4i+2j}$, and $3j \leq 4i$ gives $k \mid q$. By Lemma 3.13 every root of the union costs at least $pqt = u^{12i-3j}$. The size monomials have exponents at most $8i$, attained by $zq = u^{8i}$; the forward moment monomials have exponents among $12i - 4j$, $8i + j$, $4i + 6j$, $8i$, $4i + j$ and $4i - j$, all at most $12i - 4j$ exactly when $5j \leq 4i$, with $12i - 4j$ attained by pq^2; and the reverse moments contain $qp^2 = u^{12i-2j}$. Each statistic is a sum of a bounded number of such monomials with bounded coefficients, which gives the Θ's with absolute constants and $\kappa = 3/2 - j/(2i) \in [11/10, 3/2]$. For real κ, the floor changes $4j$ by less than 4, as in Corollary 3.11; at $\kappa = 3/2$ take $j = 0$. The upper bounds are Theorem 3.3 and Corollary 3.5(1), whose gap $(3 - 2\kappa)/8$ is at most $1/8$ and equals $1/10$ at $\kappa = 11/10$. □

Remark 3.15 (the window $\kappa \in [1, 11/10)$)**.** (a) *A lower bound on the window.* Adding v^{11} disjoint isolated triangles to the three rotations of $L(v^5, v^3, v^4, v^2, v, 1)$ keeps the instance fully reduced and gives $N = \Theta(v^{11})$, $B_2 = \Theta(v^{11})$ and $C \geq v^{12}$: gap $1/11$ at slope 1 (exactly, at $v = 2$: $N = 17424$, $C = 17840$, $B_2 = 33872$; at $v = 3$: $N = 1044171$, $C = 1321272$, $B_2 = 2894211$). Scaling the core parameters in the same way gives gap $\kappa/11$ at every $\kappa \in [1, 11/10]$, so on the window the fully reduced gap lies in $[\kappa/11, (3 - 2\kappa)/8]$. (b) *The limit of this design.* For a union of three rotated live gadgets with arbitrary valid parameters, $\min_i p_i q_i t_i \leq C \leq 3N + \min_i p_i q_i t_i$ (the other two roots of a gadget cost at most $3N_g/2$ and $2N_g$, and the non-critical part of root x at most $3N_g$). In exponent form, with $c = P + Q + T$ the designated cost exponent and m the average forward moment exponent, two linear identities over the gadget parameters give $c \leq 12m/11$ and $c \leq 3/8 + 3m/4$, so $c - m \leq \min(m/11, (3 - 2m)/8) \leq 1/10$, with equality only at $m = 11/10$, and a controlling reverse moment gives no gap at all. So this design cannot close the window or exceed $1/10$; doing so needs another design or an upper bound that uses triangle support, not only pairwise support. An exponent program over the design that bounds size exponents by 1 without forcing one to equal 1 has value $1/11$ at budget slope 1, but its optimum has actual slope $11/10$; the value $1/11$ at actual slope 1 is attained only with the padding of (a).

[Proved] (exact checks in Appendix G)

An earlier draft used a different fully reduced gadget, with gap $1/12$ and an explicit instance with $C > N + B_2$ already at $N \approx 2.4 \cdot 10^{11}$ ($\mathcal{L}(v)$ first exceeds 1 at $v = 14$, $N \approx 1.0 \cdot 10^{12}$); it is kept in Appendix F.

3.5. Separations from fixed orders and fixed roots

A strategy has *worst-case gap* δ against a comparator K if its cost is at most $N^{\delta+o(1)}K$ on all instances and this is attained. By Theorem 3.3 and Theorem 3.6, per-binding choice has gap exactly $1/8$ against $N + B_2$ (at $B_2 \asymp N$). The families below use three distinct relations.

Additional families. GMR(m): X of size m^3, Y of size m, Z_0 of size m^4, Z_1 of size m; $R = X \times Y$, $S = Y \times Z_1$, and $T = Z_0 \to X$ biregular with $d_T^- = m$ on X and $d_T^+ = 1$ on Z_0. GMIN(m), for $m \geq 2$, has parts X_0, X_1, X_2, X_3 of sizes m^3, m^5, m^2, m; Y_0, Y_1, Y_2, Y_3 of sizes m, m^3, m^2, m^5; Z_0, Z_1, Z_2, Z_3 of sizes m^5, m^2, m, m^3; and blocks $R = X_0 \times Y_0 \cup (X_1 \to Y_1) \cup X_3 \times Y_2$, $S = Y_0 \times Z_1 \cup Y_1 \times Z_2 \cup (Y_3 \to Z_3)$, $T = (Z_0 \to X_0) \cup Z_2 \times X_2 \cup Z_3 \times X_3$, where $X_1 \to Y_1$ and $Y_3 \to Z_3$ are biregular with out-degree 1, and $Z_0 \to X_0$ is biregular with $d_T^- = m^2$ on X_0 and $d_T^+ = 1$ on Z_0; all other blocks are complete.

Theorem 3.16 (fixed orders, fixed roots, and two levels of adaptivity)**.**

1. (one fixed GAO) *For $a \geq 2$, $\mathcal{F}_a$ has $M_R = a^3$, $M_S = M_T = a^2$, $C = 2a^2$, $\mathrm{Out} = a^2$, $N = 3a^2$, $B_2 = a^{8/3}$, and is pairwise semijoin-reduced. So the GAOs whose second relation is R cost $\Theta\big(N^{1/2}\big) \cdot C$ and $\Theta\big(N^{1/6}\big) \cdot (N + B_2)$. On general inputs,* GMR(m) *has $M_R = m^5$, $M_S = M_T = 0$, $C = 0$, $N = 2m^4 + m^2$ and $B_2 \leq m^4$, so those GAOs cost $M_R/(N + B_2) \geq m/4 = \Theta\big(N^{1/4}\big)$ times $N + B_2$. Rotations give the same for the other second relations.*
2. (best of the six GAOs) *For $a \geq 2$, $\mathcal{G}_a$ has $\min(M_R, M_S, M_T) = a^3 + 2a^2$, $C = 6a^2$, $N = 9a^2$, and is pairwise semijoin-reduced; so the best GAO per instance costs $\Theta\big(N^{1/2}\big) \cdot C$. For $m \geq 2$,* GMIN(m) *has $M_R = M_S = M_T = m^6$, $C = m^5 + m^3 \leq N$, $N = 3\big(m^5 + m^4 + m^3\big)$ and $\alpha = \beta = \gamma = 3m^5$ (and larger reverse norms), so $B_2 = 3m^5$ and $\min_\pi M_\pi/(N + B_2) \geq m/9 = \Theta\big(N^{1/5}\big)$ while $\min_\pi M_\pi/C \geq m/2$.* GMIN($m$) *has no triangles and is not semijoin-reduced; its padding (Lemma 3.8) is pairwise semijoin-reduced with $\mathrm{Out} = \Theta(N)$, $C = \Theta(N)$, $N + B_2 = \Theta\big(m^5\big)$ and $\min_\pi M_\pi \geq m^6$. So the best GAO loses $\Theta\big(N^{1/5}\big)$ against both $N + B_2$ and $C + N$, on semijoin-reduced inputs with $C = \Theta(N)$.*
3. (fixed root) *For $a \geq 2$, $\mathcal{H}_a$ has $C_x = a^3 + a^2$, $C_y = C_z = a^2$, $\mathrm{Out} = 0$, $N = 4a^2$; its padding is pairwise semijoin-reduced, keeps $C_x \geq a^3$ and has $C + N = \Theta\big(a^2\big)$. So root x with per-binding option choice costs $\Theta\big(N^{1/2}\big) \cdot (C + N)$, on semijoin-reduced inputs as well. Against $N + B_2$, a single $G_8(m)$ has $C_x = m^9 + m^6$, $C = C_y = m^5$ and $N + B_2 = \Theta\big(m^8\big)$, so a fixed root loses $\Theta\big(N^{1/8}\big)$ against $N + B_2$, and $\Theta\big(N^{1/2}\big)$ against C alone. Rotations give the same for roots y and z.*

[Proved]

Proof. (1) $\mathcal{F}_a$: $X = \{x_i\}$, $Y = \{y_j\}$, $Z = \{z_{i,k}\}$ for $i, j, k \in [a]$; $R = X \times Y$; $S = \{(y_i, z_{i,k})\}$; $T = \{(z_{i,k}, x_i)\}$. Degrees: $d_R^+ = d_R^- = a$, $d_S^+(y) = a$, $d_S^-(z) = 1$, $d_T^+(z) = 1$, $d_T^-(x) = a$. A triangle (x_i, y_j, z) needs $z = z_{j,k}$ with $(z, x_i) \in T$, so $j = i$ and $\mathrm{Out} = a^2$. Every endpoint of every tuple has an incident tuple in both other relations (the instance is pairwise semijoin-reduced). $M_R = a^2 \cdot \min(a, a) = a^3$; $M_S = a^2 \cdot \min(1, a) = a^2$; $M_T = a^2 \cdot \min(a, 1) = a^2$. Root x: $c_R = a(1 + a)$ and $c_T = a(1 + 1) = 2a$, so $C_x = 2a^2$; similarly

$C_y = C_z = 2a^2$ (root y: options cost $a(1+a)$ and $2a$; root z: both cost 2). Forward squared degree sums are a^3, a^3, a^2 and reverse ones a^3, a^2, a^3, so $B_2 = a^{8/3}$. Then $M_R/C = a/2 = \Theta\big(N^{1/2}\big)$ and $M_R/(N+B_2) = a^3/\big(3a^2+a^{8/3}\big) = \Theta\big(N^{1/6}\big)$. GMR($m$): every R-tuple has $d_S^+(y) = m = d_T^-(x)$, so $M_R = m^4 \cdot m$; S ends in Z_1, where $d_T^+ = 0$, and T starts in Z_0, where $d_S^- = 0$, so $M_S = M_T = 0$; for root z every binding has an empty option, so $C = 0$. Norms: $\alpha = m^3 \cdot m^2$, $\beta = m \cdot m^2$, $\gamma = m^4$, so $B_2 \le \big(m^{12}\big)^{1/3}$.

(2) $\mathcal{G}_a$ is the disjoint union of $\mathcal{F}_a$ and its two rotations, hence semijoin-reduced. Each M_π equals a^3 on one copy and a^2 on the other two, so $M_\pi = a^3 + 2a^2$ for every π; by Lemma 3.1 every GAO costs at least this. In $\mathcal{F}_a$ all of C_x, C_y, C_z equal $2a^2$, so in each copy, whatever its rotation, every root costs $2a^2$, and $C = 6a^2$. GMIN(m): M_R collects $m^4 \cdot \min(m^2, m^2)$ from $X_0 \times Y_0$ ($d_S^+ = |Z_1|$ on Y_0, $d_T^- = m^2$ on X_0) and 0 from the other blocks ($d_T^- = 0$ on X_1, $d_S^+ = 0$ on Y_2). Likewise $M_S = m^6$ from $Y_1 \times Z_2$ ($d_T^+ = |X_2|$, $d_R^- = m^2$) and $M_T = m^6$ from $Z_3 \times X_3$ ($d_R^+ = |Y_2|$, $d_S^- = m^2$); every other block meets a zero degree. Root x: X_0 takes option T (m^2 unit-cost tuples from Z_0, where $d_S^- = 0$; option R costs $m(1+m^2)$), giving $m^3 \cdot m^2$; X_3 takes option R (m^2 unit-cost tuples into Y_2, where $d_S^+ = 0$), giving $m \cdot m^2$; X_1 ($d_T^- = 0$) and X_2 ($d_R^+ = 0$) have an empty option. So $C_x = m^5 + m^3$. Symmetrically, $C_y = m^3 + m^5$ (Y_0, option S into Z_1, where $d_T^+ = 0$; Y_1, option R from X_1, where $d_T^- = 0$) and $C_z = m^3 + m^5$ (Z_2, option T into X_2; Z_3, option S from Y_3, where $d_R^- = 0$). Sizes: $|R| = m^4 + m^5 + m^3$, $|S| = m^3 + m^4 + m^5$, $|T| = m^5 + m^3 + m^4$. Forward norms: each of X_0, X_1, X_3 contributes m^5 to α ($m^3 \cdot m^2$, $m^5 \cdot 1$, $m \cdot m^4$), and likewise Y_0, Y_1, Y_3 to β and Z_0, Z_2, Z_3 to γ. The reverse norms are larger ($\alpha' \ge m \cdot m^6 = m^7$ from Y_0, and similarly for β', γ'), so $B_2 = 3m^5$. We also checked every closed form exactly for $m = 2, ..., 12$ with two separately written evaluators. Then $m^6/(6m^5 + 3m^4 + 3m^3) \ge m/9$ for $m \ge 2$. No triangle exists, and, e.g., $X_3 \times Y_2$ is an R-block whose Y_2 has no S-tuple. For the padding, Lemma 3.8 gives $\min_\pi M_\pi \ge m^6$, $N' \le 7N$, $C' \le C + 9N = O(m^5)$, squared norms $O(m^5)$, and $C' \ge \mathrm{Out}' \ge m^5$ (all of Y_3 is padded).

(3) $\mathcal{H}_a$: $X = \{x_i\}$, $Y_1 = \{y_i\}$, $Y_2 = \{y_i'\}$, $Z_1 = \big\{z_{i,k}\big\}$, $Z_2 = \{z_i'\}$ for $i, k \in [a]$; $R = X \times Y_1$; $S = \big\{\big(y_i, z_{i,k}\big)\big\} \cup Y_2 \times Z_2$; $T = Z_2 \times X$. Since S maps Y_1 only into Z_1 and T leaves only Z_2, Out = 0. Root x: $c_R(x) = a(1 + \min(a, a))$ and $c_T(x) = a(1 + \min(a, a))$, so $C_x = a^3 + a^2$. Root y: for $y \in Y_1$, option S costs $a(1 + \min(a, 0)) = a$ (since $d_T^+ = 0$ on Z_1); for $y \in Y_2$ option R is empty; so $C_y = a^2$. Root z: for $z \in Z_1$ option T is empty; for $z \in Z_2$ option S costs $a(1 + \min(0, a)) = a$; so $C_z = a^2$. By Lemma 3.8 the padded instance is pairwise semijoin-reduced, has $C_x' \ge a^3 + a^2$, $C' \le C_y + 9N = 37a^2$ and $N' \le 28a^2$; we also measured $C_x'/(C' + N') = 0.61, 0.97, 1.69, 3.14, 6.05$ for $a = 4, 8, 16, 32, 64$. $G_8(m)$: see the proof of Theorem 3.6; its forward norms are $m^8, 2m^8, 2m^8$, so $B_2 \le 4^{1/3} m^8$. □

All quantities stated for $\mathcal{F}_a, \mathcal{G}_a, \mathcal{H}_a$ were also recomputed by brute force for $a \in \{4, 8, 16, 32\}$, and those for GMR and GMIN for $m = 2, 3, 4$ and $m = 2, ..., 12$, respectively. A single fixed GAO always costs at least the best of the six, so Theorem 3.16(2) also gives gap at least 1/5 for one fixed GAO on semijoin-reduced inputs, superseding the $N^{1/6}$ of $\mathcal{F}_a$ there. The families of Theorem 3.16 and of Theorem 3.6 are complementary: on $\mathcal{F}_a, \mathcal{G}_a, \mathcal{H}_a$ (and padded $\mathcal{H}_a$) $C = \Theta(N)$, so their $\Theta\big(N^{1/2}\big)$ losses are against $C + N$, the maximum

possible since every GAO costs at most $N + \sqrt{|R|\ |S|\ |T|}$; the families G_8, GMR and GMIN separate against $N + B_2$, but against C only GMIN has $C = \Theta(N)$. Table 1, in Section 1, summarizes.

Theorem 3.17 (best of the six GAOs: exact gap 1/5)**.** *On every instance*

$$\min(M_R, M_S, M_T)^5 \le |R|\ |S|\ |T| \cdot \min(\alpha\beta\gamma, \alpha'\beta'\gamma'),$$

i.e. $\min_\pi M_\pi \le (|R|\ |S|\ |T|)^{1/5} B_2^{3/5} \le (N/3)^{3/5} B_2^{3/5}$. *The product form (the displayed fifth-power inequality) holds with equality on every complete tripartite instance $K_{a,b,c}$; the relaxation to $(N/3)^{3/5}$ is tight only when in addition $|R| = |S| = |T|$. Hence* $\min_\pi M_\pi \le \big(2^{2/5}/5\big) N^{1/5}(N + B_2) \approx 0.264 N^{1/5}(N + B_2)$, $\min_\pi M_\pi \le 12^{-3/5}(N + B_2)^{6/5}$, *and* $\min_\pi M_\pi \le 3^{-3/5} N^{6/5}$ *when $B_2 < N$. With* GMIN(m), *where* $\min_\pi M_\pi/(N + B_2) = m^3/(6m^2 + 3m + 3)$, *and its padding (Theorem 3.16(2)), the best of the six GAOs has worst-case gap exactly* 1/5 *against $N + B_2$, on general and on pairwise semijoin-reduced inputs, and worst-case cost* $\Theta\Big((N + B_2)^{6/5}\Big)$.

[Proved] (on paper; exact checks in Appendix G)

Proof. Write $a_x = d_R^+(x)$, $t_x = d_T^-(x)$ and $P(x) = \sum_{y \in R(x,\cdot)} \min\big(d_S^+(y), t_x\big)$, so $M_R = \sum_x P(x)$. Then $P(x) \le a_x t_x$ and, by step 1 of the proof of Lemma 3.2, $P(x) \le \sqrt{a_x \beta}$; so $P(x) \le (a_x t_x)^{3/5} (a_x \beta)^{1/5} = \beta^{1/5} a_x^{4/5} t_x^{3/5}$. Hölder with exponents $(5/2, 5/3)$ gives $M_R \le \beta^{1/5} \Big(\sum_x a_x^2\Big)^{2/5} \Big(\sum_x t_x\Big)^{3/5} = |T|^{3/5}\ \alpha^{2/5}\beta^{1/5}$, and by rotation $M_S \le |R|^{3/5}\ \beta^{2/5}\gamma^{1/5}$ and $M_T \le |S|^{3/5}\ \gamma^{2/5}\alpha^{1/5}$. The minimum is at most the cube root of the product, $(|R|\ |S|\ |T|\ \alpha\beta\gamma)^{1/5}$; reflection gives the primed product. On $K_{a,b,c}$ every $M_\pi = abc$, $|R|\ |S|\ |T| = (abc)^2$ and both moment products are $(abc)^3$. With $v = B_2/N$, $(N/3)^{3/5} B_2^{3/5}/(N + B_2) = 3^{-3/5} N^{1/5} v^{3/5}/(1 + v)$, maximal at $v = 3/2$; and $NB_2 \le (N + B_2)^2/4$. A GAO with second relation π costs $|\pi| + M_\pi \le N + M_\pi$ (Lemma 3.1). For GMIN(m), which is the union of three rotations of the gadget E below with $(L, a, h) = (m^3, m, m^2)$, Theorem 3.16(2) gives $B_2 = 3m^5$ exactly and $M_\pi = m^6$. □

The gadget $E(L, a, h)$. $|X| = L$, $|Y| = a$, $|Z_1| = h$, $Z_0 = \big\{z_{x,j} : x \in X, j \in [h]\big\}$, $R = X \times Y$, $S = Y \times Z_1$, $T = \big\{(z_{x,j}, x)\big\}$. Then $(|R|, |S|, |T|) = (La, ah, Lh)$, $(\alpha, \beta, \gamma) = (La^2, ah^2, Lh)$, $(\alpha', \beta', \gamma') = (aL^2, ha^2, Lh^2)$, $(M_R, M_S, M_T) = (Lah, 0, 0)$, $\big(C_x, C_y, C_z\big) = (Lh, ah, 0)$ and Out $= 0$; GMR$(m) = E\big(m^3, m, m\big)$.

Theorem 3.18 (one fixed GAO: exact gap 1/4 on general inputs)**.** *On every instance $M_R^4 \le |R|\ |T|^2\ \alpha\beta \le |R|\ |T|\ \alpha\beta\gamma$ and $M_R^4 \le |R|\ |S|^2\ \alpha'\gamma' \le |R|\ |S|\ \alpha'\beta'\gamma'$, and likewise for M_S, M_T by rotation. Hence, for any prescribed driver, $M_R \le (N/2)^{1/2} B_2^{3/4} \le \Big(3^{3/4}/\big(4\sqrt{2}\big)\Big) N^{1/4}(N + B_2)$, and the coefficient $3^{3/4}/\big(4\sqrt{2}\big) \approx 0.403$ is asymptotically sharp ($E(u^3, 6u, 6u)$: $N = 12u^4 + 36u^2$, $B_2 = 36u^4$, $M_R = 36u^5$). With* GMR(m), *where $M_R/(N + B_2) = m^3/(3m^2 + 1)$, the gap of one fixed GAO against $N + B_2$ is exactly* 1/4 *on general inputs, with worst-case cost* $\Theta\Big((N + B_2)^{5/4}\Big)$. *On pairwise semijoin-reduced inputs it lies in* $[1/5, 1/4]$ *(lower bound: padded* GMIN, *Theorem 3.16(2)) and remains open.*

[Proved] (on paper; exact checks in Appendix G)

Proof. Keep $a_x, t_x, P(x)$ and put $b_y = d_S^+(y)$. Terms with $t_x = 0$ have $P(x) = 0$, and weighted Cauchy–Schwarz gives $M_R^2 \le |T| \sum_{x:t_x>0} P(x)^2/t_x$. Since $\min(b,t) \le \sqrt{bt}$, $P(x)^2/t_x \le \left(\sum_{y\in R(x,\cdot)} \sqrt{b_y}\right)^2 \le a_x \sum_{y\in R(x,\cdot)} b_y$. With $v_y = \sum_{x:(x,y)\in R} a_x$ this sums to $\sum_y b_y v_y \le \sqrt{\beta}\left(\sum_y v_y^2\right)^{1/2}$, and $v_y^2 \le d_R^-(y) \sum_x a_x^2 = d_R^-(y)\alpha$, so $\sum_y v_y^2 \le |R|\ \alpha$ and $M_R^2 \le |T|\ \sqrt{|R|\ \alpha\beta}$; $\gamma \ge |T|$ gives the second form. Grouping by y with weights b_y instead, $M_R^2 \le |S| \sum_y V_y^2/b_y$ with $V_y = \sum_{x:(x,y)\in R} \min(b_y, t_x)$, $V_y^2/b_y \le d_R^-(y) \sum_{x:(x,y)\in R} t_x$; summing over y gives $\sum_x t_x w_x$ with $w_x = \sum_{y\in R(x,\cdot)} d_R^-(y)$, $w_x^2 \le a_x \alpha'$ and $\sum_x t_x^2 = \gamma'$, so $M_R^2 \le |S|\ \sqrt{|R|\ \alpha'\gamma'}$; $\beta' \ge |S|$. Since $|R|\ |T|, |R|\ |S| \le N^2/4$, choosing the orientation that realizes B_2^3 gives $M_R \le (N/2)^{1/2} B_2^{3/4}$, and $v^{3/4}/(1+v)$ is maximal at $v = 3$. The sharp family and GMR are instances of E (GMR: $B_2 = m^4$ exactly). □

Corollary 3.19 (a fixed root against $N + B_2$)**.** *For every instance and every fixed root, e.g. x, $C_x \le |R| + M_R \le N + \left(3^{3/4}/\left(4\sqrt{2}\right)\right) N^{1/4}(N + B_2)$. So the worst-case gap of a fixed root against $N + B_2$ lies in $[1/8, 1/4]$, the lower end from $G_8(m)$ (Theorem 3.16(3)).*

[Proved]

Proof. By Lemma 3.1, $C_x \le \sum_x c_R(x) = |R| + M_R$, and Theorem 3.18 bounds M_R for the prescribed driver R. Roots y and z use drivers S and T. □

Remark 3.20 (what this does not show)**.** The G_8 separation of a fixed root is instance-wise: on G_8, $C + N \asymp N + B_2 \asymp m^8$, and the fixed root loses $N^{1/8}$ against either. That exponent equals the per-binding worst-case gap against $N + B_2$, so it does not show that a fixed root has a worse *worst-case* exponent against $N + B_2$; that exponent lies in $[1/8, 1/4]$ (Corollary 3.19). The orders are settled: the best of six has gap exactly $1/5$ (Theorem 3.17), and one fixed GAO exactly $1/4$ on general inputs and between $1/5$ and $1/4$ on semijoin-reduced ones (Theorem 3.18).

Proposition 3.21 (a single predicate)**.** *For $P \subseteq V \times V$ consider the cyclic self-join $P(a,b) \wedge P(b,c) \wedge P(c,a)$, the instance $R = S = T = P$ on $X = Y = Z = V$; its input size is $N = 3|P|$.*

1. (tagged embedding) *Let (R, S, T) be an instance on disjoint X, Y, Z and $P = R \cup S \cup T$ on $V = X \sqcup Y \sqcup Z$. Then $\mathrm{Out}_P = 3\ \mathrm{Out}$, $N_P = 3N$, all three roots cost $C_x + C_y + C_z$, every $M_\pi^P = M_R + M_S + M_T$, every forward moment of P is $\alpha + \beta + \gamma$ and every reverse moment $\alpha' + \beta' + \gamma'$, so $B_{2,P} = \min(\alpha + \beta + \gamma, \alpha' + \beta' + \gamma')$.*
2. (the lower bounds survive) *One $G_8(m)$ embedded this way has exactly the values of $\mathcal{U}(m)$: $N = 3(2m^8 + m^7 + m^6 + m^5)$, $B_2 = 5m^8$, $C = m^9 + m^8 + m^6 + m^5$; so Theorem 3.6 and the exponent $9/8$ of Corollary 3.7 hold for one directed predicate. Padding keeps the embedding tagged and pairwise semijoin-reduced. $\mathcal{F}_a$ gives $N = 9a^2$, $C = 6a^2$, $\mathrm{Out} = 3a^2$ and every GAO cost $a^3 + 5a^2$, a $\Theta\left(N^{1/2}\right)$ loss of every order against $C + N$ on a semijoin-reduced instance; $\mathrm{GMIN}(m)$ gives every GAO cost $3m^6 + 3(m^5 + m^4 + m^3)$ against $N + B_2 = 18m^5 + 9m^4 + 9m^3$, the gap $1/5$. The low-slope results do not transfer: every directed P has $B_{2,P} = \min\left(\sum_v d_P^+(v)^2, \sum_v d_P^-(v)^2\right) \ge |P| = N_P/3$, so no growing single-predicate family has slope below 1 (e.g. $\kappa = 19/20$ is impossible), and only the $9/8$ case at $B_2 \asymp N$ and the order separations above are claimed for one predicate.*

3. (no fixed-root or fixed-order separation) *For the cyclic self-join, renaming $a \to b \to c \to a$ maps the query to itself, so $C_a = C_b = C_c$ and $M_R = M_S = M_T$: a fixed root costs exactly C, and all six GAOs cost the same. The separations of Theorem 3.16(3) and between one GAO and the best of six cannot occur for this query.*
4. (symmetric predicate) *If $P = P^\top$ (an undirected graph with degrees d), the two options at every binding walk the same neighbours at the same per-edge costs, all six moments equal $\sum_v d(v)^2 = B_2$, and $C = |P| + \sum_{(u,v)\in P} \min(d(u), d(v)) \le |P| + B_2 \le N + B_2$. The first inequality is an equality for a single undirected edge ($|P| = B_2 = 2$, $C = 4$); the relaxation to $N + B_2$ is not ($N = 6$). There is no $N^{1/8}$ gap for undirected single-relation triangles.*

[Proved] (exact checks in Appendix G. Parts (1)–(2) were found by two model-based reviews of the second draft, the GPT-6-Astra Pro review and a separate GPT-6 Pro session; parts (3)–(4) by the latter; Section 9.)

Proof. (1) Every tuple of P goes $X \to Y$, $Y \to Z$ or $Z \to X$, so a directed triangle of P is a triangle of (R, S, T) read from one of its three vertices, and d_P^+, d_P^- restrict to the degrees of the relation leaving, resp. entering, each part. A binding $a \in X$ of the self-join has options walking $P(a, \cdot) = R(a, \cdot)$ and $P(\cdot, a) = T(\cdot, a)$, with per-edge costs $1 + \min(d_S^+(y), d_T^-(a))$ and $1 + \min(d_R^+(a), d_S^-(z))$: exactly the options of root x. Bindings in Y and Z reproduce roots y and z, so $C_a^P = C_x + C_y + C_z$, and by the symmetry in (3) the same holds for roots b, c. The M and moment identities are the same restriction. (2) Substitute the values of the proof of Theorem 3.6 ($C_x + C_y + C_z = m^9 + m^6 + m^5 + m^8$, forward sums $m^8 + 2m^8 + 2m^8$, reverse sum $m^{12} + 2m^{11} + 3m^{10} + m^7$), of Theorem 3.16(1) for $\mathcal{F}_a$ ($C_v = 2a^2$, $M = a^3 + 2a^2$, $|P| = 3a^2$) and of Theorem 3.16(2) for GMIN. Padding adds tagged triangles, and a vertex of the padded P has an in- and an out-tuple, which is pairwise reduction for the self-join. (3) The renaming permutes the three atoms, the three roots, and the three drivers. (4) A binding v walks $P(v, \cdot) = P(\cdot, v)$ in both options with per-edge cost $1 + \min(d(v), d(u))$; summing, $C = |P| + \sum_{(v,u)\in P} \min(d(v), d(u)) \le |P| + \sum_{(v,u)\in P} d(v) = |P| + \sum_v d(v)^2$. □

The tagged union of padded $G_8(m)$ has three components in the role-expanded instance (one per rotation), so a per-component root choice could escape it; adding one tuple $x_0 \to x_1$ between two vertices of X makes it connected. For $m = 2, 3$ the result is pairwise semijoin-reduced and connected, with $(N, C, B_2, \mathrm{Out}) = (6891, 4277, 3873, 1560)$ and $(167187, 112924, 98515, 39447)$ (computed exactly, `artifacts/hillclimb/checks/selfjoin_connected.py`), and the extra tuple changes the squared sums by $O(m^4)$, so the exponent stays 9/8. What remains open for self-joins is the case of two edges sharing a predicate, where renaming the variables is not a symmetry.

3.6. Sorted intersections

The lower bounds above charge the shorter run (Section 2). An engine that stores sorted runs intersects them by comparisons, leapfrog or galloping, and can be much cheaper when identifiers are contiguous. We now let the adversary choose the identifiers and charge comparisons. Every lower bound against $N + B_2$ survives without logarithmic loss, as do the fixed-root and padded-GMIN bounds against $C + N$ (the other $\Theta(N^{1/2})$ losses against $C +$

N are established only up to a factor $\log N$, Remark 3.27); the upper bound of Theorem 3.3 survives up to a factor $(\log N)^{1/4}$, and that factor is necessary. The single-intersection ingredients are classical: comparison certificates and alternation lower bounds for sorted-set intersection [DLM00, BK02, BK08], including randomized lower bounds [BK02], and doubling (galloping) search [BY76]. What is new here is their simultaneous realization, under one global labeling, inside the join constructions of the previous subsections, and the resulting quarter-logarithm tradeoff for per-binding choice.

Definition 3.22 (sorted fresh-call comparison model)**.** An intersection call receives two strictly increasing runs of opaque keys and their lengths. It may compare any two keys (three-way or Boolean predicates) and do index arithmetic; key arithmetic, hashing and addressing by key value are unavailable. Every key comparison costs one unit; a plan is additionally charged one unit per visited outer tuple. A call is *fresh*: it knows the internal order of its two runs and nothing else (no comparison results, certificates, aliases or rank hints from other calls). The routine A is fixed before the adversary assigns term identifiers (one global total order on each of X, Y, Z); the root and the per-binding option may be chosen with full knowledge of the labeling. Q_A is the minimum, over roots and per-binding options, of the resulting cost (comparisons plus outer visits) of a correct run of A. For a zero-error randomized routine, every call draws its own random tape, Q_A is the same minimum for the realized tapes (so the plan may be chosen after seeing them), and we bound $E[Q_A]$, the expectation of this realized minimum; this is at most the minimum over plans of their expected costs.

Relabelling permutes rows and columns of A_R, A_S, A_T, so sizes, degrees, singular values, N, B_2, U and Out are unchanged: nothing in Section 4 or Section 5 depends on the labeling.

Lemma 3.23 (comparison certificate)**.** *Let A, B be disjoint strictly increasing runs and $t(A, B)$ the number of adjacent pairs of their merged order whose members lie in different runs. Every correct fresh intersection routine makes at least $t(A, B)$ key comparisons on (A, B). If the runs alternate, $a_0 < b_0 < ... < a_{s-1} < b_{s-1}$, then $t = 2s - 1$, and ternary merge makes exactly $2s - 1$ comparisons.*

[Proved] (the classical certificate argument of [DLM00, BK02], stated in our model)

Proof. If an adjacent opposite-run pair is never compared, identify its two keys. Both runs stay strictly increasing (the keys lie in different runs), no key lies between them, so every other comparison has the same outcome, and the routine answers as before; but the intersection is now nonempty. □

Lemma 3.24 (one order for all rows)**.** *For $A = \{a_k : k < s\}$ and pairwise disjoint private sets $B_h = \{b_{h,k} : k < s\}$, the single order $a_k, b_{0,k}, b_{1,k}, ...$ for $k = 0, ..., s - 1$ restricts to a perfect alternation on $A \cup B_h$ for every h. Orders on disjoint term domains, and on vertex-disjoint components, concatenate to one global order.*

[Proved]

Theorem 3.25 (the 9/8 lower bound in the sorted model)**.** *Realize $G_8(m)$ with $X = \{(g, h) : g < m^3, h < m\}$, $Z_2 = \{(g, t) : g < m^3, t < m^3\}$, T complete between $Z_{2,g} =$*

$\{(g,t)\}$ *and* $X_g = \{(g,h)\}$ *for each* g, m^4 *private* Z_0*-predecessors per* x, *a private* S*-predecessor set* $Y_{2,z}$ *of size* m^2 *for each* $z \in Z_2$, $R = X \times Y_0$ *and* $S \supseteq Y_0 \times Z_1$ *(degrees exactly as in Table 3). Order* Z *as: for* $t < m^3$, $Z_1[t], Z_2[0,t], ..., Z_2[m^3-1,t]$; *then all of* Z_0. *Order* Y *as: for* $k < m^2$, $Y_0[k]$, *then* $Y_{2,z}[k]$ *for every* $z \in Z_2$. *Then, simultaneously for all* x *and* z, *every call* $(Z_1, T(\cdot,x))$ *needs at least* $2m^3-1$ *comparisons and every call* $\big(Y_0, Y_{2,z}\big)$ *at least* $2m^2-1$. *For every eligible routine* A *and every* x, *option* R *costs at least* $2m^5$ *and option* T *at least* m^4+2m^5, *with equality for ternary merge; the other roots of* G_8 *cost* m^5 *and* m^8. *On* $\mathcal{U}(m)$ *every root has exact optimal ternary cost* $2m^9+m^8+m^5$, *a lower bound on* Q_A *for every* A; *with* $N = 3\big(2m^8+m^7+m^6+m^5\big)$ *and* $B_2 = 5m^8$,

$$Q_A/(N+B_2) \geq m/7 \geq (N/9)^{1/8}/7.$$

So no c, *and no* $c = O\big(\log^k N\big)$, *satisfies* $Q_A \leq c(N+B_2)$ *for comparison-based sorted intersection (plain merge, leapfrog, galloping with any step schedule) under an adversarial identifier assignment. Padding (Lemma 3.8) with all fresh keys placed before all original keys keeps every hard run and its certificate, so the* $\Omega\big(N^{1/8}\big)$ *gap holds on pairwise semijoin-reduced inputs with* $\Theta(N)$ *triangles.*

[Proved] (on paper; exact checks in Appendix G)

Proof. On $Z_1 \cup T(\cdot,x)$, with $x \in X_g$, the order restricts to $Z_1[0], Z_2[g,0], Z_1[1], Z_2[g,1], ...$ followed by the private Z_0-keys of x, which lie in the same run as their predecessor: $t = 2m^3-1$ (Lemma 3.23). On $Y_0 \cup Y_{2,z}$ the order alternates by Lemma 3.24: $t = 2m^2-1$. Option R at x makes m^2 calls $(Z_1, T(\cdot,x))$ and visits m^2 outer tuples: $m^2(1+2m^3-1) = 2m^5$. Option T visits m^4 tuples into Z_0 (empty S-run, no comparison) and makes m^3 calls $\big(Y_0, Y_{2,z}\big)$: $m^4 + m^3(1+2m^2-1)$. At roots y and z the cheap options of the proof of Theorem 3.6 meet empty runs, at cost one per visited tuple. In $\mathcal{U}(m)$ every root contains one copy of root x of G_8, so it costs at least $m^4 \cdot 2m^5$, plus m^5+m^8 from the other copies. $N + B_2 \leq 14m^8$ for $m \geq 2$. For padding, the padded vertices are Y_2, Z_0 and Z_1 (each has tuples in only one relation); X, Y_0, Z_2 are not padded, and every new tuple touches a fresh vertex, so the runs Z_1, $T(\cdot,x)$, $R(x,\cdot) = Y_0$ and $S(\cdot,z) = Y_{2,z}$ of original vertices are unchanged; the critical root still costs at least $2m^9$, while $N' \leq 7N$ and $B_2' = O(m^8)$ (Lemma 3.8). □

Theorem 3.26 (slopes, separations and fixed roots in the sorted model)**.**

1. (slopes) *Fix* $i \geq j \geq 1$, $u \geq 2$, *and put* $n = u^{4i}$, $a = u^{4i-2j}$, $b = u^{4i-j}$, $d = u^{4i-3j}$, $H = u^{3j}$. *Partition* X *into* H *groups of size* d *and* Z_2 *into* H *groups of size* b, *make the paired* T*-blocks complete, give every* $z \in Z_2$ *a private* a*-element* S*-predecessor set, and order by Lemma 3.24 (*Z_1 *against every* Z_2 *group,* Y_0 *against every* $Y_{2,z}$*). Both options at every* x *cost exactly* $2ab$ *for ternary merge and at least* $2ab$ *for every eligible routine; the union of three rotations has exact optimal cost* $Q = 2u^{12i-3j} + u^{8i} + u^{8i-3j}$, $N = 3(u^{8i} + u^{8i-j} + u^{8i-2j} + u^{8i-3j})$ *and* $B_2 = 3u^{12i-4j} + u^{8i}$, *so* $Q = \Theta\Big(N^{3/8} B_2^{3/4}\Big)$ *at* $B_2 = \Theta(N^\kappa)$, $\kappa = 3/2 - j/(2i)$, *with no logarithmic factor.*
2. (fixed GAOs) *In* $\mathcal{F}_a$ *with* $z_{i,k}$ *ordered by* (k,i) *every off-diagonal* R*-tuple* $\big(x_i, y_j\big)$ *is an alternating pair of length-*a *runs, so the* R*-driven GAOs need* $a(a-1)(2a-1)$ *compar-*

isons in total, with $N = 3a^2$, $B_2 = a^{8/3}$: loss $\Theta(N^{1/6})$ against $N + B_2$ on a pairwise semijoin-reduced instance. In GMR(m) *with the common Z_1 layered against every private T-predecessor set, all m^4 R-tuples need $2m-1$: loss $\Theta(N^{1/4})$. In* GMIN(m) *with the three layerings (Z_1 against the Z_0-fibres, X_2 against the X_1-fibres, Y_2 against the Y_3-fibres; disjoint domains, one order) each driver has m^4 hard tuples of $2m^2 - 1$ comparisons and every other tuple meets an empty run: the best of the six GAOs loses $\Theta(N^{1/5})$ against $N + B_2$ and against adaptive execution plus N. Padded with fresh keys first,* GMIN *has $N' = 12m^5 + 3m^4 + 3m^3 + 9m^2$, $B_2' = 8m^5 + 3m^2$,* Out$' = 3m^5 + 3m^2$*, unchanged hard tuples, and cheap options that are $O(1)$-comparison searches for a fresh key placed before all original keys; so adaptive execution costs $\Theta(N')$ while every GAO costs $\Theta(m^6)$.*

3. (fixed root) *In $\mathcal{H}_a$ with Y_1, Y_2 alternated and the common Z_2 layered against every private Z_1-fibre, each option at every x makes a alternating calls on length-a runs, aggregate certificate $a^2(2a-1)$; a fixed root x costs $\Theta(a^3)$ and root y costs $\Theta(a^2)$, and fresh-first padding leaves the certificates of the original X unchanged: the $\Theta(N^{1/2})$ loss of a fixed root against adaptive cost plus N holds on semijoin-reduced inputs.*

All lower bounds hold for every eligible routine; merge attains them.

[Proved] (on paper; exact checks in Appendix G)

Proof. Each item applies Lemma 3.23 to the stated alternations, which Lemma 3.24 realizes in one global order (the three rotated copies, and the three layerings of GMIN, live on disjoint domains). In (1), option R at x makes a calls of $2b-1$ comparisons plus a visits and option T makes b calls of $2a-1$ plus b visits, both $2ab$; roots y and z cost ab and u^{8i} (empty runs), and the statistics are those of the proof of Corollary 3.11. In (2), the counts are $a(a-1)$ tuples times $2a-1$, $m^4(2m-1)$ and $m^4(2m^2-1)$; the statistics of GMR and GMIN are those of Theorem 3.16. A fresh key placed before all original keys is found or excluded by galloping with $O(1)$ comparisons. In (3), option R at x walks Y_1 and intersects the private fibre $S(y_i, \cdot)$ with the common $T(\cdot, x) = Z_2$, and option T walks Z_2 and intersects $R(x, \cdot) = Y_1$ with $S(\cdot, z') = Y_2$. □

Remark 3.27 (the $\Theta(N^{1/2})$ ratios against C)**.** In $\mathcal{F}_a$ and $\mathcal{G}_a$ the cheap options are singleton membership tests in length-a runs; a fresh singleton search costs up to $\lceil \log_2(a+1) \rceil$ comparisons while its certificate is 1 or 2. So in the sorted model the $\Theta(N^{1/2})$ loss of Theorem 3.16(1)–(2) against $C + N$ is established only between $\Omega(N^{1/2}/\log N)$ and $O(N^{1/2})$; the ratios against $N + B_2$ in Theorem 3.26 are unaffected.

Upper bound. For runs of lengths p, q put $s = \min(p, q)$, $l = \max(p, q)$ and the *price* $h(p, q) = s(1 + \lceil \log_2(l/s) \rceil)$, with $h = 0$ if a run is empty, and $L_N = 1 + \lceil \log_2 N \rceil$. Doubling galloping (walk the shorter run, gallop from the current position in the longer one) intersects the two runs with $O(h(p, q))$ comparisons. h is computable from the two degrees: $\lceil \log_2(l/s) \rceil = \text{bitlength}(\lfloor (l-1)/s \rfloor)$.

Lemma 3.28 (galloping and the AGM cap)**.** *$h(p, q) \le L_N \cdot s$ when $l \le N$, and $h(p, q) \le \sqrt{9pq/2} < 3\sqrt{pq}$; below we use the round constant 3. Consequently*

$\sum_{(x,y)\in R} h(d_S^+(y), d_T^-(x)) \le 3\|A_R\|_2\sqrt{|S|\ |T|} \le 3\sqrt{|R|\ |S|\ |T|}$*: the fixed-driver galloping kernel costs* $O\big(N + \|A_R\|_2\sqrt{|S|\ |T|}\big)$ *comparisons, and* $C \le N + \sqrt{|R|\ |S|\ |T|}$ *of Lemma 3.1 survives galloping with constant* 3 *and no logarithm.*

[Proved] (on paper; exact checks in Appendix G)

Proof. The first bound is $\lceil \log_2(l/s) \rceil \le \lceil \log_2 N \rceil$. For the second, if $l = s$ then $h = s = \sqrt{pq}$; otherwise put $r = l/s > 1$ and $k = \lceil \log_2 r \rceil \ge 1$, so $r > 2^{k-1}$ and $h^2/(pq) = (k+1)^2/r < (k+1)^2/2^{k-1} \le 9/2$: the values are $4, 9/2, 4$ at $k = 1, 2, 3$, and they decrease from $k = 2$ on because $(k+2)^2 \le 2(k+1)^2$ for $k \ge 2$. Then $\sum h \le 3\sum_{(x,y)\in R} \sqrt{d_T^-(x)}\sqrt{d_S^+(y)} \le 3\|A_R\|_2\sqrt{|T|\ |S|}$ as in the proof of Proposition 5.2. □

Theorem 3.29 (sorted-model analogue of Theorem 3.3)**.** *Let* $\hat{C}$ *be the per-binding cost in which every edge is priced* $1 + h(\cdot,\cdot)$ *instead of* $1 + \min(\cdot,\cdot)$ *(same roots, options and minimum). Then*

$$\hat{C} \le N + 3^{3/4} L_N^{1/4} (|R|\ |S|\ |T|)^{1/8} \min(\Lambda, \Lambda')^{1/2} \le N + 3^{3/4} L_N^{1/4} (N/3)^{3/8} B_2^{3/4}.$$

The prices are computable from degrees in $O(N)$ *reads; selecting the best priced root and options and executing by galloping costs* $O\big(\hat{C}\big)$ *comparisons. Hence the sorted-model per-binding cost is* $O\Big(N + (\log N)^{1/4}(|R|\ |S|\ |T|)^{1/8}\min(\Lambda, \Lambda')^{1/2}\Big)$*. The price-then-run scheduler of Proposition C.2 runs unchanged with the prices* $1 + h$*, so (1)–(3) of Proposition C.2 hold with* C *replaced by* $\hat{C}$*, in per-edge price units, without sampling.*

[Proved] (on paper; exact checks in Appendix G)

Proof. At root x put $a = d_R^+(x)$, $b = d_T^-(x)$, $\hat{P}(x) = \sum_{y\in R(x,\cdot)} h(d_S^+(y), b)$, $\hat{Q}(x) = \sum_{z:(z,x)\in T} h(a, d_S^-(z))$ and $w(x) = \sum_{z:(z,x)\in T} \sqrt{d_S^-(z)}$. By Lemma 3.28, $\hat{P}(x) \le L_N \sum_y d_S^+(y) \le L_N\sqrt{a\beta}$ and $\hat{Q}(x) \le 3\sqrt{a}w(x)$, so $\min\big(\hat{P}, \hat{Q}\big) \le \hat{P}^{1/4}\hat{Q}^{3/4} \le 3^{3/4} L_N^{1/4} \beta^{1/8} a^{1/2} w(x)^{3/4}$, and steps 4–5 of the proof of Lemma 3.2 give $\hat{C}_x \le |R| + |T| + 3^{3/4} L_N^{1/4} \alpha^{1/4} \beta^{1/8} (\gamma\ |S|)^{3/8}$. Rotation and reflection give the other roots and the primed forms, and the proof of Theorem 3.3 (geometric mean of the nonnegative bounds) gives the claim. Lockstep pricing (Lemma C.1) needs only per-edge prices of at least 1 revealed one at a time. □

Corollary 3.30 (sorted-model analogue of Theorem 3.4)**.** *For every instance with* $N > 0$*,* $\hat{C} \le N + 3^{7/8} L_N^{1/4} N^{-1/8} B_2^{5/4}$*; in integers,* $N^3\big(\hat{C} - N\big)_+^{24} \le 3^{21} L_N^6 \min(\alpha\beta\gamma, \alpha'\beta'\gamma')^{10}$*.*

[Proved]

Proof. The proof of Theorem 3.29 bounds the priced minima at each root by $3^{3/4} L_N^{1/4}$ times the multiplicative terms of Lemma 3.2. Repeating the proof of Theorem 3.4 with these bounds multiplies its conclusion by $3^{3/4} L_N^{1/4}$: $\hat{C} - N \le 3^{3/4} L_N^{1/4} B_2^{5/4} (N/3)^{-1/8}$, and the 24th power is the integer form. □

Whether the factor $L_N^{1/4}$ is needed below slope 1 is open (Section 8).

Lemma 3.31 (decision trees on disjoint runs)**.** *Fix a correct deterministic fresh-call routine and disjoint runs of lengths* p, q*, and refine its comparison tree by replacing every Boolean*

comparison with the three-way comparison of the same two keys (the routine ignores the extra information, so every comparison count is unchanged). A leaf of the refined *tree that answers "disjoint" is reached by at most one of the $\binom{p+q}{p}$ shuffles of the two runs. Hence fewer than 2^D shuffles are decided with fewer than D comparisons.*

[Proved]

Proof. At a refined leaf, if the partial order generated by the three-way outcomes and the two internal orders is not total, some opposite-run pair is incomparable, and the input that makes those two keys equal is consistent with every outcome on the path, so it reaches the same leaf with a nonempty intersection; hence the leaf determines the merged order. On disjoint inputs every three-way comparison has two possible outcomes, so fewer than 2^D refined leaves lie at depth below D. The refinement is needed: in the *unrefined* Boolean tree the routine "test $a = b$; output a if true" on singletons reaches its "false" leaf from both shuffles $a < b$ and $b < a$. □

Theorem 3.32 (the quarter logarithm is necessary)**.** *Fix $i \geq j \geq 1$ and an integer $e \geq 1$; put $M = 2^e$, $L = M^4$, $u = 2^L$, and $n = |X| = u^{4i}/M^4$, $a = |Y_0| = u^{4i-2j}M^2$, $p = |Z_1| = u^{4i-j}/M$, $q = u^{4i}M^4$, $k = pL$, $d = u^{4i-3j}M$, $c = |Z_2| = u^{4i+2j}/M^2$ (all integers; $n/d = c/k = u^{3j}/M^5$). Take $R = X \times Y_0$, $S \supseteq Y_0 \times Z_1$, a private a-element S-predecessor set $Y_{2,z}$ for every $z \in Z_2$, complete T-blocks pairing d vertices of X with k vertices of Z_2, and q private dead Z_0-predecessors for every x (so $ca = nq = u^{8i}$, $cd = nk$, no triangles). For the union of three rotations, $N = 3(2u^{8i} + u^{8i-j}/M + u^{8i-2j}/M^2 + u^{8i-3j}M) \in [6u^{8i}, 15u^{8i}]$, each forward squared-degree sum is $3u^{12i-4j} + 2u^{8i}$ and each reverse sum exceeds it, so $B_2 = 3u^{12i-4j} + 2u^{8i} \in [3u^{12i-4j}, 5u^{12i-4j}]$. For every eligible deterministic routine A there is one global labeling ℓ_A with*

$$Q_A \geq naD = (M/2)u^{12i-3j}, \quad D = pL/2.$$

Since $M = (\log_2 u)^{1/4} = \Theta\big((\log N)^{1/4}\big)$ and $N^{3/8}B_2^{3/4} = \Theta(u^{12i-3j})$, this matches Theorem 3.29 up to a constant on every slope $\kappa = 3/2 - j/(2i) \in [1, 3/2)$. At $i = j$, $B_2 = \Theta(N)$ and $\sup_\ell Q_A = \Omega\big(N^{9/8}(\log N)^{1/4}\big)$, so the worst-case sorted-model scale at $B_2 \asymp N$ is $\Theta\big(N^{9/8}(\log N)^{1/4}\big)$. For a zero-error randomized routine there is likewise one labeling on which the expected realized *minimum over plans (Definition 3.22) satisfies $E[Q_A] \geq naD/2 = (M/4)u^{12i-3j}$; a fortiori every plan's expected cost is at least this. No universal constant gives $Q_A \leq c\Big(N + (|R|\ |S|\ |T|)^{1/8}\min(\Lambda, \Lambda')^{1/2}\Big)$ in the sorted model; the ratio grows as $(\log N)^{1/4}$.*

[Proved] (on paper; exact checks in Appendix G)

Proof. Labeling. Order Y by Lemma 3.24 so that Y_0 alternates with every $Y_{2,z}$; order Z as $Z_1 \cup Z_0$ first and Z_2 last, the dead keys of each x shuffled into the $p + 1$ gaps of Z_1: in each gap, concatenate the keys assigned to it in a fixed order of x, so that independent shuffles for all x are realized by one order. The rotated copies use disjoint domains. *Option T.* At every x it makes k calls $\big(Y_0, Y_{2,z}\big)$ with $2a - 1$ comparisons each (Lemma 3.23): at least $ka = 2aD$. *Option R.* At x it makes a calls $(Z_1, T(\cdot, x))$; the Z_2 part of $T(\cdot, x)$ lies after Z_1, and the hard part is the shuffle of the q dead keys with Z_1. There are $\binom{p+q}{p} \geq (q/p)^p = \big(u^j M^5\big)^p \geq u^p =$

2^{pL} shuffles, so by Lemma 3.31 a uniform shuffle makes a prescribed call cost less than D with probability at most $2^D/2^{pL} = 2^{-pL/2}$. There are $3na$ critical R-calls in the union and $\log_2(3na) < 2 + 8iL < 16iL \le D$ (as $p \ge u^{2i} \ge 32i$), so the union bound leaves a labeling, i.e. a choice of shuffles, on which every critical R-call costs at least D. *Total.* Every root of the union contains a critical copy, where both options cost at least aD per x: $Q_A \ge naD$. *Randomized routines.* Fix random tapes for all $3na$ critical R-calls at once, and let E be the event that at least one of them costs less than D. Each tape makes the routine deterministic, so for a uniform shuffle ω and independent tapes, $\Pr_{\omega,\,\text{tapes}}[E] \le 3na \cdot 2^{-pL/2} < 1/2$, since $\log_2(6na) < 3 + 8iL < 16iL \le D$. Hence some labeling has $\Pr_{\text{tapes}}[E] < 1/2$. Outside E every critical call costs at least D, so every root and every choice of options, even one made after seeing the tapes, costs at least naD; therefore $E[Q_A] \ge naD/2$. (Bounding each plan's expected cost separately would not suffice: two plans with costs $(0, 2)$ and $(2, 0)$ under two equally likely tapes both have expectation 1, while their realized minimum is 0.) *Statistics.* Per gadget $|R| = u^{8i-2j}/M^2$, $|S| = u^{8i-3j}M + u^{8i}$, $|T| = u^{8i-j}/M + u^{8i}$, the forward sums are u^{12i-4j}, $u^{12i-4j} + u^{8i}$, $u^{12i-4j} + u^{8i}$, and the reverse sums are u^{12i-2j}/M^6, $u^{12i-5j}M^3 + u^{12i-2j}M^2$ and $u^{12i}M^4 + 2u^{12i-j}M^3 + u^{12i-2j}M^2$; in the union each reverse moment contains $u^{12i}M^4$, which exceeds the forward moment $3u^{12i-4j} + 2u^{8i}$. □

The quantifiers are "for every routine there is a labeling"; a single labeling hard for all routines cannot exist. The explicitly interleaved G_8 and $F_{i,j}$ have no logarithmic overhead (Theorem 3.25, Theorem 3.26); the quarter logarithm appears only in the worst case over labelings against a fixed routine. At the smallest member with $e = 4$ ($i = j = 1$) the integer form of Theorem 3.3, $(C - N)_+^8 \le |R|\ |S|\ |T| \min(\alpha\beta\gamma, \alpha'\beta'\gamma')^2$, already fails for Q_A: with $N = 3(2^{524289} + 2^{458748} + 2^{393208} + 2^{327684})$, $B_2 = 5 \cdot 2^{524288}$ and $Q_A \ge Q_0 = 2^{589827}$, the quotient $(Q_0 - N)^8/((N/3)^3 B_2^6)$ has integer floor 134: it equals $(2097152/15625)(1 - \delta)^8/(1 + \varepsilon)^3$ with $\delta = N/Q_0 < 2/u$ and $\varepsilon < 1/u$, $u = 2^{65536}$, so it lies strictly between $2096544/15625 > 134$ and $2097152/15625 < 135$ (no instance is materialized). The parameters form a doubly exponential sequence; at $i = j = 1$ one may instead take $u = 2^t$ for every integer t with $M^4 \le t < 16M^4$, keeping $L = M^4$, since the shuffle entropy is then at least $pt \ge pL$; consecutive t give sizes within a constant factor, and isolated edges fill intermediate sizes with $B_2 = \Theta(N)$, so the bound holds for all large N, not only along a sparse sequence (an observation of the GPT-6-Astra Pro review of the third draft, Section 9).

Remark 3.33 (scope: fresh calls)**.** The lower bounds are for fresh calls. In unpadded G_8 one semijoin pass (Z_1 has no T-tuples, so $Y_0 \times Z_1$ and then $X \times Y_0$ are deleted) leaves only cheap structure, and all $d_R^+(x)$ calls at a binding x are the identical pair $(Z_1, T(\cdot, x))$, so memoizing calls reduces the critical root to m^4 distinct calls. Both are outside the model, and Proposition C.4 shows that degree metadata can determine an instance. So these are not lower bounds for unrestricted comparison-based join algorithms; algorithms that reuse certificates across calls, and algorithms using different orders on value ranges (ADOPT), remain open (Section 8).

4. Spectral worst-case output bounds

This section and the next treat the statistics (r, σ) of sizes and operator norms, the level at which per-instance rotation suffices. The upper bound below is textbook: its Hölder terms are the $(\infty, 2, 2)$ case of the Schatten–Hölder inequality, and its degree terms are the degree bounds of [ANOS24] with maximum degrees bounded by σ^2. The content of Theorem 4.1 is its first sentence: the nine terms of U are exactly the norm-feasibility conditions of one family of block instances, so that one family witnesses the lower bound in every regime. Its use for planning is Corollary 5.3: the best rotation of the kernel does work at most U. Section 4.5 extends the characterization to the labeled 4-cycle, where complete blocks no longer suffice.

Theorem 4.1 (spectral AGM for the triangle)**.** *Let (r, σ) be regular. The nine terms of U are exactly the norm-feasibility conditions of one block family: for $V > 0$, the real block shape $(\max(1, V/r_S), \max(1, V/r_T), \max(1, V/r_R))$ of Section 4.2 satisfies the three norm constraints if and only if $V \le U$, and at $V = U$ its relaxed value, with real parameters and no floors, is exactly U. Flooring this shape loses less than a factor 4:*

$$U(r, \sigma)/4 < W(r, \sigma) \le U(r, \sigma).$$

The upper bound holds for every admissible instance with constant 1. The lower bound is witnessed by the block instance $\mathrm{Blk}(a, b, c, m)$ with

$$a = \lfloor \max(1, U/r_S) \rfloor, \quad b = \lfloor \max(1, U/r_T) \rfloor, \quad c = \lfloor \max(1, U/r_R) \rfloor,$$
$$m = \lfloor \min(r_R/(ab), r_S/(bc), r_T/(ca)) \rfloor.$$

[Proved; Lean-checked] (Appendix D. The feasibility statement for $V \neq U$ (feasible for $V < U$, infeasible for $V > U$) is on paper only; it is immediate from the identity in the proof of Lemma 4.3. The factor 4 is best possible, Theorem 4.7.)

4.1. Upper bound

Lemma 4.2 (degree at most σ^2)**.** *For any 0/1 matrix A, every row and column degree is at most $\|A\|_2^2$ and at most the number of ones of A.*

[Proved; Lean-checked] (`rowDeg_le_opNorm_sq`, `colDeg_le_opNorm_sq`.)

Proof. For row x, $\left\|A^\top e_x\right\|_2^2 = \deg(x)$ and $\left\|A^\top e_x\right\|_2 \le \left\|A^\top\right\|_2 = \|A\|_2$. Columns are symmetric. The second bound is trivial. □

Proof of the upper bound of Theorem 4.1. Fix an admissible instance; by Lemma 4.2 and clipping every degree of relation i is at most s_i.

Degree terms. Counting by R-tuples, $\mathrm{Out} = \sum_{(x,y)\in R} |S(y, \cdot) \cap T(\cdot, x)|$, and each summand is at most $\min(d_S^+(y), d_T^-(x)) \le \min(s_S, s_T)$. Hence $\mathrm{Out} \le D_{R,S}$ and $\mathrm{Out} \le D_{R,T}$. Counting by S- and T-tuples gives the other four.

Hölder terms. For $z \in Z$ let $t_z \in \mathbb{R}^X$, $t_z(x) = [(z, x) \in T]$, and $u_z \in \mathbb{R}^Y$, $u_z(y) = [(y, z) \in S]$. Then

$$\mathrm{Out} = \sum_z \langle t_z, A_R u_z\rangle \le \|A_R\|_2 \sum_z \|t_z\|\|u_z\| \le \|A_R\|_2 \left(\sum_z \|t_z\|^2\right)^{1/2} \left(\sum_z \|u_z\|^2\right)^{1/2} = \|A_R\|_2 \sqrt{|T|\ |S|},$$

by $|\langle t, Au\rangle| \le \|A\|_2 \|t\|\|u\|$ and Cauchy–Schwarz. Since also $\|A_R\|_2 \le \sqrt{|R|}$, this is at most H_R. Equivalently, by trace duality, $\mathrm{tr}(A_R \cdot A_S A_T) \le \|A_R\|_2 \|A_S A_T\|_{S_1} \le \|A_R\|_2 \|A_S\|_F \|A_T\|_F$. H_S and H_T follow by rotation.† □

4.2. Lower bound: one block family for every regime

Block instance. For integers $a, b, c, m \ge 1$, $\mathrm{Blk}(a, b, c, m)$ consists of m vertex-disjoint blocks, each with a x's, b y's, c z's and all three bipartite pieces complete: $A_R = I_m \otimes J_{a\times b}$ and likewise for S, T. Then

$$|R| = mab, \quad |S| = mbc, \quad |T| = mca, \quad \|A_R\|_2^2 = ab, \quad \|A_S\|_2^2 = bc, \quad \|A_T\|_2^2 = ca, \quad \mathrm{Out} = mabc.$$

(Lean: `card_blk`, `opNorm_blk_eq`, `out_blk`; feasibility needs only "≤", which the Schur test $\|A\|_2^2 \le \Delta_{\mathrm{row}}\Delta_{\mathrm{col}}$ gives, `opNorm_adj_le_schur`.) So Blk is admissible iff $ab \le s_R$, $bc \le s_S$, $ca \le s_T$ and $m \le \min(r_R/(ab), r_S/(bc), r_T/(ca))$.

Lemma 4.3 (the real shape is feasible)**.** *Let* $\hat a = \max(1, U/r_S)$, $\hat b = \max(1, U/r_T)$, $\hat c = \max(1, U/r_R)$. *Then* $\hat a\hat b \le s_R$, $\hat b\hat c \le s_S$, $\hat c\hat a \le s_T$.

[Proved; Lean-checked] (`Stats.real_shape_feasible`.)

Proof. For $p, q > 0$, $\max(1,p)\max(1,q) = \max(1,p,q,pq)$. So $\hat a\hat b = \max(1, U/r_S, U/r_T, U^2/(r_S r_T))$, and each entry is at most s_R: $1 \le s_R$ by regularity; $U \le D_{S,R} = r_S s_R$; $U \le D_{T,R} = r_T s_R$; $U^2 \le H_R^2 = s_R r_S r_T$. Rotating the labels, $\hat b\hat c \le s_S$ uses $1, D_{T,S}, D_{R,S}, H_S$ and $\hat c\hat a \le s_T$ uses $1, D_{R,T}, D_{S,T}, H_T$. With any $V > 0$ in place of U the same identity shows that the shape is feasible iff V is at most each of the nine terms, i.e. iff $V \le U$. □

Thus each of the nine terms is exactly one feasibility condition of the minimal block shape; this is why U is the answer (Theorem 4.1, first sentence).

Lemma 4.4 (the relaxed value is U)**.** *With* $m^* = \min\big(r_R/(\hat a\hat b), r_S/(\hat b\hat c), r_T/(\hat c\hat a)\big)$ *we have* $m^* \ge 1$ *and* $m^*\hat a\hat b\hat c = U$.

[Proved; Lean-checked] (`Stats.real_shape_value`, `Stats.real_shape_attains_U`.)

Proof. $m^* \ge 1$ because $\hat a\hat b \le s_R \le r_R$ and similarly. Next, $m^*\hat a\hat b\hat c = \min\big(\hat c r_R, \hat a r_S, \hat b r_T\big) = \min_j \max\big(r_j, U\big)$. This equals U because $U \ge \min_j r_j$: each $D_{i,j} \ge r_i$ (as $s_j \ge 1$) and $H_R \ge \sqrt{r_S r_T} \ge \min(r_S, r_T)$, and similarly for H_S, H_T. □

Proof of the lower bound of Theorem 4.1. Put $a = \lfloor \hat a \rfloor$, $b = \lfloor \hat b \rfloor$, $c = \lfloor \hat c \rfloor$, all ≥ 1. Flooring shrinks products, so $ab \le s_R$, $bc \le s_S$, $ca \le s_T$ still hold. Let $\mu = \min(r_R/(ab), r_S/(bc), r_T/(ca))$. Then $\mu \ge \min(r_R/s_R, r_S/s_S, r_T/s_T) \ge 1$, using the three

† A tempting nuclear-norm shortcut, $\|A\|_* \le |A|\ /\|A\|_2$, is false: for $A = \left(\begin{smallmatrix}1 & 1\\ 1 & 0\end{smallmatrix}\right)$, $\|A\|_* = \sqrt{5} \approx 2.236$ while $|A|\ /\|A\|_2 = 3/\varphi \approx 1.854$; the reverse inequality $|A| = \|A\|_F^2 \le \|A\|_2\|A\|_*$ always holds. The valid bound $\mathrm{Out} \le \sigma_R\sigma_S\|A_T\|_* \le \sigma_R\sigma_S r_T$ is dominated by $D_{T,R}$ or $D_{T,S}$, since $\min(s_R, s_S) \le \sigma_R\sigma_S$.

norm conditions and $s_i \le r_i$. Also $\mu abc = \min(cr_R, ar_S, br_T)$; each argument floors one coordinate and $\lfloor x \rfloor > x/2$ for $x \ge 1$, so $\mu abc > \min\big(\hat{c}r_R, \hat{a}r_S, \hat{b}r_T\big)/2 = U/2$ by Lemma 4.4. With $m = \lfloor \mu \rfloor > \mu/2$ the size conditions hold ($mab \le \mu ab \le r_R$, etc.) and $\mathrm{Out} = mabc > U/4$. □

Remark 4.5 (regime witnesses)**.** If $U = D_{R,S} = r_R s_S$ then $b = 1$ (as $D_{R,S} \le H_S$ iff $r_R s_S \le r_T$) and $c = \lfloor s_S \rfloor$; when $r_R s_S \le r_S$ this is a "fan" in which every R-tuple owns s_S private z's. If $U = H_R$ then $a = \left\lfloor \sqrt{s_R r_T / r_S} \right\rfloor$ and $b = \left\lfloor \sqrt{s_R r_S / r_T} \right\rfloor$; the $\max(1, \cdot)$ is not needed, because $U = H_R$ forces $r_S \le s_R r_T$ and $r_T \le s_R r_S$. Here the R-blocks are tight for s_R and the S- and T-blocks are slack. The tempting variant $c = s_S/b$ is not admissible in general (for $r = (100, 100, 100)$, $s = (1, 100, 1)$ it gives $ca = 100 > s_T$); only the general formula of Theorem 4.1 is claimed.

Remark 4.6 (the balanced block is not optimal)**.** Take $s_i = 2^{12}$ for all i, $r_R = 2^{12}$, $r_S = r_T = 2^{16}$. Then $U = H_S = H_T = 2^{20}$, and the construction gives $a = b = 16$, $c = 256$, $m = 16$, so $\mathrm{Out} = 2^{20} = U$ exactly. Balanced blocks ($a = b = c$, all three norm constraints tight) reach only 2^{18}. The optimal shape leaves some norm constraints slack.

4.3. The factor 4 is sharp

Theorem 4.7 (the constant 4 is best possible)**.** *For every real $t \in [1, 2)$ the statistics $r = (r_R, r_S, r_T) = (t^2, t, t^2)$ and $\sigma = \left(\sqrt{t}, 1, \sqrt{t}\right)$ are regular, $U = t^2$, and $W = 1$. Consequently*

$$\inf\{W(r, \sigma)/U(r, \sigma) : (r, \sigma) \text{ regular}\} = 1/4,$$

and the infimum is not attained.

[Proved; Lean-checked] (Appendix D, also in the two-sided form $\inf W/U = 1/4$. This family, in the rational form below, was found by a model-based review: the GPT-5.6-sol Pro review of the first draft, Section 9.)

Proof. Here $s_R = s_T = \min(t, t^2) = t$ and $s_S = \min(1, t) = 1$. The three Hölder terms are $H_R = \sqrt{t \cdot t \cdot t^2}$, $H_S = \sqrt{1 \cdot t^2 \cdot t^2}$ and $H_T = \sqrt{t \cdot t^2 \cdot t}$, all equal to t^2; the degree terms are $D_{R,S} = D_{S,R} = D_{S,T} = D_{T,S} = t^2$ and $D_{R,T} = D_{T,R} = t^3 \ge t^2$. So $U = t^2$. Let an instance be admissible. Then $|S| \le t < 2$, so $|S| \le 1$, and by Lemma 4.2 every out-degree of T is at most $\|A_T\|_2^2 \le t < 2$, so at most 1. Counting by S-tuples, $\mathrm{Out} = \sum_{(y,z) \in S} |\{x : (x, y) \in R, (z, x) \in T\}| \le |S| \cdot \max_z d_T^+(z) \le 1$. A single triangle ($|R| = |S| = |T| = 1$, all norms 1) is admissible, so $W = 1$ and $W/U = t^{-2} \to 1/4$ as $t \to 2$. With Theorem 4.1 ($W > U/4$ for every regular (r, σ)) the infimum is $1/4$ and is not attained. □

The form of the family given in that review keeps every statistic rational: with $(p_1, q_1) = (7, 5)$, $p_{i+1} = 3p_i + 4q_i$, $q_{i+1} = 2p_i + 3q_i$ one has $p_i^2 - 2q_i^2 = -1$, and $t_i = p_i^2/q_i^2 = 2 - 1/q_i^2$ gives $\sigma_R = \sigma_T = p_i/q_i$ and $U/W = t_i^2 = 2401/625, 2825761/707281, \ldots \to 4$ (recomputed in exact rational arithmetic by `artifacts/verify/chk_exact.py`). On this family the block construction of Theorem 4.1 has $(a, b, c, m) = (1, 1, 1, 1)$ and is itself optimal; the loss of the factor 4 is in U, not in the construction: no instance at all beats U/t^2, so no better choice of blocks could improve the constant of Theorem 4.1.

4.4. A single norm constraint

The following special case, in which only R's norm is constrained and $|R|$ is free, is a consistency check on Theorem 4.1.

Corollary 4.8 (single norm constraint)**.** *Let $\sigma \geq 1$ and $1 \leq r_S \leq r_T$. Then*

$$\sup\Big\{\mathrm{Out} : \|A_R\|_2 \leq \sigma, |S| \leq r_S, |T| \leq r_T\Big\} \in (U_1/4, U_1], \quad U_1 = \min(\sigma^2 r_S, \sigma\sqrt{r_S r_T}, r_S r_T).$$

[Proved]

Proof. Any instance in the class is admissible for $r_R = |R|$ (which may be increased freely), $\sigma_R = \sigma$, $\sigma_S = \sqrt{r_S}$, $\sigma_T = \sqrt{r_T}$ (the last two are no constraint, by clipping). Take $r_R \geq \max(\sigma^2, r_S r_T)$. Then $s_R = \sigma^2$, $s_S = r_S$, $s_T = r_T$, and the nine terms are $H_R = \sigma\sqrt{r_S r_T}$, $H_S = H_T = \sqrt{r_R r_S r_T} \geq r_S r_T$, $D_{R,S} = r_R r_S \geq r_S r_T$, $D_{R,T} = r_R r_T \geq r_S r_T$, $D_{S,R} = r_S \sigma^2$, $D_{S,T} = D_{T,S} = r_S r_T$, $D_{T,R} = r_T \sigma^2 \geq r_S \sigma^2$. Hence $U = U_1$ for every such r_R. Since U is nondecreasing in r_R, U_1 bounds every instance (Theorem 4.1, upper bound), and the lower bound of Theorem 4.1 at $r_R = \max(\sigma^2, r_S r_T)$ gives an instance with $\mathrm{Out} > U_1/4$. □

The three regimes ($\sigma^2 \leq r_T/r_S$: star-shaped R; $r_T/r_S \leq \sigma^2 \leq r_S r_T$: disjoint $K_{a,b}$ with $ab = \sigma^2$; $\sigma^2 \geq r_S r_T$: a single complete block) are all specializations of the block formula. Because disjoint copies do not change σ, once σ is known the AGM bound can be off by an unbounded factor.

The next corollary measures norms for two relations only, the case of a store that keeps σ for some predicates and not for others.

Corollary 4.9 (two measured norms)**.** *Let $|R| \leq r_R$, $|S| \leq r_S$, $|T| \leq r_T$, $\|A_R\|_2 \leq \sigma_R$ and $\|A_S\|_2 \leq \sigma_S$, with no norm bound on T and all five bounds at least 1. Put $s_R = \min(\sigma_R^2, r_R)$, $s_S = \min(\sigma_S^2, r_S)$ and*

$$U_2 = \min\big\{\sqrt{s_R r_S r_T}, \sqrt{s_S r_R r_T}, r_R s_S, r_S s_R, r_T s_R, r_T s_S\big\} = \min\big\{H_R, H_S, D_{R,S}, D_{S,R}, D_{T,R}, D_{T,S}\big\}.$$

Then $U_2/4 < W_2 \leq U_2$, where W_2 is the supremum of Out *over the class. The upper constant 1 is attained, and the lower constant $1/4$ is best possible: for $1 \leq t < 2$ the statistics $r = (t^2, t, t^2)$, $\sigma_R = \sqrt{t}$, $\sigma_S = 1$ have $U_2 = t^2$ and $W_2 = 1$. The witness is* $\mathrm{Blk}(a, b, c, m)$ *of Theorem 4.1 with U replaced by U_2.*

[Proved] (on paper; exact checks in Appendix G)

Proof. Every T with $|T| \leq r_T$ has $\|A_T\|_2^2 \leq |T| \leq r_T$ (Section 2), so the class is the admissible class of Theorem 4.1 with $\sigma_T = \sqrt{r_T}$, which is regular, and $s_T = r_T$. With $s_T = r_T$ three of the nine terms are dominated: $H_T = \sqrt{r_R r_S r_T} \geq H_R$, $D_{R,T} = r_R r_T \geq r_T s_R = D_{T,R}$ and $D_{S,T} = r_S r_T \geq r_T s_S = D_{T,S}$. So $U = U_2$, and Theorem 4.1 gives both bounds and the witness. For the sharpness family, $s_R = t$, $s_S = 1$ and the six terms are $(t^2, t^2, t^2, t^2, t^3, t^2)$, so $U_2 = t^2$. An admissible instance has $|S| \leq t < 2$, so $|S| \leq 1$, and every column degree of R is an integer at most $\|A_R\|_2^2 \leq t < 2$ (Lemma 4.2), so at most 1; counting by S-tuples, $\mathrm{Out} \leq 1$, whatever T is. One triangle is admissible, so $W_2 = 1$, and $W_2/U_2 = t^{-2} \to 1/4$ as $t \to 2$. □

With only σ_R measured and all three sizes known ($s_S = r_S$, $s_T = r_T$), the nine terms reduce in the same way to

$$U_1' = \min\{\sqrt{s_R r_S r_T},\, r_S s_R,\, r_T s_R,\, r_R r_S,\, r_R r_T,\, r_S r_T\},$$

and $U_1'/4 < W \le U_1'$. Corollary 4.8 is the case $r_R \ge \max(\sigma^2, r_S r_T)$ with $r_S \le r_T$, where the two r_R-terms and $r_T s_R$ drop out.

Proposition 4.10 (every one of the nine terms is necessary)**.** *For every integer $n \ge 2$ and each of the nine terms of U there are regular statistics (r, σ), all powers of n, at which that term is the unique minimum of U, the block of Theorem 4.1 is admissible with* $\mathrm{Out} = U$ *exactly (so $W = U$), and deleting the term from the minimum raises it by the factor n. With $(r; s)$ written as exponent vectors $(e_R, e_S, e_T; f_R, f_S, f_T)$, i.e. $r_i = n^{e_i}$ and $s_i = n^{f_i}$, witnesses are H_R: $(4,4,4;2,4,4)$ with $\mathrm{Blk}(n,n,n,n^2)$ and $U = n^5$ (the other eight terms are at least n^6); $D_{R,S}$: $(2,6,6;2,2,4)$ with $\mathrm{Blk}(1,1,n^2,n^2)$ and $U = n^4$ (the others at least n^5); and, by the action of S_3 on the labels, which leaves U invariant, H_S: $(4,4,4;4,2,4)$, H_T: $(4,4,4;4,4,2)$, $D_{R,T}$: $(2,6,6;2,4,2)$, $D_{S,R}$: $(6,2,6;2,2,4)$, $D_{S,T}$: $(6,2,6;4,2,2)$, $D_{T,R}$: $(6,6,2;2,4,2)$, $D_{T,S}$: $(6,6,2;4,2,2)$. The same holds for the six terms of U_2 in Corollary 4.9.*

[Proved] (on paper; exact checks in Appendix G)

Proof. For H_R: $s_R = n^2$, $r = (n^4, n^4, n^4)$, $s_S = s_T = n^4$ give $H_R = \sqrt{n^2 n^8} = n^5$, $H_S = H_T = D_{S,R} = D_{T,R} = n^6$ and $D_{R,S} = D_{R,T} = D_{S,T} = D_{T,S} = n^8$. $\mathrm{Blk}(n,n,n,n^2)$ has sizes n^4, squared norms $n^2 \le s_i$ and $\mathrm{Out} = n^5$. For $D_{R,S}$: $r = (n^2, n^6, n^6)$, $s = (n^2, n^2, n^4)$ give $D_{R,S} = n^4$, $H_S = n^5$, $H_T = D_{R,T} = n^6$, $H_R = n^7$ and $D_{S,R} = D_{T,R} = D_{T,S} = n^8$, $D_{S,T} = n^{10}$. $\mathrm{Blk}(1,1,n^2,n^2)$ has sizes $(n^2, n^4, n^4) \le r$, squared norms $(1, n^2, n^2) \le s$ and $\mathrm{Out} = n^4$. In both cases $\mathrm{Out} = U$ and the upper bound of Theorem 4.1 gives $W = U$. Relabelling by S_3 permutes the terms (Section 2) and the witnesses. For U_2 the six blocks of `artifacts/hillclimb/P3-spectral/report.md` (same shapes) work verbatim. □

So the regimes of Remark 4.5 are all realized: no term of U can be dropped, even for constant-factor accuracy.

Remark 4.11 (exact sizes)**.** For regular (r, σ) with integer r_i, $\sup\big\{\mathrm{Out} : |i| = r_i, \|A_i\|_2 \le \sigma_i \text{ for all } i\big\} = W(r, \sigma)$. An admissible instance pads to exact sizes with isolated edges on fresh endpoints: each adds a norm-1 block-diagonal component (so $\|A_i\|_2$ becomes $\max\big(\|A_i\|_2, 1\big) \le \sigma_i$) and no triangle, while the exact-size class is a subclass. The same holds for the two-norm class of Corollary 4.9.

[Proved]

4.5. The labeled 4-cycle

The same questions for the 4-cycle $R(x,y) \wedge S(y,z) \wedge T(z,w) \wedge Q(w,x)$, with $\mathrm{Out} = \mathrm{tr}\big(A_R A_S A_T A_Q\big)$, have a larger answer. The labels R, S, T, Q are in cyclic order, so $\{R, T\}$ and $\{S, Q\}$ are the opposite pairs. Sizes r_e, norms σ_e, the clipped $s_e = \min(\sigma_e^2, r_e)$, admissibility, regularity and the worst case $W_4(r, \sigma)$ are defined as for the triangle. Complete four-partite blocks no longer suffice: with R, S, Q complete $n \times n$ and T a perfect matching,

Table 5: The 38 monomials of U_4. Indices i, j, k, l are distinct labels; "opposite" means at distance two around the cycle. The family A splits into two orbits of the dihedral group of the cycle ($\{j,k\}$ the two neighbours of i: 4 terms; otherwise: 8 terms), so there are ten orbits.

Family	Monomial	Condition	Number
A	$r_i s_j s_k$	$\{j,k\} \subseteq$ the other three labels	12
B	$r_i \left(s_j s_k s_l\right)^{1/2}$	—	4
C	$\left(r_i r_j s_k s_l\right)^{1/2}$	$\{i,j\}$ opposite	2
C'	$\left(r_i r_j s_k s_l\right)^{1/2}$	$\{i,j\}$ adjacent	4
D	$r_i r_j$	$\{i,j\}$ opposite	2
E	$\left(r_i r_k s_i s_j s_k\right)^{1/2}$	i,j,k consecutive	4
F	$\left(r_i r_j^2 s_i s_k s_l\right)^{1/3}$	$\{i,j\}$ opposite	4
G	$\left(r_j r_k r_l s_i s_k s_l\right)^{1/3}$	$\{i,j\}$ opposite	4
H	$\left(r_i r_j\right)^{1/2} \left(s_R s_S s_T s_Q\right)^{1/4}$	$\{i,j\}$ opposite	2

$\mathrm{Out} = n^3$, while any admissible union of complete blocks for the same statistics ($s_T = 1$ forces the Z- and W-parts of every block to size 1) has output at most $|R| = n^2$.

Theorem 4.12 (spectral AGM for the 4-cycle)**.** *Let $U_4(r,s)$ be the minimum of the 38 monomials of Table 5. Every admissible instance has* $\mathrm{Out} \le U_4$*, with constant* 1 *in each of the 38 inequalities, and for regular statistics*

$$U_4/16 < W_4(r,\sigma) \le U_4.$$

Each of the 38 terms is the unique minimum of U_4 on an open set of regular statistics, where an arc instance (below) attains it, so none can be dropped even for constant-factor accuracy.

[Proved] (on paper, not in Lean; the identification of the dual vertices is an exact finite enumeration. Verified by two separately run exact-arithmetic rebuilds; Appendix G.)

The new ingredient is a bound on a "lifted" Gram matrix indexed by the tuples of the middle relation.

Lemma 4.13 (lifted Gram bound)**.** *Put $\rho = A_R^\top A_R$ and $\tau = A_T A_T^\top$, and let K be the matrix indexed by the tuples $s = (y,z)$, $s' = (y',z')$ of S with $K_{ss'} = \sqrt{\rho_{yy'}\tau_{zz'}}$. Then $\lambda_{\max}(K) \le \|A_R\|_2 \|A_S\|_2 \|A_T\|_2$.*

[Proved]

Proof. If R, S or T is empty there is nothing to prove. On each connected component c of the bipartite graph of S choose positive vectors p, q with $A_S q = \sigma_c p$ and $A_S^\top p = \sigma_c q$ on that component, where $\sigma_c \le \|A_S\|_2$ (Perron–Frobenius). For $s = (y,z)$ put $\theta_s = \sqrt{q_z/p_y}$. For a nonnegative vector u on S let $U_y = \sum_z u_{yz}\theta_{yz}$ and $V_z = \sum_y u_{yz}/\theta_{yz}$. By Cauchy–Schwarz and $\sum_{z \in S(y,\cdot)} q_z = \sigma_c p_y$, $U_y^2 \le \sigma_c \sum_z u_{yz}^2$, so $\|U\|^2 \le \|A_S\|_2 \|u\|^2$, and likewise for V. For $\kappa > 0$ the arithmetic–geometric mean inequality gives $K_{ss'} \le \left(\kappa \theta_s \theta_{s'} \rho_{yy'} + \kappa^{-1} (\theta_s \theta_{s'})^{-1} \tau_{zz'}\right)/2$, hence $u^\top K u \le \left(\kappa U^\top \rho U + \kappa^{-1} V^\top \tau V\right)/2 \le \left(\kappa \|A_R\|_2^2 + \kappa^{-1} \|A_T\|_2^2\right) \|A_S\|_2 \|u\|^2/2$. Take $\kappa = \|A_T\|_2 / \|A_R\|_2$. Since K is symmetric and nonnegative, $\lambda_{\max}(K)$ is attained at a nonnegative u. □

Proof of the upper bound of Theorem 4.12. As for the triangle, every degree of relation e is at most s_e and $\|A_e\|_2^2 \le s_e$. By rotation and reflection it suffices to prove one member of each family. *A, B.* An R-tuple (x,y) extends to at most $d_S^+(y)\Delta_T^+$, $d_S^+(y)d_Q^-(x)$ and $d_Q^-(x)\Delta_T^-$ cycles, and to at most $1_{S(y,\cdot)}^\top A_T 1_{Q(\cdot,x)} \le \|A_T\|_2\sqrt{d_S^+(y)d_Q^-(x)}$; summing over R gives $r_R s_S s_T$, $r_R s_S s_Q$, $r_R s_T s_Q$ and $r_R\big(s_S s_T s_Q\big)^{1/2}$. *C, C′.* $\mathrm{Out} \le \|A_R A_S\|_F \big\|A_T A_Q\big\|_F \le \|A_R\|_2\|A_S\|_F\|A_T\|_2\big\|A_Q\big\|_F$, and $\mathrm{Out} \le \|A_R A_S A_T\|_F\big\|A_Q\big\|_F \le \|A_R\|_2\|A_S\|_2\|A_T\|_F\big\|A_Q\big\|_F$. *D.* A cycle is determined by its R- and T-tuples. *E.* Forgetting Q, Out is at most the number of paths $x \to y \to z \to w$, which is $(d_R^-)^\top A_S d_T^+ \le \|A_S\|_2\|d_R^-\|\big\|d_T^+\big\|$, and $\|d_R^-\|^2 \le \Delta_R^-\,|R| \le s_R r_R$, $\big\|d_T^+\big\|^2 \le s_T r_T$. *F.* Let $M = A_R A_S A_T$, so M_{xw} counts paths. Expanding two of the three factors of M_{xw}^3 over pairs of S-tuples, $\sum_{x,w} M_{xw}^3 = \sum_{s,s' \in S} 1_{X_{ss'}}^\top M 1_{W_{ss'}}$, where $X_{ss'}$ is the set of common R-predecessors of y, y' and $W_{ss'}$ that of common T -successors of z, z'. Each term is at most $\|M\|_2\sqrt{|X_{ss'}|\;|W_{ss'}|} \le \|A_R\|_2\|A_S\|_2\|A_T\|_2 K_{ss'}$, so by Lemma 4.13 $\sum M^3 \le |S|\;\|A_R\|_2^2\|A_S\|_2^2\|A_T\|_2^2$, and Hölder over the Q-tuples gives $\mathrm{Out}^3 \le |Q|^2\sum_{(w,x)\in Q} M_{xw}^3 \le r_Q^2 r_S s_R s_S s_T$. *H.* The $0/1$ matrix B indexed by $S \times Q$ with $B_{(y,z),(w,x)} = A_R(x,y)A_T(z,w)$ has $1^\top B 1 = \mathrm{Out}$ and $\big(BB^\top\big)_{ss'} = 1_{W_{ss'}}^\top A_Q 1_{X_{ss'}} \le \big\|A_Q\big\|_2 K_{ss'}$ entrywise, so $\|B\|_2^2 \le \big\|A_Q\big\|_2^2\lambda_{\max}(K) \le \prod_e \|A_e\|_2$ and $\mathrm{Out} \le \|B\|_2\sqrt{|S|\;|Q|}$. *G.* Split R into classes R_{ab} by $d_Q^-(x) \in [2^a, 2^{a+1})$ and $d_S^+(y) \in [2^b, 2^{b+1})$ (tuples with a zero degree lie on no cycle), and let $h = A_Q A_{R_{ab}} A_S$ and X_{ab}, Y_{ab} the vertices it touches. Its cycles number at most $\min\big(\|h\|_1, \sqrt{|T|}\|h\|_F\big) \le \big(|T|\;\|h\|_1\|h\|_F^2\big)^{1/3}$, with $\|h\|_1$ the entry sum, and $\|h\|_1 \le 4\cdot 2^{a+b}\;|R_{ab}|$, $\|h\|_F^2 \le \big\|A_Q\big\|_2^2\|A_S\|_2^2\;|R_{ab}|$, $|R_{ab}|^2 \le \|A_R\|_2^2\;|X_{ab}|\;|Y_{ab}|$, $2^a\;|X_{ab}| \le |Q|$ and $2^b\;|Y_{ab}| \le |S|$. So each class has at most $\big(4r_T r_Q r_S s_Q s_S s_R\big)^{1/3}$ cycles, and there are at most $(1+\log_2|Q|)(1+\log_2|S|)$ classes. The n-fold Kronecker power of the instance is a $0/1$ instance whose output, sizes and norms are the n-th powers; applying the bound to it and taking n-th roots removes the factor $4^{1/3}$ and the logarithms as $n \to \infty$. □

Arc instances. Call the vertex classes X, Y, Z, W around the cycle. An *arc* is a nonempty set of consecutive classes; we use the nine arcs $\{X\}, \{Y\}, \{Z\}, \{W\}$, $\{X,Y\}, \{Y,Z\}, \{Z,W\}, \{W,X\}$ and $V = \{X,Y,Z,W\}$. Given integers $n_A \ge 1$, a vertex of class v is a tuple of coordinates, one in $[n_A]$ for every arc A containing v, and an edge of the cycle joins two vertices exactly when they agree on every arc containing both classes. Each relation is then a Kronecker product of identity matrices (arcs containing both ends), all-ones vectors (arcs containing one end) and scalars, so

$$|e| = \prod_{A\cap e\neq\varnothing} n_A, \quad \|A_e\|_2^2 = \prod_{|A\cap e|\,=1} n_A, \quad \mathrm{Out} = \prod_A n_A.$$

Proof of the lower bound of Theorem 4.12. With $\nu_A = \log n_A \ge 0$, the best real arc instance solves the linear program: maximize $\sum_A \nu_A$ subject to $\sum_{A\cap e\neq\varnothing}\nu_A \le \log r_e$ and $\sum_{|A\cap e|\,=1}\nu_A \le \log s_e$ for every edge e. By LP duality its value is the minimum, over the vertices of the dual polyhedron, of the corresponding combinations of $\log r_e$ and $\log s_e$. Exact enumeration gives 50 vertices; 12 are dominated by others whenever $1 \le s_e \le r_e$, and the remaining 38 are the terms of Table 5. (Adding the four arcs of three classes changes nothing: moving such a factor to V keeps sizes and output and lowers two norms.) So the

value is $\log U_4$; let $n_A \geq 1$ be an optimal real solution. Floor the eight proper arcs and set $m = \lfloor m^* \rfloor$ with $m^* = \min_e r_e / \prod_{A \neq V, A \cap e \neq \varnothing} \lfloor n_A \rfloor$. The instance is admissible: floors only lower the norms, which do not involve V, and the sizes are at most r_e by the choice of m. Exactly three proper arcs miss each edge, so $m^* \prod_{A \neq V} \lfloor n_A \rfloor = \min_e r_e \prod_{A \cap e = \varnothing} \lfloor n_A \rfloor > \min_e r_e \prod_{A \cap e = \varnothing} n_A/8 \geq U_4/8$, the last step because $r_e \geq \prod_{A \cap e \neq \varnothing} n_A$. Since $m^* \geq n_V \geq 1$, $m > m^*/2$, and $\mathrm{Out} > U_4/16$. *Necessity.* For each kept vertex an exact rational point of regular statistics where it is the unique minimum is recorded in the artifacts; minimality persists on an open cone around it, and the arc instance supported on the vertex's tight arcs attains the term with equality there. Scaling along the cone makes the ratio of the next term to it unbounded. □

Proposition 4.14 (the best constant lies in $[8, 16]$)**.** *For $1 \leq t \leq L$, the statistics $r = (L, t, L, L)$ and $s = (t, t, t, L)$ have $U_4 = \min\left(t^3, \sqrt{Lt^3}\right)$. For $1 \leq t < 2$ and $L = 8$ they are regular, $U_4 = t^3$ and $W_4 = 1$; hence $\sup U_4/W_4$ over regular statistics lies in $[8, 16]$.*

[Proved] (the envelope by an exact exponent comparison of all 38 terms; Appendix G)

Proof. Every term of Table 5 at these statistics is $L^a t^b$ with $a \geq 0$, $a + b \geq 2$ and $3a + b \geq 3$; comparing exponents at $L = t$ and $L = t^3$, every term is at least $\sqrt{Lt^3}$ for $L \leq t^3$ and at least t^3 above, and $r_S s_R s_T = t^3$ and $\left(r_S r_Q s_R s_T\right)^{1/2} = \sqrt{Lt^3}$ attain the two values. For $t < 2$ and $L = 8 \geq t^3$: $|S| \leq t$, so S has at most one tuple (y, z); the column degree of y in R and the row degree of z in T are integers at most $s_R = s_T = t < 2$; so $\mathrm{Out} \leq 1$, and one cycle is admissible. □

The condition $L \geq t^3$ is needed: for $L < t^3$ the family has $U_4 = \sqrt{Lt^3} < t^3$. If norms are given only for a set P of relations, put $s_e = r_e$ off P; Theorem 4.12 holds unchanged, and the obstruction 8 persists whenever $|P| \geq 2$ (for adjacent $P = \{R, S\}$ use $r = (8, 8, 8, t)$, $s = (t, t, 8, t)$). The constant 16 comes from the rounding: flooring an arbitrary optimal real arc instance can lose a factor approaching 16 (take $n_Z = n_W = n_{\{Z,W\}} = n_V = t \to 2$, all others 1: $U_4 = t^4$ and the floored instance has one cycle). Improving it needs a better choice among optima or instances outside the arc family; on small integer grids the best integer arc instance is within a factor 2.6 of U_4. The 4-cycle instance with $r = (1, n, n, n)$ and $s = (1, n, 1, n)$, which defeats complete blocks, is the single arc $\{Z, W\}$ with $n_{\{Z,W\}} = n$, and attains $U_4 = n$.

5. Per-instance rotation: kernel optimality for (r, σ)

Lemma 5.1 (nested neighbourhoods)**.** *Fix R and the degree vectors d_S^+ and d_T^-. Order Z once, and let S' give each y the first $d_S^+(y)$ elements of Z and T' give each x the first $d_T^-(x)$ elements. Then S' has the same out-degrees d_S^+ as S, T' has the same in-degrees d_T^- as T, $|S'| = |S|$, $|T'| = |T|$, and $M_R = \mathrm{Out}(R, S', T')$.*

[Proved; Lean-checked] (`kernelWork_eq_out_nest`, `rowDeg_nestS`, `colDeg_nestT`, `card_nestS`, `card_nestT`.)

Proof. Two prefixes of one order intersect in the shorter prefix, so $|S'(y, \cdot) \cap T'(\cdot, x)| = \min(d_S^+(y), d_T^-(x))$. Sum over R. □

So the kernel's work is achieved as *output* by an instance with the same R, the same out-degrees of S, the same in-degrees of T and the same sizes; the in-degrees of S' and out-degrees of T' change in general. This does **not** give $M_R \le W(r,\sigma)$ for the original norms: nesting can raise operator norms. The 2×2 identity has $\|\cdot\|_2 \le 1$, while its nested version has $\|\cdot\|_2^2 \ge 2$ (Lean: `nestS_can_raise_opNorm`). We therefore combine Lemma 5.1 only with bounds that use no norm of S' or T'.

Proposition 5.2 (kernel work against the nine terms)**.** *On every admissible instance,* $M_R \le \min\big(H_R, D_{R,S}, D_{R,T}, D_{S,R}, D_{T,R}\big)$*, and by rotation* $M_S \le \min\big(H_S, D_{S,T}, D_{S,R}, D_{T,S}, D_{R,S}\big)$ *and* $M_T \le \min\big(H_T, D_{T,R}, D_{T,S}, D_{R,T}, D_{S,T}\big)$.

[Proved; Lean-checked] (`kernelWork_le_terms`, `kernelWork_le_opNorm_mul_sqrt`.)

Proof. H_R: with $\min(p,q) \le \sqrt{pq}$, $u_x = \sqrt{d_T^-(x)}$ and $v_y = \sqrt{d_S^+(y)}$, $M_R \le u^\top A_R v \le \|A_R\|_2 \|u\|\|v\| = \|A_R\|_2 \sqrt{|T|\,|S|}$. $D_{R,S}, D_{R,T}$: Lemma 4.2. $D_{S,R}$: $M_R \le \sum_{(x,y)\in R} d_S^+(y) = \sum_y d_R^-(y) d_S^+(y) \le s_R r_S$. $D_{T,R}$: $M_R \le \sum_x d_R^+(x) d_T^-(x) \le s_R r_T$. □

Corollary 5.3 (per-instance rotation is worst-case optimal)**.** *Every one of the nine terms appears in at least one of the three lists of Proposition 5.2. Hence on every admissible instance* $\min(M_R, M_S, M_T) \le U$*, and if* (r,σ) *is regular then* $U < 4W(r,\sigma)$.

[Proved; Lean-checked] (`kernel_min_le_U`; the second inequality is Theorem 4.1. Without regularity, U can be positive while $W = 0$, e.g. $r = (0.5, 1, 1)$, $\sigma = (1,1,1)$.)

The three M_π are computable exactly from degree arrays in $O\big(\sum_i r_i\big)$ time, or the rotation owning the argmin term of U can be chosen from statistics alone. Replacing s_j by the corresponding measured maximum degree in the six degree terms $D_{i,j}$ keeps every bound valid (the Hölder terms need the norm). A kernel running the chosen rotation with $O(1+\min)$ intersections does $O\big(\sum_i r_i + W\big)$ work for regular statistics.

Corollary 5.4 (two measured norms suffice for the planner)**.** *In the setting of Corollary 4.9, every instance in the class satisfies* $M_R \le \min\big(H_R, D_{R,S}, D_{S,R}, D_{T,R}\big)$ *and* $M_S \le \min\big(H_S, D_{S,R}, D_{R,S}, D_{T,S}\big)$*. Every term of* U_2 *occurs in one of the lists, so* $\min(M_R, M_S) \le U_2 < 4W_2$*: a planner drives* R *when a minimizing term of* U_2 *is* H_R*,* $D_{S,R}$ *or* $D_{T,R}$*, and* S *when it is* H_S*,* $D_{R,S}$ *or* $D_{T,S}$*. The restriction to the two measured relations is necessary. As a budget: with* R *and* S *perfect matchings on* k *vertices per part and* $T = K_{k,k}$*,* $s_R = s_S = 1$ *and* $r = (k,k,k^2)$ *give* $U_2 = W_2 = \mathrm{Out} = M_R = M_S = k$ *while* $M_T = k^2$*, so* M_T/W_2 *is unbounded (but* $M_T \le N$ *there). Including the input: with* $T = K_{k,k}$*,* R *giving each* x *a private out-star of* k *vertices and* S *giving each* z *a private in-star of* k *further vertices,* $N = 3k^2$*,* $s_R = s_S = k$*,* $U_2 = k^{5/2}$ *and* $M_T = k^3$*, so* $M_T/(N+U_2) = \Theta\big(k^{1/2}\big)$.

[Proved] (on paper; exact checks in Appendix G)

Proof. Proposition 5.2 with $s_T = r_T$; $D_{R,T} = r_R r_T \ge D_{T,R}$ and $D_{S,T} \ge D_{T,S}$ are redundant. In the example an R-tuple (x_i, y_i) has $d_S^+(y_i) = 1$ and $d_T^-(x_i) = k$, and an S-tuple (y_i, z_i) has $d_R^-(y_i) = 1$ and $d_T^+(z_i) = k$, so $M_R = M_S = k$; every T-tuple (z,x) has $d_R^+(x) = d_S^-(z) = 1$, so $M_T = k^2$; the six terms are $\big(k^{3/2}, k^{3/2}, k, k, k^2, k^2\big)$ and $\mathrm{Out} = k$. In the star example, disjoint stars $K_{1,k}$ have squared norm k, so $r = (k^2, k^2, k^2)$, $s_R = s_S = k$, and

the six terms are $\big(k^{5/2}, k^{5/2}, k^3, k^3, k^3, k^3\big)$; every T-tuple (z, x) has $d_R^+(x) = d_S^-(z) = k$, so $M_T = k^3$. □

The Hölder term also survives sorted intersections: by Lemma 3.28 the galloping kernel driven by R costs $O\big(N + \|A_R\|_2 \sqrt{|S|\ |T|}\big)$ comparisons, with constant 3 and no logarithm.

Proposition 5.5 (no fixed rotation is optimal; nor is the sum)**.** *(a) There are instances with $W \le 1$ and $M_R = k$. (b) There are instances with $M_R / \big(\sum_i r_i + U\big) = \Theta\big(\sqrt{k}\big)$ while $M_S = M_T = 0$. By relabelling, the same holds for the fixed rotations S and T, and for $M_R + M_S + M_T$.*

[Proved] (on paper; not formalized.)

Proof. (a) $R = \{(x_i, y_0)\}_{i \le k}$, $S = \{(y_0, z_0)\}$, $T = \{(z_i, x_i)\}_{i \le k}$ with $z_i \ne z_0$. Then Out $= 0$, $M_R = k$, $M_S = M_T = 0$, and $s = (k, 1, 1)$ so $W \le U \le D_{S,T} = 1$. (Here $M_R \le r_R$, so this refutes only $O(W)$.) (b) $R = K_{k,k}$; S gives each y a private out-star of k z's; T gives each x a private in-star of k further z's. Then $r = (k^2, k^2, k^2)$ and, since a disjoint union of stars $K_{1,k}$ has norm $\sqrt{k}$, $s = (k^2, k, k)$. The nine terms are $H_R = k^3$, $H_S = H_T = k^{5/2}$, $D_{S,R} = D_{T,R} = k^4$ and k^3 for the other four, so $U = k^{5/2}$. Every R-tuple has $\min(d_S^+, d_T^-) = k$, so $M_R = k^3$; every S-tuple ends at a z with $d_T^+(z) = 0$ and every T-tuple starts at a z with $d_S^-(z) = 0$, so $M_S = M_T = 0$. Hence $M_R / \big(3k^2 + k^{5/2}\big) = \Theta\big(\sqrt{k}\big)$ (numerically 0.80, 1.37, 2.29, 3.70, 5.82 for $k = 4, 8, 16, 32, 64$). □

Both families have Out $= 0$ and are not semijoin-reduced: one semijoin pass empties them. Reduction does not remove the loss in (b): padding it (Lemma 3.8) gives a pairwise semijoin-reduced instance with $M_R \ge k^3$ (costs only grow), sizes $\Theta(k^2)$, and $s_S, s_T = O(k)$ by the Schur test, so $U \le H_S = O\big(k^{5/2}\big)$ and the $\Theta\big(\sqrt{k}\big)$ loss against $\sum_i r_i + U$ remains (measured ratios 0.56, 0.84, 1.30 for $k = 8, 16, 32$).

6. Engine and experiments: summary

This section states what the engine does with the results above and what the measurements do and do not show; Appendix A and Appendix B give the setups, tables and caveats. The engine is Leatherback, which will be open-sourced; at the time of writing the triangle operator and both admission stages described below are on a development branch, not on its main branch. The paper repository (Appendix H) holds the engine-run scripts and raw data.

The operator. Leatherback's triangle operator answers a run of three triple patterns forming a triangle with constant predicates, in any of the eight edge directions. It returns exactly the bag the greedy left-deep plan would produce, or declines before any answer is visible, and allocates nothing per row. For each tuple of the driver E_1 it intersects two sorted runs from the shorter side, so its candidate budget is M_π; the entries it actually walks are at most M_π (a Lean bridge proves this for a replica, Appendix D), and its total work adds scan, galloping and seek overheads. It does not implement per-binding choice or price-then-run.

Planning. For each candidate driver the planner reads 64 evenly strided rows, estimates $\widehat{M} = \lfloor |E_1| \cdot \mathrm{mean}(\min(d_2, d_3)) \rfloor$ from CSR degrees and picks the driver minimizing $2|E_1| + \widehat{M}$; a gate then admits the operator only when its estimate beats greedy's first 2-path (stage 1), and a second stage in calibrated units re-prices declined triangles on warm stored views (the *rescue*, Appendix B). Corollary 5.3 certifies the exact-M choice and the statistics-owner choice, **not** the sampled estimator, and no fixed sampling grid can be certified:

Proposition 6.1 (no fixed sampling grid is within a constant factor)**.** *Fix $k \geq 1$. Let a planner price each candidate driver E with $n = |E|$ rows by reading the rows at a set $P(n)$ of $1 \leq k_n \leq k$ positions of E's scan order, chosen in advance for each n, setting $\widehat{M}_E = \left\lfloor n \sum_{q \in P(n)} \min(d_2, d_3)/k_n \right\rfloor$, and choosing the argmin of $2n + \widehat{M}_E$, ties to the first pattern. (The stride $\lfloor sn/64 \rfloor$ and the golden-ratio grid that Leatherback's other planners sample with are two such rules.) For every $m > k$ there are an instance I_m with $N = 3m^2 + 2$ and an integer $1 \leq j \leq k$ such that:*

1. *$\widehat{M}_R = 0$, the planner picks R, and on a warm store the gate admits it for every ratio $\rho \leq (m+1)/2$, and the floor does whenever $N \geq 2048$;*
2. *$M_R = m \cdot \min_\pi M_\pi = m \cdot \mathrm{Out} = m^2(m-j)$, and the operator driven by R walks at least $(m-1)m(m-j)$ entries;*
3. *if $m \geq 2k$, then $(2|R| + M_R)/\min_\pi(2|E_\pi| + M_\pi) \geq m/8$.*

So on I_m with $m \geq 2k$ the chosen driver costs $\Theta\big(N^{1/2}\big)$ times the best one, in budget, in walked entries and in the planner's own objective; no fixed bounded sample count and no fixed grid avoids this.

[Proved] (on paper; proof in Appendix B, exact checks in Appendix G)

Three planner quantities. *Output upper bounds* (AGM, the degree bound, B_2, U) bound $|\mathrm{Out}|$ and size buffers; by themselves they bound no executor's work. *Work bounds for the chosen executor*: for the kernel driven by π the walked entries are at most M_π, and $\min_\pi M_\pi \leq U$ (Corollary 5.3), while B_2 is not a work bound for this kernel even with the best driver (GMIN, Theorem 3.16(2)) or with per-binding choice (Corollary 3.7). *Estimated execution costs* ($2|E_1| + \widehat{M}$ against greedy's sampled first 2-path, and the rescue's calibrated costs) are estimates, not bounds: $\widehat{M}$ can be 0 when $M_\pi = \Theta\big(N^{3/2}\big)$ (Proposition 6.1).

What the measurements show (Appendix A, Appendix B). All rates are for selected canonical triangle queries over the largest predicates of each dataset, with intersecting supports and nonzero output; they are not weighted by any workload, exclude zero-answer queries, and one dataset (sci50) is 50 relabelled copies of one graph. First-row and tail latency and metered kernel work were not measured.

- *Spectral bound (3,124 queries).* U is below AGM by a median factor 9.6, but tighter than the best of AGM, the cyclic ℓ_2 bound and the degree bound on only 6.5% of queries, and $U_{\mathrm{cert}} \geq \ell_2$ on 81.2% of them: our pre-registered success criterion is not met as registered. U is useful as one more term in a minimum, and for its guarantee (Corollary 5.3), not as a replacement estimator.
- *Driver choice (1,241 knowledge-graph queries, six policies).* The driver matters: the worst forced driver is up to 132 × slower than the best. But choosing it by the exact

planner objective $2|E_1| + M_\pi$ instead of the 64-row estimate changes the driver on only 3–6% of queries, the median and p90 of the exact pick's forced-time gain are exactly 1, and the exact policy is 36–90 × slower end to end because of its degree passes. So the sampled planner did as well as exact counting on this workload, although it has no worst-case guarantee (Proposition 6.1).

- *Admission (same queries).* Stage 1 declined 66–73% of queries, and on many of them the operator would have been faster. A rescue stage, fitted to these same queries, cuts total time by 20.6%, 48.8% and 32.5% against a stage-1-only planner run in the same process with the policy order shuffled every round (run C). On queries whose plan the rescue leaves unchanged, the two arms agree to a median ratio of 1.000–1.005: there the extra rescue tests stay within run-to-run noise (descriptively; we did no uncertainty analysis), and no rescued query is more than 1.2 × slower. A model-run audit (Appendix G) found that the first implementation's candidate pruning was not equivalent to the searched rule; the fixed implementation that run C measures decides as the rule over every candidate, which changed one decision (a CoDEx-L query now declined). On three synthetic control stores the rescue changes no decision; a 1.06 × slowdown on a 54 ms `COUNT(*)` seen in a single earlier run did not reproduce over 10 alternating process pairs (median 0.995).

7. Related work and novelty

Worst-case optimal joins, and where the kernel comes from. The AGM bound [AGM13], NPRR [NPRR18], Leapfrog Triejoin [Vel14], Ngo's survey [Ngo18] and "Skew strikes back" [NRR13] establish the size-only theory; Minesweeper [NNRR14] and geometric resolution [ANRR15] go beyond worst case; Joglekar and Ré use degree uniformization [JR16]; Free Join [WWS23] unifies binary and WCOJ plans; Kara et al. maintain triangle counts under updates [KNNOZ19]. The kernel of Section 5 is not new. For a fixed driver R, its work M_R is, up to a semijoin of $R(x,\cdot)$ with $\pi_Y S$, the intersection cost of Generic Join on the triangle with the order x, y, z, i.e. of Algorithm 2 of [NRR13], whose bound $M_R \le \sqrt{|R|\ |S|\ |T|}$ is classical [NRR13, App. B]; Birler, Kemper and Neumann implement the same per-tuple smaller-side intersection as the Expand3 operator [BKN24]. What Section 5 adds is the bound of M_π by the spectral and clipped-degree terms (Proposition 5.2) and, with Theorem 4.1, the worst-case optimality of choosing the driver per instance for the statistics (r, σ) (Corollary 5.3).

Per-binding and adaptive orders. Choosing the evaluation strategy separately for each binding is prior art, and we do not claim it. Algorithm 1 of [NRR13] chooses, for each binding of the first variable, between two evaluation options using $O(1)$ degree products ("the power of two choices"). In compact WCOJ indexes for graph databases, Arroyuelo et al. bind next the variable with the fewest candidates and note that, after the first variable is bound, the next variable to bind may differ for each binding [ABGHNR24, Sec. 5]; Arroyuelo et al. [ABFGN25, Sec. 6.1] study such adaptive variable elimination orders systematically, recomputing the second variable for each binding from the sizes of the candidate ranges. ADOPT [WTKO23] splits the value space into hypercubes and runs

Leapfrog Triejoin on each under an attribute order chosen by reinforcement learning; it proves AGM worst-case optimality (under the assumption that the number of episodes is constant) and convergence to the attribute order that minimizes the number of steps. SkinnerDB [TWM+19] learns join orders at run time, and Graphflow [MS19] fixes the first two query vertices, so for a triangle it has no per-binding freedom. Not all of these guarantees are size-only. Arroyuelo et al. treat each k-nearest neighbour clause as a virtual relation in which every x has at most k partners, and bound the output by a linear program that extends Ngo's bound for acyclic degree constraints; the bound is tight when the constraint graph is acyclic. Their Leapfrog Triejoin over the Ring, binding variables in a topological order of that graph, runs in time $O(Q^* \, |Q| \log N)$ for that program's value Q^* when the constraints are acyclic or form a single 2-cycle [ABGHNR24, Sec. 4, Lemma 3 and Thms. 2–3]. Their Example 4 is a triangle with one relation replaced by a k-NN clause: the degree constraint lowers the output bound from AGM's $O\big(N^{3/2}\big)$ to $O(kN)$, and the order y, z, x can need $N^{3/2}$ variable eliminations where y, x, z needs kN. So variable order under degree constraints is analysed there. The differences from this paper are the statistic and the class: their degree bound is a constant k known in advance for a similarity relation, the data relations enter only through their sizes, and their fewest-candidates rule is not claimed worst-case optimal for other constraint cycles. None of these works bounds the cost of per-binding choice against degree-sequence or ℓ_2 statistics of the data relations, and we found no statement of the cyclic ℓ_2 bound of Theorem 3.3, its lower bound, or the separations of Theorem 3.16. Our per-binding class (Section 3) differs from these selectors in two ways that matter for the statements. First, the option at a binding is the one of smaller exact cost $c_P(u)$, not the one with fewer candidates; so Theorem 3.3 bounds this exact-cost choice (and price-then-run, Proposition C.2), not candidate-count heuristics. The lower bounds Theorem 3.6 and Theorem 3.10 bound the best per-binding choice, so they apply to every selector in the class. Second, an intersection is charged at the length of the shorter run, as with hash or bitmap probing (the Expand3 operator and our engine); a leapfrog or galloping intersection of sorted runs can cost less than the shorter run when identifiers are contiguous. With adversarial identifiers our lower bounds against $N + B_2$ hold for every fresh comparison-based intersection routine, and the worst case at $B_2 \asymp N$ gains exactly $(\log N)^{1/4}$ (Section 3.6).

Adaptive sorted-set intersection. Demaine, López-Ortiz and Munro [DLM00] gave comparison lower bounds for adaptive intersection, union and difference of sorted sets in terms of the size of a proof (certificate) of the answer; Barbay and Kenyon [BK02, BK08] analysed intersection by the alternation of the sets, with deterministic and randomized lower bounds; and doubling search is the unbounded search of Bentley and Yao [BY76]. Lemma 3.23 and Lemma 3.31 are these arguments for two runs in our model, and the galloping price h is the standard doubling-search cost. What Section 3.6 adds is that one global labeling makes all the critical calls of a join construction simultaneously hard (Lemma 3.24, Theorem 3.25), and the resulting tradeoff for per-binding choice: an upper bound with the factor $(\log N)^{1/4}$ and its necessity (Theorem 3.29, Theorem 3.32).

Partitioned algorithms. Algorithms that meet ℓ_p bounds partition the data. Jayaraman, Ropell and Rudra [JRR21] bucket each relation by the power-of-two degree of its source attribute and, for every combination of buckets, run the three spanning acyclic (path) plans of the directed triangle in parallel, stopping at the first to finish; for $p \in (1, 2]$ this runs in time linear in the bound of one orientation (their Thm. 4.4, with a factor $2^{(p+1)|V|}$, $O(1)$-time B-tree access in the RAM model, and $O(N \log N)$ index construction). Running it on both orientations meets B_2 up to that factor. Abo Khamis et al. reduce ℓ_p constraints to PANDA [ANS17] by partitioning each relation into parts with near-uniform degrees, losing a polylogarithmic factor [ANOS24, Thm. 2.6]; PANDAExpress [ANS25, Cor. 8.2] handles ℓ_p constraints in time $O((N + B) \log N)$ with data-dependent hyperplane partitions, and shows that a constant number of axis-parallel (threshold) partitions does not suffice for some disjunctive Datalog rules. Jaguar [AC26a] evaluates Boolean conjunctive queries under degree constraints in time $O\big(N^{\text{subw}\,+\varepsilon}\big)$ by heavy/light partitioning guided by violations of submodularity; it does not treat ℓ_p constraints. Guanaco [AC26b] evaluates Boolean conjunctive queries in time with exponent just above the submodular width by enforcing consistency, partitioning relations until they are globally uniform, and joining pairs of relations. Like Jaguar and PANDAExpress, it partitions relations to control the discrepancy of their degrees, the lever that the unpartitioned class of Section 3 lacks. Capelli, Irwin and Salvati [CIS25] use one *fixed* variable order and branch on the bits of values; their algorithm is worst-case optimal up to polylogarithmic factors whenever the class of instances is prefix closed for that order, which covers cardinality constraints and acyclic degree constraints. Our Corollary 3.7 is a statement about the *unpartitioned* class of Section 3 (one root per instance, an option per binding, smaller-side intersections). It is consistent with, and explains, the degree partitioning of [JRR21, ANOS24, ANS25]; it does not say that adaptive algorithms in general cannot meet B_2.

Fixed orders. Ngo's survey records that fixed-order algorithms are optimal under acyclic degree constraints [Ngo18]. Under ℓ_2 constraints, Jayaraman, Ropell and Rudra show that no single spanning acyclic subquery meets the ℓ_2 bound [JRR21, Sec. 2.4 and Ex. B.1]: the path join $R \bowtie S$ can output $L^{5/2}$ tuples while the triangle has at most L^2, a gap of $N^{1/4}$ at $|R| = L^2$. That example concerns path plans, which do not intersect with the closing relation. The separations of Theorem 3.16 concern Generic Join *with* smaller-side intersections, a fixed root with per-binding options, and the best of the six orders per instance, on pairwise semijoin-reduced inputs. ADOPT converges to the best single attribute order, against which Theorem 3.16(2) gives a loss of $N^{1/5}$ relative to $N + B_2$ in our cost model; but ADOPT may use different orders on different value ranges, and we have not shown that our families defeat a constant number of value-range cubes under an adversarial assignment of identifiers.

Norm-based bounds and the operator norm. Abo Khamis, Nakos, Olteanu and Suciu bound join sizes by ℓ_p-norms of degree sequences [ANOS24]; their Eq. (3) gives our degree comparator and their Eq. (4) is the cyclic ℓ_2 bound B_2 (one orientation). LpBound [ZMA+25] turns these bounds into a pessimistic estimator with stored norms $p \in \{1, ..., 10, \infty\}$. The degree sequence bound [DSBC23], SafeBound [DSB23] and the

ambidextrous bounds of Lin and Wang [LW25] use degree sequences; MNC [SBE+19] estimates the sparsity of matrix products. None of these uses the operator norm of the incidence matrix, and the operator norm is not a function of degree sequences, even labeled ones: the 0/1 matrices with rows $\{1,2\},\{1\},\{3\}$ and with rows $\{2,3\},\{1\},\{1\}$ both have row sums $(2,1,1)$ and column sums $(2,1,1)$, but squared operator norms $\left(3+\sqrt{5}\right)/2$ and 2. So the class of Theorem 4.1 is not an ℓ_p class. Conversely, the six degree terms of U are Eq. (3) of [ANOS24] with maximum degrees replaced by the bound σ^2 of Lemma 4.2.

Matrix and spectral inequalities. The Hölder term is the case $(\infty,2,2)$ of the Schatten–Hölder inequality $|\operatorname{tr}(ABC)| \le \|A\|_{S_p}\|B\|_{S_q}\|C\|_{S_r}$, $1/p+1/q+1/r=1$ [Bha97, Ch. IV], combined with $\|A\|_2^2 \le |A|$. The other cases with $p,q,r \ge 2$ add nothing for our statistics: $\|A_i\|_{S_p}^p \le \|A_i\|_2^{p-2}\,|A_i|$, so the bound is at most $\prod_i \sigma_i^{1-2/p_i} r_i^{1/p_i}$, which is log-linear in $(1/p_R, 1/p_S, 1/p_T)$ and so is minimized at a vertex of $\left\{\theta \in [0,1/2]^3 : \sum_i \theta_i = 1\right\}$, that is, at a Hölder term (with σ_i^2 in place of s_i). The degree terms use the 0/1 entries (Lemma 4.2) and have no counterpart in this family. Hatami [Hat08] characterizes the graphs H whose homomorphism densities $t_H(\cdot)^{1/|E(H)|}$ are norms through a Hölder-type inequality; even cycles give Schatten norms, but Hölder graphs are bipartite, so the triangle lies outside that theory, and no lower bounds under mixed size and norm constraints are given. Collins and Yuan [CY26] bound multi-leg partial traces of tensor products of matrices under operator-norm constraints alone, by powers of the tensor-factor dimension; with one leg the problem is trivial, and neither sizes nor 0/1 entries enter. For a simple graph with m edges, t triangles and spectral radius λ_1, the folklore bound $6t = \operatorname{tr}(A^3) = \sum_i \lambda_i^3 \le \lambda_1 \sum_i \lambda_i^2 = 2m\lambda_1$ is our Hölder term with $R = S = T$ the symmetric edge relation, $r_i = 2m$, $\sigma_i = \lambda_1$ and $\mathrm{Out} = 6t$. Bollobás and Nikiforov [BN07] bound clique counts from below by the spectral radius; Nikiforov's survey [Nik11] covers spectral Turán problems; Ning and Zhai [NZ21] bound the number of triangles in terms of n, m and λ_1; Rivin [Riv01] bounds cycle counts by the number of edges; Alon, Yuster and Zwick [AYZ97] count short cycles; EigenTriangle [Tso08] estimates triangle counts from top eigenvalues. All of these concern one symmetric graph. The Schur test [Sch11, HS78], the signless-Laplacian facts [CRS10] and Ryser's interchange theorem [Rys57] are classical. SPARQL WCOJ engines include [HRRS19, AHN+21, GHN26].

Novelty, stated carefully. The proofs are elementary, and the analytic ingredients (Schatten–Hölder, degree bounds, the smaller-side kernel, per-binding choice, racing alternatives) are known. To our knowledge, the following statements are new; we regard (iii) and (iv) as the main contributions. (i) Under joint size and operator-norm constraints on three labeled relations, the worst-case triangle output satisfies $U/4 < W \le U$ for the nine-term U, the factor 4 is asymptotically sharp (Theorem 4.7, with a family found by a model-based review, Section 9), and one block family witnesses the lower bound in every regime (Theorem 4.1). (ii) The work of the smaller-side kernel driven by the best rotation is at most U, so per-instance choice of the driver is worst-case optimal for (r,σ), while no fixed driver is (Corollary 5.3, Proposition 5.5). (iii) In the cost model of Section 3, the best per-binding choice costs at most $N + (|R|\,|S|\,|T|)^{1/8}\min(\Lambda,\Lambda')^{1/2}$ (Theorem 3.3), and this is tight: at

$B_2 \asymp N$ the worst-case exponent is 9/8 (Theorem 3.6), and together with Theorem 3.4 the bound is attained up to a constant factor at every real slope in $[9/10, 3/2]$ (Corollary 3.11, Theorem 3.12); the 9/8 lower bound at $B_2 \asymp N$ also holds on pairwise semijoin-reduced, connected inputs and for a single directed predicate (Proposition 3.21; below slope 1 no single-predicate family exists, since $B_{2,P} \geq N_P/3$), and on fully reduced inputs the gap lies in $[1/10, 1/8]$, exact for $\kappa \geq 11/10$ (Theorem 3.14). We claim these bounds for this cost model only; per-binding choice itself is prior art, and partitioned algorithms meet B_2 [JRR21, ANS25]. (iv) The separations of Theorem 3.16 for Generic Join with smaller-side intersections: against $N + B_2$, one fixed order has gap exactly 1/4 (GMR, Theorem 3.18) and the best of the six orders exactly 1/5 (GMIN, also after padding to a semijoin-reduced family with $C = \Theta(N)$; Theorem 3.17); against $C + N$, a fixed order, the best of six orders and a fixed root each lose $N^{1/2}$; against $N + B_2$ a fixed root loses between $N^{1/8}$ and $N^{1/4}$ (Corollary 3.19). A polynomial gap for single path plans was observed in [JRR21, Ex. B.1]. (v) A deterministic scheduler that selects a root of cost exactly C in $C + \min(6C, 2N) + 4n + O(1)$ per-edge units (Proposition C.2); running alternatives in parallel and stopping at the first to finish is the multiplexing of [JRR21], and what is new is the exact-cost selection and the bound. (vi) The Ferrers profiles of Proposition C.4, an elementary consequence of Ryser's theorem; they concern *labeled* degree sequences, and say nothing about the coarser classes of sorted degree sequences used by [DSBC23, DSB23, ZMA+25]. (vii) The sorted fresh-call comparison model of Section 3.6: the lower bounds against $N + B_2$ of (iii) and (iv) under one adversarial labeling, and the matching quarter logarithm (Theorem 3.32); the single-call certificate and doubling-search arguments are classical [DLM00, BK02, BY76]. (viii) For the labeled 4-cycle, the 38-term U_4 with $U_4/16 < W_4 \leq U_4$, the lifted Gram bound Lemma 4.13 and the arc instances (Theorem 4.12); Hatami's Schatten-norm results for even cycles [Hat08] concern one symmetric matrix and no size constraints.

We read in full, or in all sections bearing on these claims: AGM13, NPRR18, Vel14, NRR13, Ngo18, JR16, ANOS24, JRR21, ZMA+25, CIS25, ANS25, AC26a, WTKO23, ABGHNR24, ABFGN25, BKN24, Hat08, CY26, WWS23, TWM+19, NNRR14, ANRR15 and KNNOZ19. We read only the abstracts of BN07, NZ21 and AC26b, and did not check SBE+19, LW25, Riv01, AYZ97 or Nik11 in full. We cite DLM00, BK02, BK08 and BY76 for the techniques named above on the basis of their abstracts and a summary in a model-based review (Section 9), not a full reading. Searches for operator-norm or spectral join-size bounds, for extremal traces of products of nonnegative or 0/1 matrices under operator-norm and Frobenius constraints, and for tensor-network contraction bounds found nothing closer than the works above.

8. Open problems

1. **Per-binding cost on fully reduced inputs.** When every tuple lies in a triangle, the worst-case gap against $N + B_2$ lies in $[1/10, 1/8]$ and equals the general value $(3 - 2\kappa)/8$ at every slope $\kappa \in [11/10, 3/2]$ (Theorem 3.14). On the window $\kappa \in [1, 11/10)$ it lies in $[\kappa/11, (3 - 2\kappa)/8]$ (Remark 3.15), and no union of rotated live gadgets can do better than 1/10. Is the answer 1/10? Raising the lower end needs another design;

lowering the upper end needs a bound that uses triangle support, not only pairwise support. [Open]

2. **The quarter logarithm below slope 1.** In the sorted model the upper bound $\hat{C} \leq N + 3^{7/8} L_N^{1/4} N^{-1/8} B_2^{5/4}$ holds (Corollary 3.30). Is the factor $(\log N)^{1/4}$ necessary on $(9/10, 1)$, against the family of Theorem 3.12, as it is on $[1, 3/2)$ (Theorem 3.32)? [Open]
3. **Minimal partitioning.** Is a *constant* number of root classes per variable, chosen from degree metadata, enough to reach $O(N + B_2)$? This would be the least partitioning that closes the $N^{1/8}$ gap of Corollary 3.7. A computation in a relaxed exponent model (vertex classes and degrees as powers of N), not reproduced here, suggests that at most 7 parts per variable matter. PANDAExpress shows that a constant number of axis-parallel partitions does not suffice for some disjunctive Datalog rules [ANS25, Sec. 3]; whether this happens for the ℓ_2 triangle is open. [Open]
4. **Partition-free algorithms in general.** Corollary 3.7 is a lower bound for the per-binding cost model of Section 3, and Theorem 3.25 for fresh comparison-based intersection calls. Does *every* partition-free algorithm, for a suitable definition of the class, fail to run in $O(N + B_2)$, or $\tilde{O}(N + B_2)$? Semijoin pruning and certificate reuse already escape the fresh-call model (Remark 3.33). [Open]
5. **Value-range orders.** Section 3.6 settles sorted intersections for fresh calls: the lower bounds survive, and the worst case at $B_2 \asymp N$ is $\Theta\big(N^{9/8}(\log N)^{1/4}\big)$. Do they survive against algorithms such as ADOPT that use different orders on a constant number of value ranges [WTKO23]? [Open]
6. **A fixed root against $N + B_2$, and one GAO on reduced inputs.** Determine the worst-case gap of a fixed root, which lies in $[1/8, 1/4]$ (Corollary 3.19), and the gap of one fixed GAO on pairwise semijoin-reduced inputs, which lies in $[1/5, 1/4]$ (Theorem 3.18). [Open]
7. **Self-joins with two equal predicates.** Proposition 3.21 settles one predicate: the $9/8$ lower bound and the order separations survive, a fixed root cannot be separated for the cyclic self-join, and a symmetric predicate has $C \leq N + B_2$. Which separations survive when exactly two of the three edges share a predicate, where renaming the variables is not a symmetry? [Open]
8. **A joint spectral and polymatroid bound.** Combine σ_i, maximum degrees $\Delta_i \leq \sigma_i^2$ and ℓ_p-norms of degree sequences in one bound with a matching lower bound. Replacing s_j by Δ_j in the degree terms keeps the upper bound valid; the matching lower bound under joint (σ, Δ) constraints, and the integration of σ into the polymatroid framework of [ANOS24], are open. The experiments show that the gain lies in combining, since U alone rarely wins on the selected nonzero canonical queries of Appendix A. [Open]
9. **Longer cycles, and the 4-cycle constant.** The 4-cycle is settled up to the constant: $U_4/16 < W_4 \leq U_4$ (Theorem 4.12), and the best constant lies in $[8, 16]$ (Proposition 4.14). Is it 8? For k-cycles, $k \geq 5$, the arc instances and the lifted Gram argument appear to extend (we have not checked this), the list of needed terms (the vertices of the dual of the arc program) has not been computed, and queries with several cycles are open. [Open]

10. **Bags.** Equality constraints $|R_i| = r_i$ are settled (Remark 4.11) and the galloping logarithm is analysed (Section 3.6); bag semantics is not. **[Open]**
11. **The sampled planner.** Proposition 6.1 answers the question for fixed sampling grids: the gap to the best driver can be $\Theta(N^{1/2})$. Does a planner reading $O(1)$, or $O(\log N)$, *uniformly random* rows per candidate pick, with constant probability, a driver within a constant factor of $\min_\pi(2|E_\pi| + M_\pi)$? Do grids chosen from the data (for example, one row per distinct key), or semijoin-reduced inputs, avoid the gap? **[Open]**

9. Use of AI tools

Generative AI was used throughout this work: to search for and prove theorems, to write and run the exact-arithmetic checks and the Lean proofs, to implement and measure the engine operator, to draft and revise the text, and to review it. The author directed the work, checked the results he relies on, and is responsible for every claim; no AI system is an author. The form of this statement follows the AI-use disclosure that closes [GHN26].

Who did what. Orchestration, implementation and drafting: Claude Code with Claude Opus 5.5 (Anthropic), in one session that also launched Claude subagents for theorem search, proofs, internal refereeing, verification and audits, and that wrote the Lean proofs and the engine code. Solvers and verifiers for the results added in the third and fourth drafts: Claude Fable 5.1 agents at maximum effort and Codex CLI agents (OpenAI) with GPT-6-Astra at "ultra" effort, run as paired solver and verifier lanes; a Codex CLI agent also searched for the admission rule of Appendix B and audited its implementation. Reviews: four internal drafts preceded this version. The first was reviewed by GPT-5.6-sol Pro (OpenAI) and by Claude Fable 5.1; the second by GPT-6-Astra Pro, by Claude Fable 5.1, and in a separate GPT-6 Pro session run by the author; the third by GPT-6-Astra Pro. Before posting, Weaver, TJC's internal work agent, reviewed the third draft and supplied the outside checks of Appendix G. All of these reviews were model-based; no human referee has reviewed this work.

Credits and checks. Where the text credits a review (Theorem 4.7, Corollary 3.11, Proposition 3.21, Remark C.3 and the remark after Theorem 3.32), the credited content came from that review; Table 10 gives the provenance of every group of results. Model output entered the paper only after the checks of Appendix G: exact rebuilds by agents that did not share the solvers' code, line-by-line proof audits, and, for the results tagged Lean-checked, the Lean kernel.

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split, train+valid+test); CoDEx-L (github.com/tsafavi/codex); sci50, a local synthetic RDF store made of 50 relabelled copies of a 369,661-triple scientists knowledge graph; the copies share their predicates, literals and 134 non-namespace IRIs (objects only). SHA-256 digests of the downloaded files are recorded with the experiment.

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Appendices

A. Experiments

Setup. Data are listed in [Data]. For each dataset we took the largest predicates (wikidata5m: 24, CoDEx-L: 36, sci50: 81; each SNAP graph is one predicate, also symmetrized) and enumerated every triangle query

```
?x p1 ?y .  ?y p2 ?z .  ?z p3 ?x
```

with each predicate possibly transposed, canonical up to rotation and whole-cycle reversal, keeping queries whose variables' supports intersect and whose output is nonzero: 3,124 queries (wikidata5m 1,781; CoDEx-L 871; sci50 456; SNAP 16). The 16 SNAP queries are too few to support any claim on their own; their column in Table 6 is descriptive only. All rates below are conditional on this selection: canonical queries over the chosen large predicates, with intersecting supports and nonzero output. They are not weighted by any workload and exclude zero-answer queries. The code, the results file, the logs, the dataset URLs with SHA-256 digests, the commands and the Python environment are in `artifacts/e2/` of the paper repository (Appendix H); re-running the commands on the six SNAP graphs and CoDEx-L reproduces every column except timings; a checker audits the results file for internal consistency, and the summary script, run on the committed results file and spectral data, rebuilds the summary exactly. The sci50 file can be rebuilt from its one-copy input, which is its own first 369,661 lines; the repository records the SHA-256 digests of both files and the replication recipe. We write a query as `p1 . p2 . p3`, listing the predicates on the R, S, T edges, with `^T` marking a transposed predicate. **sci50 is not an independent knowledge graph**: it is 50 relabelled copies of one scientists knowledge graph of 369,661 triples. Copy $k > 0$ renames every IRI of the base namespace, so the copies share their predicates and the terms the renaming leaves alone: 42,993 literals and 134 IRIs outside the namespace, all in object position only ($2{,}481{,}777 = 50 \times 48{,}773 + 43{,}127$ terms). The copies are therefore not vertex-disjoint. Since no shared term is a subject, a triangle has at most one shared vertex and its other two vertices lie in one copy, so every size and every Out is exactly 50 times the copy's (checked on all 456 rows). σ is not always the copy's: a relation whose objects are all shared is 50 row blocks over the same columns, and its σ is $\sqrt{50}$ times the copy's (11 of the 81 predicates: dates, identifiers, `label`); the 58 with no shared object keep their σ, and the other 12 lie between (σ^2 ratio 1.62 to 12.6). On 426 of the 456 queries the binding term of U uses only unchanged σ's, so U is 50 times the copy's and AGM/U is $\sqrt{50} \approx 7.07$ times the copy's; on the other 30 the factor runs from $1/\sqrt{50}$ to 6.05. Measured on one copy, the 456 queries have median AGM/U 8.96 (minimum 1.22), against 63.4 (8.64) on sci50; ℓ_2/U has median 0.767 on both, and degree /U 0.316 against 0.307. We therefore report AGM/U with and without sci50.

Out is counted exactly and cross-checked by sparse matrix products wherever those cost at most 3×10^8 flops (2,842 of 2,842 agreed). σ comes from power iteration, cross-checked with `svds`, and is bracketed in exact rational arithmetic: below by the larger of the maximum degree and the Rayleigh quotient of the final iterate, above by the Collatz–

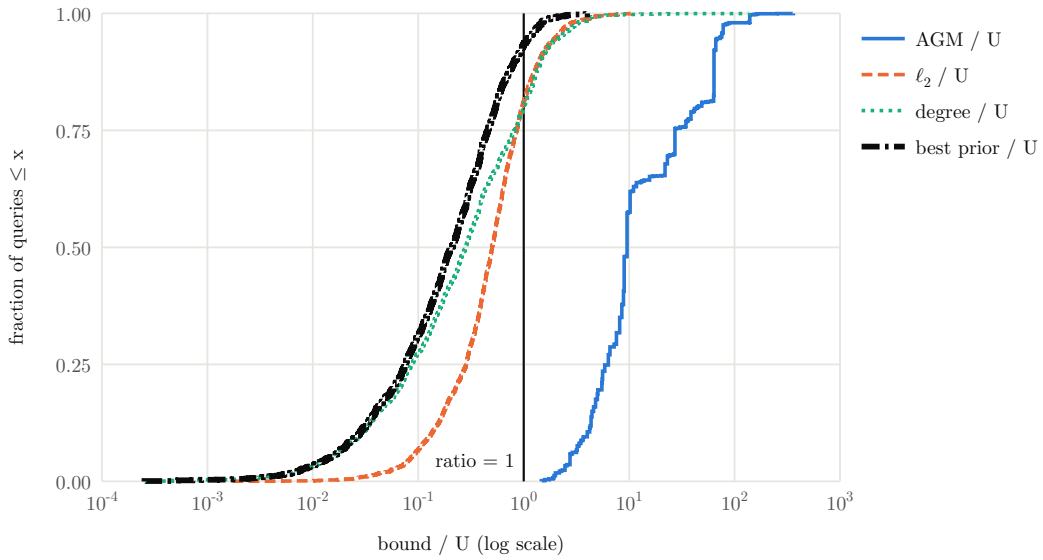


Figure 2: Distribution of bound/U over the 3,124 queries (log scale). Left of the vertical line the prior bound is tighter than U. AGM is never tighter (by construction); the best of the three prior bounds is tighter on 93.5% of the selected nonzero canonical queries.

Wielandt bound of that iterate rounded up to a positive integer vector. Every exact bracket has relative width at most 10^{-6}; on one view (wikidata5m `P106`) the floating-point iteration hit its 3,000-step cap with a float bracket of width $4.6 \cdot 10^{-6}$, and only the exact bracket reaches 10^{-6}. The *certified* σ is min(Schur bound $\sqrt{\Delta_{\text{row}}\Delta_{\text{col}}}$, Collatz–Wielandt after 5 power steps from $\mathbf{1}$), each evaluated exactly (the Schur bound as the integer $\Delta_{\text{row}}\Delta_{\text{col}}$, the Collatz–Wielandt bound as an exact fraction with the iterate rounded up to integers) and its square root rounded upward, so it is a guaranteed upper bound. An earlier computation of these certificates in floating point without outward rounding left the stored value one rounding step below the maximum-degree lower bound on six sci50 views; all certificates are recomputed here, which changes no count below. No bound fell below Out on any row. We write $M = \min(M_R, M_S, M_T)$. A separately written recomputation of 22 rows from raw data reproduced every exact quantity (sizes, Out, M_R, M_S, M_T, AGM, ℓ_2, degree) exactly and σ to 5×10^{-8}; its comparison of certified σ predates the exact certificates and is superseded by them.

Comparator caveat. "Best prior" below is min(AGM, ℓ_2, degree) with the two cyclic ℓ_2 orientations of Section 2. It omits the ℓ_3 and mixed-orientation ℓ_p bounds of [ANOS24, Eq. (5) and Section 1.1] and their LP over all ℓ_p statistics, as implemented in LpBound [ZMA+25]. Against those, U can only win less often, so the win rate below is an upper estimate.

Gap to AGM. $U \leq$ AGM always (Section 2), so only the size of the gap is empirical: the median AGM/U is 9.57 over all queries and 8.98 over the 2,668 queries outside sci50 (minimum 1.47); the sci50 median of 63.4 is $\sqrt{50}$ times the one-copy median of 8.96 (on 426 of the 456 queries AGM/U is exactly $\sqrt{50}$ times its one-copy value).

Table 6: Summary over queries; each row names its statistic. KG = wikidata5m and CoDEx-L; sci50 = 50 relabelled copies of one knowledge graph that share their literals (AGM/U is $\sqrt{50}$ times the one-copy value on 426 of the 456 queries); best prior = $\min(\text{AGM}, \ell_2, \text{degree})$; $M = \min_\pi M_\pi$.

Statistic	**All (3,124)**	**KG (2,652)**	**sci50 (456)**	**SNAP (16)**
AGM/U, median	9.57	8.98	63.4	3.73
AGM/U, minimum	1.47	1.47	8.64	2.35
ℓ_2/U, median	0.502	0.451	0.767	0.838
degree/U, median	0.282	0.275	0.307	4.43
best prior/U, median	0.209	0.199	0.291	0.838
best prior/U, maximum	3.996	3.996	3.05	2.35
queries with $U < \ell_2$	589 (18.9%)	424	163	2
queries with $U <$ best prior	204 (6.5%)	169	33	2
U/Out, median	5.4×10^4	1.0×10^5	1.4×10^3	14.8
M/U, median (maximum)	1.5×10^{-4} (0.59)	9.3×10^{-5} (0.57)	1.9×10^{-3} (0.26)	0.27 (0.59)
U_{cert}/U, p90 (maximum)	1.02 (1.14)	1.02 (1.05)	1.02 (1.06)	1.06 (1.14)
U_{Schur}/U, median	2.64	3.00	1.73	3.04

Which bound is tightest. Over the 3,124 selected queries the degree bound is the tightest on 1,902, ℓ_2 on 1,018, U on 204 and AGM on none. With the exact upper end of each σ bracket in place of the estimate, U still wins on the same 204 queries. Every one of the 204 wins of U comes from a Hölder term ($\sqrt{s_R r_S r_T}$: 121, $\sqrt{s_T r_R r_S}$: 55, $\sqrt{s_S r_R r_T}$: 28). This is forced: by Lemma 4.2 the maximum degrees satisfy $\Delta \le s$, so the degree terms of U are never smaller than the classical degree bound. The largest win is 3.996 × (e.g. wikidata5m `P17 . P47 . P17^T`, borders between countries); the largest losses are on hub predicates such as `P17` (country) and `P31` (instance of), where ℓ_2/U falls to 0.0017. A hub of degree Δ forces $\sigma^2 \ge \Delta$, and the Hölder term charges that mass to the whole relation, whereas ℓ_2 norms see the skew.

Certification. The 5-step Collatz–Wielandt bound is almost exact (median $U_{\text{cert}}/U = 1$, maximum 1.14), and the wins survive it (203 of 204), while the Schur bound alone is loose (median 2.64 ×). So a certified σ is cheap to store and to trust.

Kernel work. $M \le U$ on all 3,124 queries, as Corollary 5.3 guarantees; the median M/U is 1.5×10^{-4}. Empirically the stronger $\max(M_R, M_S, M_T) \le U$ also held on every query, which Proposition 5.2 does not guarantee. All worst-case bounds are far from real outputs: the median of the best of the four bounds over Out is about 10^4.

A pre-registered success criterion. Before running the experiment we recorded a failure condition: if the certified σ is close to σ but the certified bound is at least the ℓ_2 bound on over 90% of real predicates, the spectral bound has no planner value. The criterion was stated per predicate; we report it per predicate view and per query. A view is a dataset, a predicate and, for the six SNAP graphs, the directed or symmetrized variant. Counting a view as "no value" when $U_{\text{cert}} \ge \ell_2$ on every query that uses it, 22 of 134 views qualify (16%; merging the two variants of each SNAP graph, 16 of 128 predicates); per query, $U_{\text{cert}} \ge \ell_2$ on 81.2% (2,537 of 3,124). So the criterion as registered is not met. Against

the stronger comparator $\min(\mathrm{AGM}, \ell_2, \mathrm{degree})$, which we chose after the fact, U_{cert} is not better on 93.5% of queries (2,921 of 3,124), above the 90% threshold; 61 of 134 views (46%; 55 of 128 predicates with the SNAP variants merged) never see U_{cert} win. In spirit, then, the criterion is met against the best prior bound. Our reading, for the selected nonzero canonical queries: as a cardinality estimate, U adds modest value as one more term in a minimum with ℓ_2 and degree bounds, not as a replacement. Its main value is the optimality guarantee of Section 5, which needs no claim about tightness on real data.

B. Engine details and measurements

The triangle operator in Leatherback, which at the time of writing is on a development branch together with both admission stages below, answers a run of exactly three

triple patterns forming a triangle with three distinct variables and constant predicates, in any of the eight edge directions, the same predicate allowed on every edge. Its contract is "exact or decline, and no allocation": it returns exactly the bag the greedy left-deep path would produce (in another order), or declines before any answer is visible. The driver edge $E_1 = (x, y)$ is scanned in PSO order; the other two edges are keyed by y and by x through PSO or POS, which covers all eight directions; for each E_1 row the runs $N_2(y)$ and $N_3(x)$ are found by monotone galloping and intersected from the shorter side. Per E_1 row the operator's ideal candidate budget is $\min(|N_2(y)|, |N_3(x)|)$, so the budget summed over the driver is M_π; the entries it actually walks are at most this budget, and can be fewer, because the walk stops when the other run is exhausted. The operator's total work is the walked entries plus scan, galloping and seek overheads. The Lean bridge of Appendix D proves, for a replica of the operator, that the precharged budget equals M_π and that the walked count is at most it; with three candidates on each side it exhibits a row where the budget is 3 and the walk reads one entry.

Planning today. For each candidate driver the planner reads 64 evenly strided E_1 rows (positions $\lfloor s\ |E_1|\ /64 \rfloor$, $s < 64$), estimates $\widehat{M} = \lfloor |E_1| \cdot \mathrm{mean}(\min(d_2, d_3)) \rfloor$ from CSR degree reads, and picks the driver minimizing $2|E_1| + \widehat{M}$; the $2|E_1|$ term blocks the star example of Proposition 5.5(a). Admission has two stages. *Stage 1*: the operator runs only when the three patterns together hold at least 2048 triples (the *floor*) and $\rho\big(2|E_1| + \widehat{M} + p\big) \le n_0 + \hat{P}$ (the *gate*, $\rho = 1$), where p is the proof price (the length of the ranges the kernel would still have to certify; 0 on a warm store), n_0 is the length of the first pattern in the greedy plan's order and $\hat{P}$ is the sampled size of greedy's first 2-path. *Stage 2* (the *rescue*; at the time of writing the kernel, both stages and the policy knob below are on a development branch of Leatherback, not on its main branch): a triangle that stage 1 declined on a warm stored view is priced again in units calibrated on the knowledge-graph runs below (one unit is about 2 ns): with $L_i = \lfloor \log_2(|E_i|\ /\ |E_1|) \rfloor$ when $|E_i| > |E_1|$ and 0 otherwise, and $f = 1$ when greedy's first join is on the first pattern's object (so it probes per seed row instead of merging),

$$K_L = |E_1|\ (4 + L_2 + 3L_3) + 20\widehat{M}, \quad G_L = n_0(9 + 31f) + 26\hat{P},$$

$$K_S = |E_1|\ (8 + 2L_2 + 6L_3) + 30\widehat{M}, \quad G_S = n_0(9 + 31f).$$

If $64\hat{P} \geq n_0$, the driver minimizing K_L is admitted when it has at least 16384 rows and $3K_L \leq 2G_L$; failing that, the driver minimizing K_S is admitted when $K_S \leq G_S$. Both minimums range over all three candidates (ties to the lower position); a candidate stage 1 left unsampled is sampled only while its zero-walk cost could still change the decision, which leaves the decision equal to the rule over all three. Nothing stage 1 accepted changes. Corollary 5.3 certifies the exact-M choice and the statistics-owner choice (the rotation owning the argmin term of U), **not** the sampled estimator, and Proposition 6.1 shows that no estimator that reads a fixed set of positions can be certified.

Proof of Proposition 6.1.

Take ordered constants $x_1 < \cdots < x_m$, $y_1 < \cdots < y_m$ and $z_0 < z_1 < \cdots < z_{2m-1}$. Since $R = X \times Y$ is scanned in (x, y) order, the row at position q is $\big(x_{\lfloor q/m \rfloor + 1}, y_{(q \bmod m)+1}\big)$. Let $J = \{q \bmod m : q \in P(m^2)\}$, $j = |J|$, so $1 \leq j \leq k < m$, and call y_i *dead* when $i - 1 \in J$. Put $Z_S = \{z_0\} \cup \{z_{2i} : 1 \leq i < m\}$ and $Z_T = \{z_0\} \cup \{z_{2i+1} : 1 \leq i < m\}$: two runs of length m that share only z_0 and interleave. Let

$$R = X \times Y, \quad S = \{(y, z) : y \text{ live}, z \in Z_S\} \cup P_S, \quad T = Z_T \times X \cup P_T,$$

where P_S, P_T are tuples on fresh constants that bring S and T to $m^2 + 1$ tuples each and join nothing. The query lists its patterns as $R(x, y), T(z, x), S(y, z)$, so greedy's first join is the join of R and T on x. Then $d_S^+(y) = m$ for live y and 0 for dead y; $d_T^-(x) = m$ and $d_R^+(x) = d_R^-(y) = m$ on X and Y; $d_T^+(z) = m$ on Z_T and $d_S^-(z) = m - j$ on Z_S.

(1) Every sampled row of R has a dead y, so $\min(d_S^+(y), d_T^-(x)) = 0$ and $\widehat{M}_R = 0$. Since $|S| = |T| = |R| + 1$, $2|R| + 0 < 2|E| + \widehat{M}_E$ for $E \in \{S, T\}$, and the planner picks R. Every sampled $d_T^-(x)$ is m, so $n_0 + \hat{P} = m^2 + m^3$, and the gate $\rho \cdot 2m^2 \leq m^2 + m^3$ holds for $\rho \leq (m + 1)/2$.

(2) $M_R = \sum_{(x,y) \in R} \min(d_S^+(y), d_T^-(x)) = m(m - j)m$. An S-tuple (y, z) contributes to $M_S = \sum \min(d_T^+(z), d_R^-(y))$ only if $z \in Z_T$, i.e. $z = z_0$: $m - j$ tuples with $\min(m, m) = m$, so $M_S = m(m - j)$. Likewise $M_T = \sum \min(d_R^+(x), d_S^-(z))$ is carried by the m tuples (z_0, x), each with $\min(m, m - j) = m - j$, so $M_T = m(m - j)$. The triangles are the (x, y, z_0) with y live, so $\text{Out} = m(m - j) = \min_\pi M_\pi$. For each of the $m(m - j)$ live rows of R the operator intersects $N_2(y) = Z_S$ with $N_3(x) = Z_T$; walking either run, every element below the other run's last element is read before that run is exhausted, which is at least $m - 1$ elements.

(3) $2|R| + M_R \geq m^2(m - j) \geq m^3/2$ for $j \leq k \leq m/2$, while $2|S| + M_S = 2m^2 + 2 + m(m - j) \leq 4m^2$; divide. Finally $N = |R| + |S| + |T| = 3m^2 + 2$, so $m = \Theta\big(N^{1/2}\big)$. □

Remark B.1 (what the proposition does not cover)**.** The family is not semijoin-reduced (dead y's and the padding join nothing), and the samples are placed without looking at the data. The construction with $64 \mid m$ and the stride, where $J = \{0\}$, was first found by a Codex CLI agent (GPT-6-Astra) during the search for the admission rule (Section 9); the golden-ratio grid it suggested as a fix is covered by the proposition. Exact counting

($O\left(\sum_i r_i\right)$, the *exact* policy) removes the failure at the planning cost measured in Table 8. Randomized positions, grids chosen from the data, and semijoin-reduced inputs remain open (Section 8).

Possible extensions. (i) Store per predicate, at compaction, the certified σ together with sizes and maximum degrees, so the planner can evaluate U before reading data and use it as an admission or deadline guard. (ii) Choose the driver by exact M_π when E_1 is small after selection ($O(|R|)$ by the formula of Section 2). (iii) The per-binding variant of Section 3 with the unbuffered price-then-run scheduler of Proposition C.2 in its re-pricing variant (Proposition C.2(4), $O(1)$ working memory), which is compatible with the no-allocation contract; the earlier idea of running the three roots round-robin with private output buffers is not, and is not used.

Measurements. The operator, on a development branch of Leatherback, was compared with the greedy left-deep plan of the main branch at the time: both ReleaseFast, each built from a clean source export with a private compiler cache, three interleaved rounds of five runs, medians of 15 in-process execution times on one machine. Answers were identical in all 50 comparisons. The skewed workload is a generated graph with Zipf(1.5) degrees: the generator writes one million edge statements, which deduplicate to 101,411 distinct stored triples because the sampler repeats hub pairs. "Uniform" writes one million statements with uniform degrees (999,983 distinct). Medians are of in-process execution times; the speedups quoted in Section 1 are ratios of the displayed medians, $878.5/27.3 = 8785/273 \approx 32.2$ for rows and $941.8/8.29 = 94180/829 \approx 113.6$ for `COUNT(*)`, i.e. $114\times$ after rounding. Seeds, data hashes, the raw samples (1,200 timings), the answer comparisons and the machine (Apple M4 Max, 16 cores, 128 GiB, macOS 26.7, Zig 0.16.0) are in `artifacts/engine/` of the paper repository, with the build, data-generation and store-build commands (reconstructed from the logs after the runs). Once Leatherback is public, the repository will name the engine versions these runs correspond to.

Table 7: Engine A/B, medians of 15 runs. On skewed data the operator replaces the wedge blow-up of the left-deep plan; where the planner's gate declines, the cost is the gate itself (largest relative change: a 3.0 µs empty selective triangle became 4.9 µs).

Data	Query	Greedy	Operator	Ratio
skew, `.lb`	triangle rows (466,168)	878.5 ms	27.3 ms	0.031
skew, `.lb`	cyclic triangle rows (466,827)	881.5 ms	24.5 ms	0.028
skew, `.lb`	triangle `COUNT(*)`	941.8 ms	8.29 ms	0.009
skew, in-memory	triangle rows / `COUNT(*)`	1046 / 1047 ms	32.1 / 16.4 ms	0.031 / 0.016
uniform, `.lb`	triangle rows (gate declines)	41.3 ms	42.4 ms	1.03
uniform, `.lb`	triangle `COUNT(*)` (gate declines)	69.3 ms	68.6 ms	0.99
100k synth	4-cycle `adv_cycle_quad` (not a triangle)	381.5 µs	381.6 µs	1.00
skew, `.lb`	non-triangle chain	21.8 ms	22.9 ms	1.05

Driver policies on the knowledge-graph triangles. To test what Corollary 5.3 says about driver choice, a benchmarking-only planner knob (the default is the planner above, and unit tests pin that its plans are unchanged) runs each query under six policies: *sampled*

(the default); *exact*, which counts M_π for every candidate driver with a full degree pass charged to planning, takes the argmin of its objective, and applies the same admission rules; *forced:k* for each of the three drivers, which skips the floor and the gate; and *greedy*, with the operator disabled. There are two runs of the same harness. *Run A* (a development build of Leatherback with stage 1 only) has an exact policy that minimizes M_π alone. *Run B* (a later development build) adds the rescue, and its exact policy minimizes the planner's objective $2|E_1| + M_\pi$ and applies stage 1 and the rescue to the exact budgets, so that against *sampled* only the sampling error differs. The queries are those of Appendix A, run as `COUNT(*)`: all 871 CoDEx-L queries, and samples of 120 sci50 and 250 wikidata5m queries, stratified to equal counts per decile of $\log M$ (seed 3313). The setup is ReleaseFast on the machine above with one worker thread; each query has an untimed probe and one untimed warm-up per policy, then 5 timed rounds with the policies interleaved, and we report medians of elapsed times. In each run answers agree across the six policies in all 7,446 (query, policy) rows; the engine's exact M_π equals the independently computed values of Appendix A for every rotation of all 1,241 queries; and $\min_\pi M_\pi \le U$ holds on every query, as Corollary 5.3 requires (median $\min_\pi M_\pi/U$: $6.1 \cdot 10^{-4}$, $2.5 \cdot 10^{-3}$, $4.2 \cdot 10^{-5}$; maximum 0.57, 0.056, 0.47). In both runs one greedy row on wikidata5m records the 30 s timeout rather than a completed timing (the operator took 9.4 ms in run A); it is excluded from every greedy ratio, which therefore understate the gap.

We distinguish three time measures. The *forced time* of a driver is the elapsed time of the run with that driver forced: planning, kernel and result, not a metered count of kernel work. The *oracle* is the best forced driver, chosen after observing all three forced times; it is not a deployable policy. *Sampled choice without the gate* is the forced time of the driver the sampled planner picks, i.e. what the default planner would take if it never declined. We did not meter walked entries in these runs; M_π is the model budget, computed exactly from degree arrays.

Table 8: Driver policies on the knowledge-graph triangles of Appendix A (`COUNT(*)`, medians of 5 interleaved rounds). Run A: exact policy = argmin M_π, stage 1 only. Run B: exact policy = argmin $(2|E_1| + M_\pi)$, rescue live; in run B "pick" is the driver that ran, which on rescued queries is the rescue's choice for both policies. "Forced-time gain" is the forced time of the sampled pick divided by that of the exact pick; each row compares times within its own run. Greedy ratios exclude the one greedy timeout.

Quantity	**Run**	**CoDEx-L (871)**	**sci50 (120)**	**wikidata5m (250)**
Driver choice				
worst ÷ best forced driver (median / p90 / max)	A	2.7 / 8.6 / 47	2.19 / 6.53 / 65.7	4.99 / 28.2 / 132
fastest forced driver = argmin M_π / argmin $2\|E_1\| + M_\pi$	A	515 / 532	71 / 71	140 / 151
same	B	518 / 586	68 / 84	140 / 168
exact pick ≠ sampled pick, exact by M_π	A	405 (46.5%)	53 (44.2%)	126 (50.4%)

Quantity	**Run**	**CoDEx-L (871)**	**sci50 (120)**	**wikidata5m (250)**
exact pick $\neq$ sampled pick, exact by $2\|E_1\| + M_\pi$	B	55 (6.3%)	6 (5.0%)	8 (3.2%)
fastest forced driver = sampled / exact pick	A	517 / 501	66 / 62	147 / 139
same	B	599 / 618	86 / 83	169 / 171
forced-time gain of the exact pick (median / p90; min / max)	A	1 / 1.56; 0.093 / 7.19	1 / 1.27; 0.293 / 14.2	1 / 2.15; 0.024 / 15.2
same	B	1 / 1; 0.352 / 6.50	1 / 1; 0.295 / 5.55	1 / 1; 0.499 / 5.49
exact ÷ sampled time, end to end (median / p10)	A	26.2 / 7.9	39.1 / 7.3	69.8 / 8.7
same	B	36.5 / 10.7	62.8 / 22.7	90.1 / 12.7
Admission (stage 1)				
stage-1 declines (sampled / exact)	A	571 / 617	87 / 102	166 / 191
same	B	571 / 561	87 / 88	166 / 163
greedy ÷ operator, all queries (median / p90)	A	1.05 / 26.7	1.06 / 13.2	1.21 / 36.5
greedy ÷ operator where accepted (n; median / p90 / min)	A	300; 10.1 / 153 / 1.23	33; 8.76 / 80.9 / 1.23	83; 10.2 / 751 / 0.508
declined: greedy ÷ sampled choice without the gate (median / p90)	A	1.40 / 3.38	2.20 / 9.78	2.06 / 15.9
of those, $> 1.5\times$	A	262/571	64/87	105/166
declined: greedy ÷ oracle (median / p90)	A	1.83 / 3.92	2.8 / 14.9	3.53 / 20.2
of those, $> 1.5\times$	A	350/571	70/87	116/166

Table 8 answers three questions. *Does exact pricing change the plan?* Only when it prices something else. Minimizing M_π alone (run A) picks a different driver than the sampled planner on 44–50% of queries; minimizing the planner's own objective $2|E_1| + M_\pi$ (run B) does so on 3–6%, and on 23 of 281, 1 of 30 and 4 of 76 of the queries both policies admit at stage 1. Most of run A's disagreement was the objective, not sampling error. *Does that help?* Not on this workload, descriptively (we did no uncertainty analysis): in run B the forced-time gain of the exact pick has median and p90 exactly 1 (minimum 0.30–0.50, maximum 5.5–6.5), and the exact pick is the fastest forced driver about as often as the sampled one (618 vs 599, 83 vs 86, 171 vs 169). End to end the exact policy is 36.5–90.1 $\times$ slower at the median (run B), because its three full degree passes dominate these mostly sub-millisecond queries. So Corollary 5.3′s guarantee for the exact choice holds on every query here, and the sampled estimate does as well on this workload, although Proposition 6.1 shows it has no worst-case guarantee. *Is rotation the right lever?* The forced-driver spread says the driver matters (worst over best: median 2.2–5.0 $\times$, up to 132 $\times$), and where stage 1 admits the operator it beats the greedy plan by a median 8.8–10.2 $\times$ (one regression, 0.51 $\times$, on wikidata5m). Admission, not driver choice, was the binding constraint: stage 1 sends 66–

73% of queries to greedy. On those, the oracle would have beaten greedy by a median 1.8 –3.5 × and by more than 1.5 × on 61–80% of them, but the oracle is not a policy. The deployable comparison is the sampled choice without the gate: a median 1.40–2.20 × faster than greedy, and more than 1.5 × faster on 46–74% of the declined queries. That is the opportunity the rescue stage below goes after.

Caveats. Most CoDEx-L runs take 0.1–1 ms, which is noisy at the scale of driver choice; restricted to queries whose fastest forced run took at least 1 ms, the fastest driver in run A is the argmin of M_π on 3/7, 40/53 and 59/93 queries and of $2|E_1| + M_\pi$ on 6/7, 32/53 and 54/93. The wikidata5m process peaked at a footprint of 85.6 GB in run A and 91.7 GB in run B (the peak memory footprint reported by the operating system for the whole benchmark process), from greedy's materialized 2-paths; the operator itself allocates nothing per row. On this graph the left-deep plan therefore costs memory as well as time, and the timing ratios do not show that cost. The raw rows, queries, scripts and a summary that the scripts rebuild are in `artifacts/engine-kg/`.

The rescue stage. The two sides of stage 1 count different units: a greedy unit is a materialized 2-path row, a kernel unit a scanned edge, a seek or a walked entry. On run A, per-unit medians were 29, 62 and 57 ns for greedy and 7.4, 4.3 and 5.8 ns for the kernel, so stage 1 is several times too kind to greedy. Loosening its ratio does not separate the declined queries the kernel wins from those it loses; two structural signals do. One is the depth of the kernel's per-row seeks into E_2 (the L_i terms). The other is whether greedy's first join probes per seed row (the f term). The rule (see *Planning today*) was chosen by an offline search over run A's rows for the most total time saved subject to at most two regressions above 1.2 × per dataset and none above 1.5 × (the rule chosen had a worst slowdown of 1.26 × on those rows). Table 9 measures it in a third run.

Run C (a development build that contains the pruning fix described below) adds a bench-only policy *stage1*: the same sampled estimate and stage-1 gate, with no rescue, i.e. the planner before the rescue. It runs in the same process as the other six policies on the same queries, with 6 timed rounds, and the order of the seven policies is shuffled in every timed round (seeded by the query id). A first, fixed-order run showed a position bias: *stage1* always ran right after *sampled* and was a median 5–8% faster on queries where both run the identical plan, which produced 74, 12 and 36 spurious "regressions" above 1.2 ×; we report only the shuffled run. In run C the stage-1 decision (outcome and driver) of *sampled* equals the *stage1* arm's on every query, so the two arms differ only on the rescued queries. An earlier draft reported this comparison against an imputed baseline (run B, with the stage-1 time of a rescued query reconstructed from greedy's time): totals 189.9 → 144.8, 1004.0 → 546.7 and 1839.4 → 1270.2 ms. The measured numbers of Table 9 replace them.

Table 9: The rescue stage on the knowledge-graph triangles against the stage-1 planner measured in the same run (run C; medians of 6 rounds with the policy order shuffled per round). The median speedup is about 1 because 57–83% of queries keep their stage-1 plan; the gains sit in the rescued 17–43%. "Unchanged plan" are the queries stage 1 admits or both stages decline: both arms run the identical plan there, so their ratios measure run-to-run noise (plus, on declined

queries, the rescue's extra sampling). Worst ratio is rescue time over stage-1 time. wikidata5m includes all 250 queries (the greedy timeout affects only the greedy arm). The rule was fitted to these queries: the gains are in-sample.

Quantity (run C)	**CoDEx-L (871)**	**sci50 (120)**	**wikidata5m (250)**
admitted at stage 1 / rescued / declined	300 / 216 / 355	33 / 52 / 35	84 / 43 / 123
decline rate, stage 1 → both stages	65.6% → 40.8%	72.5% → 29.2%	66.4% → 49.2%
total ms, stage 1 → both stages (measured)	173.6 → 137.9 (−20.6%)	1041.2 → 532.6 (−48.8%)	1959.0 → 1322.3 (−32.5%)
speedup over stage 1, all queries (median / p90 / max)	1.01 / 5.30 / 24.6	1.11 / 8.37 / 30.5	1.02 / 4.80 / 153
rescued queries: total ms, stage 1 → rescue	52.8 → 17.2	837.1 → 331.4	735.6 → 99.6
rescued queries $> 1.2\times$ slower (worst ratio)	0 (1.14)	0 (1.07)	0 (0.78)
unchanged plan: both-arm time ratio (median)	1.000	1.005	1.002
$> 1.2\times$ slower (noise floor)	22 / 655	2 / 68	12 / 207
exact policy ($2\lvert E_1\rvert + M_\pi$): rescued / declined	255 / 306	59 / 29	61 / 102
answer-equality flags set across policies	6097/6097 rows	840/840 rows	1750/1750 rows

The rule is in-sample: it was searched on the same 1,240 queries it is measured on (the 1,241 less the wikidata5m query whose greedy run timed out), so Table 9 shows that it does what it was fitted to do on a fresh execution, not that it generalizes. In run B, which carried the forced-driver arms under the same rule, rescued queries ran within a median 1.02–1.15 × of the oracle (p90 1.32–2.24). The only out-of-sample check is the three synthetic stores of Table 7 (skewed, uniform, and a 100k "medium" store), where every decision and driver is unchanged and the sorted answers are identical in 21 of 21 (store, query) pairs. In a shuffled 10-round run of these stores the rescue arm over the stage-1 arm is 0.99–1.03 on every query above 1 ms; the exceptions are sub-200 µs queries (a 4.7 vs 9.0 µs pair with an identical plan that we cannot explain, and the medium store's two declined triangles, 1.04 and 1.10, which pay the rescue's sampling). An earlier draft reported a 1.058 × slowdown on the uniform `COUNT(*)` query of about 54 ms from a single old and new run. Rerun as 10 alternating process pairs of the pre-rescue and the fixed binary, 5 runs each, the per-pair ratio has median 0.995 (range 0.970–1.024; the new binary slower in 4 of 10), and the pooled medians are 52.48 and 52.58 ms (1.002, run-to-run standard deviation about 1.5 ms): the 1.058 did not reproduce. These checks show the rescue does not change decisions there; they do not test the gains.

Candidate pruning. To avoid sampling hopeless candidates, the implementation measured in run B skipped a candidate whose zero-walk cost floors already failed, and treated a candidate below the 16384-row minimum as unable to pass the large branch. The searched rule instead takes the large-cost argmin over all candidates and applies the minimum

afterwards, so a small candidate can win the argmin and veto a larger one. A model-run audit (Codex CLI with GPT-6-Astra; Appendix G) built a realizable 31,000-tuple warm-store instance (sizes $1000, 10000, 20000$, no triangles) on which that implementation admits the third driver while the all-candidate rule declines (answers were unaffected). The fix replaces the pruning: a candidate is sampled while its zero-walk floor could still take a branch's argmin within that branch's budget, so the decision is the rule over all three candidates, for at most two extra samples. A second model-run audit (a Claude subagent) re-derived this from the code, matched a port of it against the full rule on $2.8 \cdot 10^6$ exhaustive and random cases, and confirmed that the first audit's instance, added as a test, fails on the earlier implementation and passes on the fix. On the knowledge-graph queries the fix

changes one decision: codex-l-0267, which the pruned rule rescued (145 µs against greedy's 1.67 ms), is now declined (1.60 ms), because its 8,023-row candidate wins the large argmin and fails the 16384-row minimum. That is the searched rule as written; it shows the cost of applying the minimum after the argmin. By design the rescue never runs on a cold store (proof price > 0) or on the in-memory graph arm, whose probe-first join is a postings lookup. The machine carried other load throughout (load average 11–17 on 16 cores in runs A and B, 9–14 in run C); every comparison is within one run with the policies interleaved per query, and greedy's median time drifted by a factor 1.03, 1.01 and 0.98 between runs A and B. The wikidata5m process peaked at a footprint of 116 GB in run C. What is left after the rescue is smaller but not zero: in run B, on the queries both stages still decline, the sampled choice without the gate beats greedy by more than $1.5\times$ on 106, 19 and 67 of them.

The engine's intersection routine in the sorted model.

Remark B.2 (the engine's routine on relabelled G8)**.** The engine's `intersectEachBy` (`src/util/sorted_runs.zig`) walks the shorter run and, on equal lengths, its first argument; the triangle call site passes the run of the walked tuple's far endpoint first, so option R calls $(Z_1, T(\cdot, x))$ and option T calls $\left(Y_{2,z}, Y_0\right)$. On an alternating pair of length-s runs its routine (a 16-step linear prefix, then doubling, then binary search) performs $6s-3$ key predicates in the walking-first orientation ($(s-1)$ lookahead equalities, $2s-1$ disorder checks, $2s-1$ gallop comparisons and s match equalities) and $6s-5$ reversed, with work meter $3s-1$ either way. On $\mathcal{U}(m)$ relabelled as in Theorem 3.25, per x it costs $q_R(x) = 6m^5 - 2m^2$ and $q_T(x) = 6m^5 + m^4 - 4m^3$ (comparisons plus outer visits); the optimum picks T at $m = 2, 3$ and R for $m \geq 4$, and each root costs $Q(m) = m^4 \min(q_R, q_T) + m^8 + m^5 = 3104, 122715, 1631232, 12081250$ for $m = 2, ..., 5$, i.e. $Q(m) = 6m^9 + m^8 - 2m^6 + m^5$ for $m \geq 4$: an exact degree-9 polynomial against N of degree 8. These are modelled per-binding optima of the port's counts with omniscient plan choice; relation seeks, sorting and planning are excluded, and the engine itself runs one sampled driver (Appendix B), so this confirms the exponent, not an engine timing.

[Proved] (exact counts from a port of the engine routine; Appendix G)

Not reported: the per-binding scheduler of Proposition C.2 (not implemented), the Theorem 3.16 families at scale, first-row and p99 latency (the operator keeps no output buffer, but price-then-run delivers its first row only after pricing), metered kernel work split into pricing and execution (all times above are elapsed times), and any out-of-sample workload for the rescue stage.

C. Price-then-run, and degree metadata

C.1. Price-then-run: paying for the statistics

The per-binding choice needs $c^*(u)$, which is not known in advance. In a sorted-permutation store every degree is an $O(1)$ offset read, so both options can be priced without executing them.

Lockstep pricing at one binding. Let P, Q be the two options at binding u. Read the two degrees of u that the per-edge costs need. Reveal the per-edge costs of the two options one at a time; each *advance* is one neighbour read and one degree read (2 unit operations, one per-edge unit). Keep running sums t_P, t_Q. Always consider the option with the smaller running sum (ties to P; a completed option's running sum is its total): if it is complete, stop and choose it; otherwise advance it. Bindings are enumerated by merging the key lists of the two CSR indexes, at one unit per binding.

Lemma C.1 (lockstep pricing)**.** *At a binding u with option costs $c_P(u), c_Q(u)$, lockstep pricing chooses an option of cost exactly $c^*(u)$ after at most $\min\big(2c^*(u)+1,\, d_P(u)+d_Q(u)\big)$ advances, where $d_P(u), d_Q(u)$ are the numbers of tuples the two options walk. At any intermediate moment, $\min\big(t_P, t_Q\big) \le c^*(u)$ and at most $2\min\big(t_P, t_Q\big)+1$ advances have been made.*

[Proved]

Proof. Every per-edge cost is at least 1, and $t_P \le c_P$, $t_Q \le c_Q$. At any moment let $m = \min\big(t_P, t_Q\big)$. The side holding m was advanced at most m times. Each advance of the other side started from a running sum at most the first side's running sum at that time, hence at most m; these starting values are distinct integers in $[0, m]$, so there are at most $m+1$ of them. When the procedure stops, the chosen total is at most the other running sum, hence at most the other total, so it equals $c^*(u) = m$. Each advance also consumes a fresh tuple. □

The scheduler. Run one lockstep pass per root $v \in \{x, y, z\}$ over the bindings of v, interleaved as follows. Pass v maintains the lower bound

$$L_v = \sum_{\text{finished bindings } u} c^*(u) + \min\big(t_P, t_Q\big)_{\text{current binding}},$$

which never decreases and satisfies $L_v \le C_v$, with equality once pass v is finished. Fix the order $x < y < z$ of the roots. At each step, take the pass whose key $(L_v, [v \text{ unfinished}], v)$ is lexicographically smallest: the smallest L_v, ties going to a finished pass, and remaining ties to the earlier root. If it is finished, stop and call it v^*; otherwise perform one step of

it (one advance, or finishing a binding and reading the degrees of the next). Then execute the recorded option at every binding of v^*.

Proposition C.2 (price-then-run)**.** *With $O(1)$-time degree reads, the scheduler above satisfies:*

1. *$C_{v^*} = C$.*
2. *Every triangle is output exactly once.*
3. *In per-edge cost units, the total work is at most $C + \min(6C, 2N) + 4n + O(1)$, where $n = n_x + n_y + n_z \le 2N$. In particular it is at most $7C + O(n)$ and at most $C + O(N)$. In the unit operations of Section 2 it is at most $2C + \min(12C, 4N) + O(n)$.*
4. *It is deterministic and uses no sampling. Since v^* is known only when the scheduler stops, every pass records its chosen options while pricing: one bit per priced binding, at most $n \le 2N$ bits in total. A finished pass v may discard its bits once another* finished *pass v' beats it in the stop rule's order, i.e. $(C_{v'}, v') < (C_v, v)$ lexicographically: a strictly smaller total, or an equal total and an earlier root. Such a v is never v^*, because v' stays finished with $L_{v'} = C_{v'}$ and so has a smaller key than v at every later step. The non-strict rule "another finished pass has a total no larger than its own" is unsafe: two finished passes with equal totals would each discard their bits because of the other. A finished pass that is merely no longer minimal may still win, because an unfinished pass with smaller L_v can later rise above it (e.g. a finished pass with $L_A = C_A = 10$ and an unfinished one with $L_B = 9 < C_B = 20$: A wins once L_B passes 10). The bits of the losing passes are discarded at the end. The variant that re-prices v^* instead uses $O(1)$ working memory and adds at most $\min(2C, 2N) + O(n_{v^*})$ per-edge units. Neither needs an output buffer, and both serve enumeration and counting alike.*
5. *Pricing every root fully before running (no scheduler), then re-pricing and executing the root with the smallest priced total, is also correct and costs at most $2C + 15N$ unit operations with $O(1)$ working memory.*

[Proved] (simulated on random instances; Appendix G)

Proof. (1) When the scheduler stops, $C_{v^*} = L_{v^*} \le L_v \le C_v$ for every root v.

(2) Every triangle has exactly one vertex in the root variable, and either option at that vertex finds every triangle through it, once.

(3) A step of pass v is taken only when $L_v \le \min_{v'} L_{v'} \le C$. By Lemma C.1, before its last step pass v has made at most $\sum_{\text{finished } u}(2c^*(u) + 1) + 2m + 1 \le 2L_v + n_v + 1$ advances, plus three units per binding for enumeration and degree reads. So pricing costs at most $\sum_v (2C + 4n_v + O(1)) = 6C + 4n + O(1)$. Each pass is a prefix of full lockstep pricing, which by Lemma C.1 makes at most $d_P(u) + d_Q(u)$ advances per binding; each relation is walked by two roots, so all advances total at most $2N$. Execution costs $C_{v^*} = C$ per-edge units, i.e. at most $2C$ unit operations, and each advance is 2 unit operations. $n \le 2N$ because every binding with an incident tuple is an endpoint of one.

(4) Re-pricing v^* repeats its pass alone: at most $\sum_u (2c^*(u) + 1) \le 2C + n_{v^*}$ advances, and at most $|P_{v^*}| + |Q_{v^*}| \le N$, plus overhead $O(n_{v^*})$.

(5) Pricing root v fully, whose options walk relations P_v, Q_v, costs at most $\sum_u \big(3 + 2\big(d_P(u) + d_Q(u)\big)\big) \le 3n_v + 2(|P_v| + |Q_v|)$ unit operations by Lemma C.1; summed over the three roots this is at most $6N + 4N = 10N$. Re-pricing v^* costs at most $3n_{v^*} + 2(|P_{v^*}| + |Q_{v^*}|) \le 5N$, and executing it at most $2C$. The total is at most $2C + 15N$. □

Remark C.3 (constants, and why the minimum-L rule matters)**.** The constant 6 in (3) is tight for this scheduler (this argument is from the GPT-5.6-sol Pro review of the first draft, Section 9): on the union of three rotations of an instance with complete $R = X_p \times Y_q$, complete T between Z_q and X_p and empty S, every C_v equals C, and measured pricing $/C$ is 7.97, 6.50, 6.25 at $p = q = 8, 32, 64$, tending to 6. This is exact, not only measured: since all three roots have the same final cost $C = pq$, the minimum-L rule cannot stop before every L_v reaches C, and each binding's two unit-cost options of length q take $2q$ lockstep advances, so each root needs $2C$ advances and pricing needs $6C$ plus lower-order binding overhead. On random instances the worst measured $(\text{pricing} - 4n)/C$ in the runs of `artifacts/verify/thm1/rnd.py` was between 3.0 and 3.5; an outside re-implementation (Appendix G) found 3.00, 3.20 or 4.00 on the same 3,996 random instances, depending on the order in which a binding's tuples are walked (a set iteration order that varies with Python's per-process string hashing). The point of the scheduler over (5) is that pricing is bounded by $O(C + n)$ rather than $O(N)$: with $R = X \times Y$, T complete between Z and X ($m \times m$ each) and S a single stray tuple, $C = 0$ and $N = 2m^2 + 1$. Counting one unit per advance and three per binding started (enumeration and the two degree reads, as in the proof of (3)), at $m = 4, 8, 16, 32$ the scheduler's pricing was $26, 38, 62, 110$ units ($3m + 14 = \Theta\big(\sqrt{N}\big)$), where pricing every root fully cost $77, 213, 677, 2373$ ($\Theta(N)$). Here the stray tuple is (y_0, z') with z' a fresh vertex. The rule "stop at the first pass to finish" is not a substitute for the minimum-L rule: under round-robin it selected a root with $C_v > C$ on about 31% of random instances. With execution-only access (costs unknown until paid), the usual two-option adversary forces a factor close to 2; we do not prove this here. The point of Proposition C.2 is that $O(1)$ CSR degree reads make the per-binding choice cost only a constant factor, or an additive $O(N)$, beyond C.

C.2. Degree metadata alone is the wrong class

One might hope to prove optimality against the worst output over all instances sharing the labeled degree sequences of the input. That class is too fine.

Proposition C.4 (Ferrers profiles)**.** *For every $k \ge 1$ there is an instance with $N = 9k^2$ in which each of R, S, T is the unique relation with its labeled degree sequences,* $\mathrm{Out} = 0$*, and $C = k^3 + k^2$,* $\min(M_R, M_S, M_T) = k^3$*, while all six squared-degree sums equal $5k^3$, so $B_2 = 5k^3$. So neither C nor $\min_\pi M_\pi$ is bounded by $c(N + \max \mathrm{Out})$ for a constant c, the maximum taken over the instances with the same labeled degrees, whereas $C = \Theta(N + B_2)$ here.*

[Proved]

Proof. Split each of X, Y, Z into three blocks of size k (indices $0, 1, 2$) and let each relation be a union of complete block pairs: $R = X_1 \times Y_2 \cup X_2 \times Y_1 \cup X_2 \times Y_2$, $S = Y_1 \times Z_1 \cup Y_1 \times$

$Z_2 \cup Y_2 \times Z_2$, $T = Z_0 \times X_1 \cup Z_1 \times X_1 \cup Z_0 \times X_2$. The row neighbourhoods of each relation are nested (R: $\varnothing \subset Y_2 \subset Y_1 \cup Y_2$; S: $\varnothing \subset Z_2 \subset Z_1 \cup Z_2$; T: $\varnothing \subset X_1 \subset X_1 \cup X_2$), so no 2×2 interchange (pairs u_1v_1, u_2v_2 present with u_1v_2, u_2v_1 absent) exists. By Ryser's interchange theorem [Rys57] all 0/1 matrices with given row and column sums are connected by interchanges, so each relation is the unique realization of its labeled degrees. Out $= 0$: from X_1 the path goes to Y_2, then Z_2, but T enters X_1 only from $Z_0 \cup Z_1$; from X_2 it reaches $Z_1 \cup Z_2$, but T enters X_2 only from Z_0. Degrees: $d_R^+ = k, 2k$ on X_1, X_2; $d_R^- = k, 2k$ on Y_1, Y_2; $d_S^+ = 2k, k$ on Y_1, Y_2; $d_S^- = k, 2k$ on Z_1, Z_2; $d_T^+ = 2k, k$ on Z_0, Z_1; $d_T^- = 2k, k$ on X_1, X_2; all others 0. Each degree sequence is one block of k vertices of degree k and one of degree $2k$, so each squared sum is $k^3 + 4k^3$. Hence $M_R = 3k^3$, $M_S = k^3$ (only $Y_1 \times Z_1$ contributes), $M_T = k^3$ (only $Z_1 \times X_1$). Root x: on X_1, $c_R = k(1 + k)$, $c_T = k + k(1 + k)$; on X_2, $c_T = k$; so $C_x = k^3 + 2k^2$. Root y: on Y_1, $c^* = k(1 + k)$; on Y_2, option S costs k; $C_y = k^3 + 2k^2$. Root z: Z_0 has no S-tuples and Z_2 no T-tuples (cost 0); on Z_1 both options cost $k(1 + k)$; so $C_z = k^3 + k^2 = C$. □

This is why Theorem 3.3 and Theorem 3.6 measure C against B_2 rather than against the worst output per degree profile. A side remark for engines: when every slice's CSR offsets form a nested (threshold) degree sequence, the instance, and so Out, is determined by the metadata.

D. Mechanization

The Lean package `LeatherbackSpectral` (Lean 4 [dMU21] v4.32.2, mathlib [Mat20] tag v4.32.2) was written in Leatherback's Lean workspace, which is otherwise Std-only (a check there enforces that only this library imports mathlib), and is published with this paper (Appendix H). It has nine modules (`Basic`, `OpNorm`, `Upper`, `Kernel`, `Lower`, `Matrix01`, `Degenerate`, `KernelBridge`, `Sharp`; about 1,770 lines in total) and builds with `warningAsError` and `autoImplicit = false`.

Model. Vertex sets are `Fintype`s with decidable equality; relations are `Finset`s of pairs; `adj R : Matrix X Y ℝ` is the 0/1 incidence matrix; `out R S T := trace (adj R * adj S * adj T)` (with `out_eq_card` identifying it with the triangle count); the norm is mathlib's ℓ_2 operator norm on matrices (`Matrix.instL2OpNormedAddCommGroup`); `Stats` holds the six reals, `Stats.U` is the nested minimum of the nine terms, and `Admissible` is the six inequalities. `kernelWork R S T` is M_R.

What is checked.

- Upper bound: `theoremB_upper` (all nine terms, constant 1), built from `bilinear_le`, `rowDeg_le_opNorm_sq`, `colDeg_le_opNorm_sq`, `opNorm_adj_sq_le_card` (Frobenius), `opNorm_adj_le_schur` (Schur test), `out_le_opNorm_mul_sqrt` and the rotation lemmas; and the same for arbitrary 0/1 matrices (`theoremB_upper_matrix`).
- Lower bound: `theoremB_lower` (the floored block instance is admissible and $U < 4 \cdot \text{Out}$), with `card_blk`, `opNorm_blk_eq`, `out_blk`, `Stats.real_shape_feasible`, `Stats.real_shape_value`, `Stats.real_shape_attains_U`, `Stats.block_params`; and `theoremB`, their conjunction.

- Sharp constant (Theorem 4.7, module `Sharp`): `sharpStats_regular`, `sharpStats_U`, `sharpStats_out_le_one`, `sharpStats_witness`, `sharp_family`, `factor_four_sharp` (for every $\varepsilon > 0$, regular statistics with every admissible $\mathrm{Out} \le 1$, an admissible instance with $\mathrm{Out} = 1$, and $U > 4 - \varepsilon$) and `theoremB_constant_sharp` ($\inf W/U = 1/4$ in two-sided form, without forming W).
- Degenerate regime: `theoremB_degenerate` (no sign condition assumed; vacuous when a bound is negative), `admissible_exists_iff_nonneg`, `theoremB_degenerate_attained`, `opNorm_adj_empty`, and `theoremB_cases`, whose statement includes the universal upper bound together with the regular lower witness or the nonregular zero-output alternative.
- Kernel: `kernelWork_eq_out_nest` (Lemma 5.1), `out_le_kernelWork`, `kernelWork_le_terms` (Proposition 5.2), `kernel_min_le_U` (Corollary 5.3), and the counterexample `nestS_can_raise_opNorm`.
- Kernel bridge (`KernelBridge`): the engine's Std-only Lean workspace contains an executable replica of the triangle kernel with three counters. `model` is an *ideal precharged candidate budget*: before each E_1 row is processed it is incremented by $\min(|N_2(y)|, |N_3(x)|)$. `walked` counts the entries of the chosen run that the replica actually consumes, and `charged` additionally counts galloping steps and seeks. The workspace proves that `model` equals a list-based sum `kernelWorkNat`, and walked $\le$ model (`run_walked_le_model`). `kernelWorkNat_eq_kernelWork` and `kernelWorkNat_eq_kernelWork_fin` identify that sum with `kernelWork`, and `run_model_eq_kernelWork` concludes that the budget `model` equals M_R on sorted, duplicate-free, bounded edge lists, for a result sink that never stops the walk. `run_model_le_terms` and `run_model_le_opNorm` transfer the five R-driven bounds of Proposition 5.2 to `model`, and `run_walked_le_kernelWork_real` and `run_walked_le_terms` give walked $\le M_R$ and the same five bounds for `walked`. Equality walked = model is **false** in general. With $E_1 = [(0,0)]$, $E_2 = [(0,100),(0,101),(0,102)]$ and $E_3 = [(0,1),(0,2),(0,3)]$ (sorted, duplicate-free) the replica precharges model $= 3 = M_R$, but its tie rule walks E_2, reads only 100, gallops E_3 to its end and stops with walked $= 1$ (`walked_lt_model_example`, proved by `decide +kernel` in the Std-only workspace, axioms `propext` and `Quot.sound` only). So the bridge identifies M_R with a conservative budget and bounds the entries actually read by it; it proves no bound on `charged`, which includes the galloping and seek overheads.

Hygiene. No `sorry`, `admit`, `native_decide` or custom axiom occurs. `LeatherbackSpectralAudit.lean` runs `#print axioms` on 80 declarations, each inside `#guard_msgs` expecting exactly `[propext, Classical.choice, Quot.sound]`, so the build fails if any of them acquires `sorryAx` or `Lean.ofReduceBool`; the Std-only audit `LeatherbackAudit.lean` checks the replica's `run_walked_le_model`, `run_walked_le_kernelWork` and `walked_lt_model_example` in the same way.

Building it. An early archive of the Lean sources, supplied to the first model-based review (Section 9), contained only the spectral sources and `lakefile.lean`, without `lean-toolchain`, `lake-manifest.json` or the sources of the workspace's other default targets, so it could not be built on its own. The paper repository instead carries the Lean workspace with every source the spectral library and the replica import. It pins Lean with `lean-toolchain` (`leanprover/lean4:v4.32.2`) and mathlib with `lake-manifest.json` (mathlib tag `v4.32.2`,

with its dependencies). With `elan` installed, `lake exe cache get` followed by `lake build LeatherbackSpectral LeatherbackSpectralAudit` kernel-checks every spectral module and the audit; the repository's README gives the commands for the replica's audit and the full workspace check. The spectral library, both audits and the full workspace built with exit code 0 on 2026-10-09; no Lean source changed after that build.

What is not checked. W is not a Lean object: Theorem 4.1 appears as a universal upper bound and an existential instance, and its feasibility sentence is checked only at $V = U$. Not formalized: Proposition 5.5, Corollary 4.8, all of Section 3 (Lemma 3.1, Lemma 3.2, Theorem 3.3, Corollary 3.5, Theorem 3.6, Corollary 3.7, Lemma 3.8, Theorem 3.10, Theorem 3.16, Lemma C.1, Proposition C.2, Proposition C.4), Appendix E, Remark 4.5, the footnote on nuclear norms in Section 4, and the experiments. The Zig code itself is not verified: the correspondence between the Lean replica and `triangle_join.zig` is trusted, as are the Zig planner's sampled cost estimator, the $O(\sum r_i)$ overhead and the galloping log factor (analysed on paper in Section 3.6). The model has set semantics only; bag semantics is out of scope.

Other paper-only results. Also not in Lean: Corollary 4.9, Proposition 4.10, Remark 4.11, Corollary 5.4, Theorem 3.4, the revised Corollary 3.5, Lemma 3.13, Theorem 3.14, Remark 3.15, Corollary 3.11, Theorem 3.12, Theorem 3.17, Theorem 3.18, Corollary 3.19, Proposition 3.21, all of Section 3.6, all of Section 4.5 and Proposition 6.1 are proved on paper only; Appendix G describes how they were checked.

We have also written, but not formalized, `Nat`-valued Lean statements for most results of Section 3; none of them is in the package and none is tagged Lean-checked here.

Trust boundary. A reader must trust the Lean kernel, mathlib, and that the definitions above say what the paper says: chiefly that `adj` is the incidence matrix, that the norm instance is the ℓ_2 operator norm, and that `Stats.U` lists the nine terms. For the bridge, the reader must also trust that the replica models the Zig kernel. These definitions are short and are reproduced in the package's documentation.

Prior Leatherback results not claimed here. The engine's Std-only Lean workspace already proves, for the same kernel, $\mathrm{Out} \le M$, that the adaptive budget is at most either fixed orientation, and $M^2 \le |R|\ |S|\ |T|$ (`AdaptiveTriangle`, `TriangleBound`). An unrelated condensed path-counting module (`Path/CondensedCount.lean`) is not part of this paper.

E. Background: two spectral facts

We recall two facts used for planning and certification. Neither is a contribution.

Fact E.1 (the left-deep intermediate)**.** *Let $R \bowtie S$ be the intermediate of a left-deep plan, viewed as the bipartite graph from R-tuples (x, y) to S-tuples (y, z) that share y. If R and S are read as the arcs of one layered digraph $X \to Y \to Z$, this graph is the line digraph restricted to arcs from R into S. It is the vertex-disjoint union over y of complete bipartite graphs $K_{d_R^-(y), d_S^+(y)}$. Hence its biadjacency matrix has singular values $\sqrt{d_R^-(y) d_S^+(y)}$ (one*

per y with both degrees positive, the rest zero) and $|R \bowtie S| = \sum_y d_R^-(y) d_S^+(y)$, computable exactly by one merge of two degree arrays.

[**Known**] (elementary)

Proof. An R-tuple and an S-tuple are adjacent iff they share y, so the matrix is block diagonal with one all-ones block J of shape $d_R^-(y) \times d_S^+(y)$ per y. A rank-one $J_{p\times q}$ has the single nonzero singular value $\sqrt{pq}$. □

A planner can compare the exact left-deep intermediate $\sum_y d_R^-(y) d_S^+(y)$ against the bound of Corollary 5.3 when deciding between WCOJ and a binary plan. This is a heuristic switch rule, not a theorem.

Fact E.2 (signless-Laplacian bound for symmetric predicates)**.** *Let G be a simple undirected graph with at least one edge, adjacency A, vertex–edge incidence matrix E, degree matrix* Deg*, signless Laplacian $Q = \mathrm{Deg} + A = EE^\top$ with largest eigenvalue $q_1 = \lambda_1(Q)$, and line graph $L(G)$. Then $\|A\|_2 = \lambda_1(A) \le q_1/2 = (\lambda_1(L(G)) + 2)/2$.*

[**Known**] (classical [CRS10])

Proof. A is symmetric and nonnegative, so $\|A\|_2 = \lambda_1(A)$ with a unit Perron vector $v \ge 0$. Then $\lambda_1 = \sum_{ij\in E(G)} 2v_iv_j \le \sum_{ij\in E(G)} (v_i^2 + v_j^2) = v^\top \mathrm{Deg}\, v$, so $q_1 \ge v^\top Q v = v^\top \mathrm{Deg}\, v + \lambda_1 \ge 2\lambda_1$. Finally $E^\top E = 2I + A(L(G))$ has the same nonzero eigenvalues as $EE^\top = Q$, so $q_1 = 2 + \lambda_1(L(G))$. □

Our experiments do not use Fact E.2: they certify σ with Collatz–Wielandt steps on $A^\top A$ directly. The exact norm is never larger than $q_1/2$, since $q_1/2 \ge \lambda_1$, but a bound from finitely many Collatz–Wielandt steps can be: on the path with 25 vertices, five steps on $A^\top A = A^2$ from the all-ones vector give the bound 2, while $q_1/2 < 2$.

F. An earlier fully reduced gadget

An earlier draft established the fully reduced interval $[1/12, 1/8]$

with the following coordinate gadget. Theorem 3.14 supersedes its exponent, but I_v gives a smaller explicit instance with $C > N + B_2$.

The gadget I_v. For an integer $v \ge 1$ let all coordinates range over $[v]$, and take tagged classes $X = \{(A, B) : A, B \in [v]^3\}$, $Y_0 = [v]^4$, $Z_1 = [v]^5$, $Y_2 = \{(A, B, E, e)\}$ and $Z_2 = \{(A, E, F) : F \in [v]^2\}$, with $A, B, E \in [v]^3$ and $e \in [v]$. Put $f(A, B) = (A, B_1, B_2) \in Z_1$ and

$$
\begin{aligned}
R &= X \times Y_0 \cup \{((A, B), (A, B, E, e))\},\\
S &= Y_0 \times Z_1 \cup \{((A, B, E, e), (A, E, F))\},\\
T &= \{(f(x), x) : x \in X\} \cup \{((A, E, F), (A, B))\},
\end{aligned}
$$

where unspecified coordinates range freely.

Theorem F.1 (a fully reduced separation)**.** *Every tuple of I_v lies in a triangle, and* $\mathrm{Out}(I_v) = v^{10} + v^{12}$*. Let $\mathcal{U}_v$ be the disjoint union of I_v and its two rotations. Then $\mathcal{U}_v$ is fully reduced,*

$$N = 3\big(v^{12} + v^{11} + 2v^{10} + v^9 + v^6\big), \quad C = v^{15} + 3v^{12} + v^{11} + 4v^{10} + v^9 + 2v^6,$$
$$B_2 = 7v^{14} + v^7, \quad \mathrm{Out} = 3\big(v^{12} + v^{10}\big),$$

so $C/(N + B_2) \sim v/7 = \Theta\big(N^{1/12}\big)$ and $C = \Theta\Big(\big(N + B_2\big)^{15/14}\Big)$; $B_2 = \Theta\big(N^{7/6}\big)$, and $\mathcal{U}_v$ attains the exponent 5/4 *of Theorem 3.3 at slope* 7/6*. At $v = 8$, $C/(N + B_2) = 45017942/39450457 > 1$. On every fully reduced instance $B_2 \geq \mathrm{Out} \geq \max(|R|, |S|, |T|) \geq N/3$, so no fully reduced family has slope $\kappa < 1$. Hence the worst-case per-binding gap against $N + B_2$ on fully reduced inputs is at least* 1/12*.*

[Proved] (exact checks in Appendix G)

Proof. Support. An $X \times Y_0$ tuple (x, y) closes with $f(x)$; a $Y_0 \times Z_1$ tuple with any preimage of its Z_1-endpoint under f; a tuple $(f(x), x)$ with any $y \in Y_0$. A tuple $((A, B), (A, B, E, e))$ closes with any (A, E, F); a tuple $((A, B, E, e), (A, E, F))$ with (A, B); a tuple $((A, E, F), (A, B))$ with any (A, B, E, e). S joins only Y_0 to Z_1 and Y_2 to Z_2, so there are no mixed triangles and $\mathrm{Out} = v^3v^3v^4 + v^3v^3v^3v \cdot v^2 = v^{10} + v^{12}$. *Degrees.* On X: $d_R^+ = 2v^4$, $d_T^- = v^5 + 1$; on Y_0: $d_R^- = v^6$, $d_S^+ = v^5$; on Y_2: $d_R^- = 1$, $d_S^+ = v^2$; on Z_1: $d_S^- = v^4$, $d_T^+ = v$; on Z_2: $d_S^- = v^4$, $d_T^+ = v^3$. So $(|R|, |S|, |T|) = (2v^{10}, v^{12} + v^9, v^{11} + v^6)$, $(\alpha, \beta, \gamma) = (4v^{14}, 2v^{14}, v^{14} + v^7)$ and $(\alpha', \beta', \gamma') = (v^{16} + v^{10}, v^{16} + v^{13}, v^{16} + 2v^{11} + v^6)$. *Costs.* At x, $c_R(x) = v^9 + v^6 + 2v^4$ and $c_T(x) = v^9 + v^5 + v^4 + 1$, smaller by $v^6 - v^5 + v^4 - 1 \geq 0$. At Y_0 option S costs $v^6 + v^5$, at Y_2 option R costs $1 + v^2$; at Z_1 option T costs $v^5 + v$, at Z_2 option S costs $2v^4$. So $C_x(I_v) = v^{15} + v^{11} + v^{10} + v^6$, $C_y(I_v) = v^{12} + 2v^{10} + v^9$ and $C_z(I_v) = 2v^{12} + v^{10} + v^6$. *Union.* Each root of $\mathcal{U}_v$ costs the sum of the three, every forward moment is $7v^{14} + v^7$ and every reverse moment $3v^{16} + v^{13} + 2v^{11} + v^{10} + v^6$, larger for $v \geq 2$. *Obstruction.* A triangle has one tuple of each relation, so on a fully reduced instance $\mathrm{Out} \geq \max(|R|, |S|, |T|) \geq N/3$, and $\mathrm{Out} \leq B_2$ (Section 2). The upper end 1/8 is Corollary 3.5(1). □

G. How the results were checked

The proofs in this paper are on paper, except the results of Section 4 and Section 5 tagged Lean-checked (Appendix D). Every "verifier" below is a model-run agent (Section 9). Every construction found by the model-run solvers of the third and fourth drafts was rebuilt in exact integer and rational arithmetic (Python, run as `python3 -I`) by at least two verifier agents that did not run or import the solvers' scripts; each verifier also checked the proofs line by line and tried to refute the claims. For the third draft, 55 statements went through this check: 51 were accepted as stated, 4 after a repair, none rejected. For the fourth draft the fully reduced theorem, the 4-cycle theorem and the engine's rescue stage went through it: both theorems were accepted after presentation repairs (the 4-cycle sharpness threshold $L \geq t^3$ instead of $L \geq t^2$; ten dihedral orbits instead of nine; a duplicated factor in the rounding display), one auxiliary claim was refuted (the value 1/11 at slope 1, Remark 3.15),

and the engine audit refuted the claim that the implemented candidate pruning preserves the searched rule's decisions (Appendix B). These checks cover instances, closed forms and proof steps; they are evidence, not proofs, and none of these results is tagged Lean-checked.

The final review by Weaver (Section 9) re-implemented the cost model and every explicit family of Section 3 from the paper's definitions alone, in exact integer arithmetic, sharing no code with the artifacts; it reproduced every closed form it rebuilt, found no violation of the upper bounds of Section 3 and Section 4 on 5,987 random instances and 3,000-step hill climbs per ratio, found a minimum $\mathrm{Out}/U = 0.2521$ over 200,000 random regular statistics (the block witness admissible every time), and a worst $(\mathrm{pricing} - 4n)/C$ between 3.00 and 4.00 for the scheduler of Proposition C.2 on 3,996 random instances (the value depends on the tuple order, which follows Python's per-process string hashing; over 50 hash seeds it was 3.00, 3.20 or 4.00). Its scripts and recorded outputs are in the paper repository as an outside check.

Table 10 names, for each group of results, who found or proved it and who checked it; Table 11 lists the exact checks per result, with paths relative to `artifacts/`.

Table 10: Provenance. CC: the orchestrating Claude Code session (Claude Opus 5.5) and the Claude subagents it launched (theorem search, provers, referees, verifier lenses, auditors); FA: Claude Fable 5.1 agents at maximum effort; AX: Codex CLI agents with GPT-6-Astra at "ultra" effort; R1: the GPT-5.6-sol Pro review of the first draft; R2, R3: the GPT-6-Astra Pro reviews of the second and third drafts; G6: a separate GPT-6 Pro session run by the author on the second draft; WV: Weaver (final review). "Checked" lists exact rebuilds and proof audits, not the author's reading, which covers everything.

Results	**Found or proved by**	**Checked by**
Theorem 4.1, Lemma 4.2, Lemma 4.3, Lemma 4.4, Corollary 4.8; Section 5	CC; Lean proofs written by CC	Lean kernel; R1, R2 audits; WV
Theorem 4.7	family in rational (Pell) form: R1; real form and Lean proof: CC	Lean kernel; R2
Corollary 4.9, Proposition 4.10, Remark 4.11, Corollary 5.4	AX (solver); budget-only scope of the planner example: R3	two CC verifier lenses; FA verifier; R3
Theorem 4.12, Lemma 4.13, Proposition 4.14	FA (solver)	AX and FA verifiers (one more AX audit for the 4-cycle); CC rebuild of the sharpness family
Lemma 3.1, Lemma 3.2, Theorem 3.3, Theorem 3.6, Lemma 3.8, Theorem 3.10, Theorem 3.16, Proposition C.4	CC (theorem search, provers, referees); proof repair of Theorem 3.3 (geometric mean of nonnegative terms): R2 and G6	CC evaluators; R1, R2, R3 recomputations; WV
Proposition C.2, Remark C.3	CC; tightness of the constant 6 and the counterexample to the first bit-discard rule: R1	CC simulation; R2; WV
Corollary 3.11	R2	CC exact script; R3; WV
Theorem 3.4, Theorem 3.12, Theorem 3.17, Theorem 3.18	AX and FA, independently (the D gadget of Theorem 3.12 is AX's)	two CC verifier lenses; FA verifier; R3; WV
Corollary 3.19	R3	CC

Results	Found or proved by	Checked by
Lemma 3.13, Theorem 3.14, Remark 3.15	FA (solver); in Remark 3.15, the isolated-triangle padding of (a) and the cap of (b) with its added cost lemma: AX (verifier)	AX and FA verifiers (AX refuted the unpadded "1/11 at slope 1"); CC rebuild
Theorem F.1	AX (solver)	two CC verifier lenses; FA verifier; R3; WV
Proposition 3.21	(1)–(2): R2 and G6; (3)–(4): G6	CC exact scripts; R3
Definition 3.22, Lemma 3.23, Lemma 3.24, Theorem 3.25, Theorem 3.26, Remark 3.27	AX and FA, independently	two CC verifier lenses; FA verifier; R3
Theorem 3.29, Theorem 3.32, Lemma 3.31	AX; three-way refinement of Lemma 3.31, the randomized proof and the densification: R3	two CC verifier lenses (one repair); FA verifier; R3
Corollary 3.30	R3	CC
Remark B.2	AX (port of the engine routine)	two CC verifier lenses; FA verifier
Proposition 6.1	stride instance: AX; statement for every fixed grid and proof: CC	CC exact script; R3
Engine operator, planner, E2 and engine runs (Appendix A, Appendix B)	CC (implementation and runs); admission rule searched by AX over 7,056 configurations	first audit: AX; audit of the fix: CC; recomputations from the raw rows by R1 (E2) and R2 (E2 and engine)

Table 11: Exact checks behind the results proved on paper.

Result	Checks and artifacts
Theorem 3.3	not formalized; the integer form and all six per-root forms are also checked as exact integer inequalities on six runs of 4,500 random instances by `verify/thm1/rnd.py`, worst ratio 0.64 for the theorem and 0.69 per root.
Theorem 3.4	on paper. The integer form held on 5,000, 10,500 and 10,720 random and skewed instances plus 36 structured ones, in three independent runs; largest ratio in the report's run $9.6 \cdot 10^{-16}$, at $(N, C) = (211, 601)$: `hillclimb/P2-exponents/verify_new_upper.py`.
Theorem 3.10	also checked exactly at $(i, j) = (1, 1)$ for $u = 2, 3$ and at $(2, 1), (2, 2)$ for $u = 2$; ratio $C/\left(N^{3/8}B_2^{3/4}\right)$ about 0.236 to 0.312 (0.312, 0.279, 0.236, 0.267 at $(i, j, u) = (1, 1, 2), (1, 1, 3), (2, 1, 2), (2, 2, 2)$)
Corollary 3.11	exact closed forms checked for all $1 \le j \le i \le 40$ at $u = 2$, and at $\kappa = \sqrt{2}$, where $j_n = 3n - \mathrm{isqrt}(8n^2) - 1$ gives $(n, j_n) = (6, 1), (7, 1), (10, 1), (20, 3), (40, 6)$: `hillclimb/checks/slopes_check.py`. The extension is from R2 (Table 10).
Theorem 3.12	on paper. The D-statistics checked by literal enumeration on 7 triples by `d_gadget_check.py`; the union at $p = q = 1$ against $F_{1,1}(2), F_{1,1}(3)$; the closed forms evaluated exactly at $(u, p, q) = (2, 9, 10), (2, 10, 11), (2, 19, 20)$, ratios $C/(N + B_2) = 0.980, 1.96, 1004.4$: `hillclimb/P2-exponents/verify_slopes.py`, `hillclimb/checks/d_gadget_check.py`.
Theorem 3.17	on paper. 10,000 + 10,500 random instances and hill climbs, largest ratio exactly 1, at complete tripartite instances; GMIN enumerated literally for $m = 1, ..., 8$: `hillclimb/P2-exponents/best_six_verify.py`.
Theorem 3.18	on paper. All three drivers checked on the verifiers' random instance sets (10,500 and 10,720 + 7,169), largest ratio 1, attained on GMR and complete tripartite instances; GMR for $m = 1, ..., 12$ and the sharp family for $u = 1, ..., 6$ enumerated literally: `hillclimb/P2-exponents/verify_single_order.py`.
Proposition 3.21	exact checks of (1)–(2) on $G_8(2), G_8(3)$, $\mathcal{F}_a$ for $a = 2, 3, 4, 5, 8$ and GMIN$(2, 3, 4)$, and of (4) on 4,000 random symmetric graphs: `hillclimb/checks/selfjoin_check.py`.

Result	Checks and artifacts
Proposition C.2	`verify/thm1/rnd.py` simulates the scheduler on six runs of 4,500 random instances and asserts (1) and the pricing bound of (3); no violation.
Theorem 3.25	exact values $(N, B_2, Q) = (2208, 1280, 1312)$, $(48843, 32805, 46170)$, $(457728, 327680, 590848)$, $(2634375, 1953125, 4300000)$ for $m = 2, ..., 5$ by both solvers and the verifier rebuilds; padded $m = 2, 3$ with unchanged hard certificates: `hillclimb/P1-sorted/`.
Theorem 3.26	exact certificate sums for parameters $2, ..., 5$: $\mathcal{F}_a$ $6, 30, 84, 180$; GMR $48, 405, 1792, 5625$; GMIN and padded GMIN (hard tuples) $112, 1377, 7936, 30625$; $\mathcal{H}_a$ $12, 45, 112, 225$; $F_{i,j}$ at $(i, j, u) = (1, 1, 2), (1, 1, 3), (2, 1, 2), (2, 2, 2)$: $Q = 1312, 46170, 4268032, 590848$: `hillclimb/P1-sorted/separations/check.py`, `hillclimb/P1-sorted/slopes/verify_families.py`.
Lemma 3.28	bit-length identity and $h^2 \le 9pq/2$ checked exhaustively for $s < 300$, $l < 2000$: `hillclimb/checks/slopes_check.py` asserts $h^2 \le 9pq$, and on the same range the largest $h^2/(pq)$ is $2691/599 < 9/2$, at $(p, q) = (299, 599)$.
Theorem 3.29	on paper; galloping measured at most $5.97h$ in the verifier's port: `hillclimb/verified.md`, P1.7.
Theorem 3.32	on paper; degree formulas, decision-tree argument, simultaneous ordering and quantifiers re-derived by the two CC verifier lenses: `hillclimb/P1-sorted/slopes/log_normalized_proof.md` and `log_normalized_checks.py`.
Remark B.2	exact counts from a Python port checked line by line against the Zig source and against a Zig harness compiled with Zig 0.16.0: `hillclimb/P1-sorted/engine/`.
Corollary 4.9	on paper. Exact checks: exhaustive 0/1 triples on small shapes with operator norms decided by exact determinants, 44,200 instances of the sharpness family, and the grids of the archived report, 0 failures: `hillclimb/P3-spectral/`.
Proposition 4.10	exact checks: all nine witnesses materialized at $n = 2, 3, 5$ with exactly computed norms; 21,952 exponent regimes enumerated; `hillclimb/verified.md`, P3.5–P3.6.
Corollary 5.4	exact checks: 4,096 matrix triples, 73,984 budget cases, 0 failures; `hillclimb/P3-spectral/`. The rotated-star example rebuilt literally for $k = 2, 3, 4, 8$: `hillclimb/r4-fullred/editor/check_r4.py`.
Proposition 6.1	on paper; every quantity checked in exact arithmetic for the stride, the golden-ratio grid and two fixed pseudo-random grids, $m \in \{64, 65, 96, 128, 200, 256\}$.
Lemma 3.13, Theorem 3.14	Two verifiers rebuilt the live gadget literally: closed forms equal brute force on 733 parameter tuples and 60 random three-copy unions (all fully reduced); $\mathcal{L}(v)$ by brute force for $v = 2, 3, 4$ ($(N, C, \mathrm{Out}, B_2) = (8976, 12464, 5760, 29264)$, $(355266, 888246, 262440, 2441502)$, $(5263104, 22332672, 4177920, 56889600)$) and by closed forms for $v \le 60$; the general family on 100 tuples and literally at $(i, j, u) = (2, 1, 2)$, $(1, 0, 2)$, $(1, 0, 3)$, $(2, 0, 2)$; the CC editor rebuilt $\mathcal{L}(2)$, $\mathcal{L}(3)$ and $(2, 1, 2)$ from the text of Section 3.4: `hillclimb/r4-fullred/`.
Remark 3.15	(a) exact closed forms for $v = 2, ..., 5$ and literal rebuilds for $v = 2, 3$; (b) the cost lemma on 26,382 tuples, and both linear identities with exact dual certificates of an exact rational LP over the design (also with the size exponent forced to 1); `hillclimb/r4-fullred/astra/LP_REPORT.md`.
Theorem 4.12, Lemma 4.13	Two verifiers: line-by-line proof audits; exact enumeration of the dual polyhedron of the arc program (50 vertices, 38 needed with rational strict-separation certificates, identical for 9 and 13 arcs); all 38 inequalities certified with exact eigenvalue brackets on the 65,536 0/1 instances of shape $(2, 2, 2, 2)$, 91,173 near-tight instances of shape $(2, 2, 2, 3)$ and about 72,000 random, adversarial and Kronecker instances, no violation; Lemma Λ on random and annealed instances; the rounding on 11,296 grid vectors (worst loss $4.48 < 16$): `hillclimb/r4-cycle4/`.

Result	Checks and artifacts
Proposition 4.14	$U_4^{12} = t^{36}$ exactly at $L = 8$ for several $t < 2$; the envelope $\min\left(t^3, \sqrt{Lt^3}\right)$ by an exponent argument and exact checks; the adjacent variant: `hillclimb/r4-cycle4/astra/proof_exact.py`, `hillclimb/r4-fullred/editor/check_r4.py`.
Theorem F.1	Literal enumeration with a triangle-support check of every tuple for $v = 1, ..., 4$ by two separately written evaluators, and closed forms for $v \leq 8$: `hillclimb/P2-exponents/verify_fully_reduced.py`, `hillclimb/r4-fullred/fable/check_iv.py`.

H. Artifacts

The paper repository, `github.com/TJC-LP/triangle-adaptivity-paper`, holds the source of this paper and its figure scripts, the Lean workspace of Appendix D, and everything needed to check the computational claims; its `README.md` indexes it, and every `artifacts/` path in this paper is relative to it. Leatherback itself will be open-sourced; once it is public, the repository will name the Leatherback versions the engine runs used. Until then the repository holds the engine-run scripts, the raw rows and the summaries built from them.

- `artifacts/e2/` (Appendix A): the preparation, bound, run and summary scripts; `results.csv` and `summary.md` with their SHA-256 digests; run logs; per-predicate spectral data; dataset URLs with SHA-256 digests; the Python environment (3.11.9, numpy 2.4.6, scipy 1.17.1, numba 0.68.0); and the commands, reconstructed from the logs where they were not recorded. `check/csvcheck.py` recomputes the bounds and ratios and audits the results file for internal consistency, and `check/rebuild_summary.py` reruns the summary script on the committed results file and spectral data and confirms that it reproduces `results.csv` and `summary.md` exactly; `src/recertify.py` recomputes the exact σ certificates and changes only the certificate columns.
- `artifacts/verify/`: `chk_exact.py`, an exact-arithmetic check of Theorem 4.1 (integers and fractions only; exhaustive over $1 \leq s_i \leq r_i \leq 25$, 3.4×10^7 vectors, no failure) and of the family of Theorem 4.7; the evaluators and family checks for Section 3 (`g8.py`, `thm1/`); and `OUTPUTS.md`, which records every command, exit code and output with script digests.
- `artifacts/engine/` (Appendix B): data-generation seeds and digests, raw timing samples, summary and answer-comparison scripts and logs, quality-gate results and the machine description.
- `artifacts/engine-kg/` (Table 8, Table 9): the driver-policy runs on the knowledge-graph triangles, with the query files, the raw per-(query, policy) rows of runs A (`raw/`), B (`raw-rescue/`) and C (`raw-run-c/`, with the synthetic-store reruns and alternating process pairs in `controls-run-c/`), E2′s bounds per query, the scripts that convert the data, build the queries, run the harness and rebuild the summary and the tables, and `estimator_adversary.py` and `scripts/adversary.py`, which check Proposition 6.1 in exact arithmetic.
- `artifacts/hillclimb/`: the reports, scripts and recorded outputs of the model-run solvers and verifiers (Appendix G, Table 10): `verified.md` with the consolidated verdicts and `checks/` for the third draft; `r4-fullred/` (Section 3.4), `r4-cycle4/` (Section 4.5) and `r4-engine-audit/` (the audits of the rescue stage, Appendix B) for the fourth, each with the verifiers' reports and scripts.

- The Lean workspace (Appendix D), with its pinned toolchain and manifest, the spectral library, the kernel replica and both audit files.
- Weaver's outside checks: five Python scripts written from the paper's definitions alone, with their recorded outputs (Appendix G).

The raw datasets and generated graphs are not committed; each README gives the commands to download or regenerate them and their digests.